

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
BATTERY LIMIT DRAWING
PAMC TO BOILER COMPLEX

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING
& CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

REV

0000-PID-0001

A

FIRM
NUMBER
#13619

CHECK PRINT

- GENERAL NOTES
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PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
BATTERY LIMIT DRAWING
AROMATICS TO BOILER COMPLEX

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY: ENG APVL: PDN
DRAWING NO:

0000-PID-0002

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DESIGN BY:	ENG APVL: PDN
DRAWING NO: 0000-PID-0010	

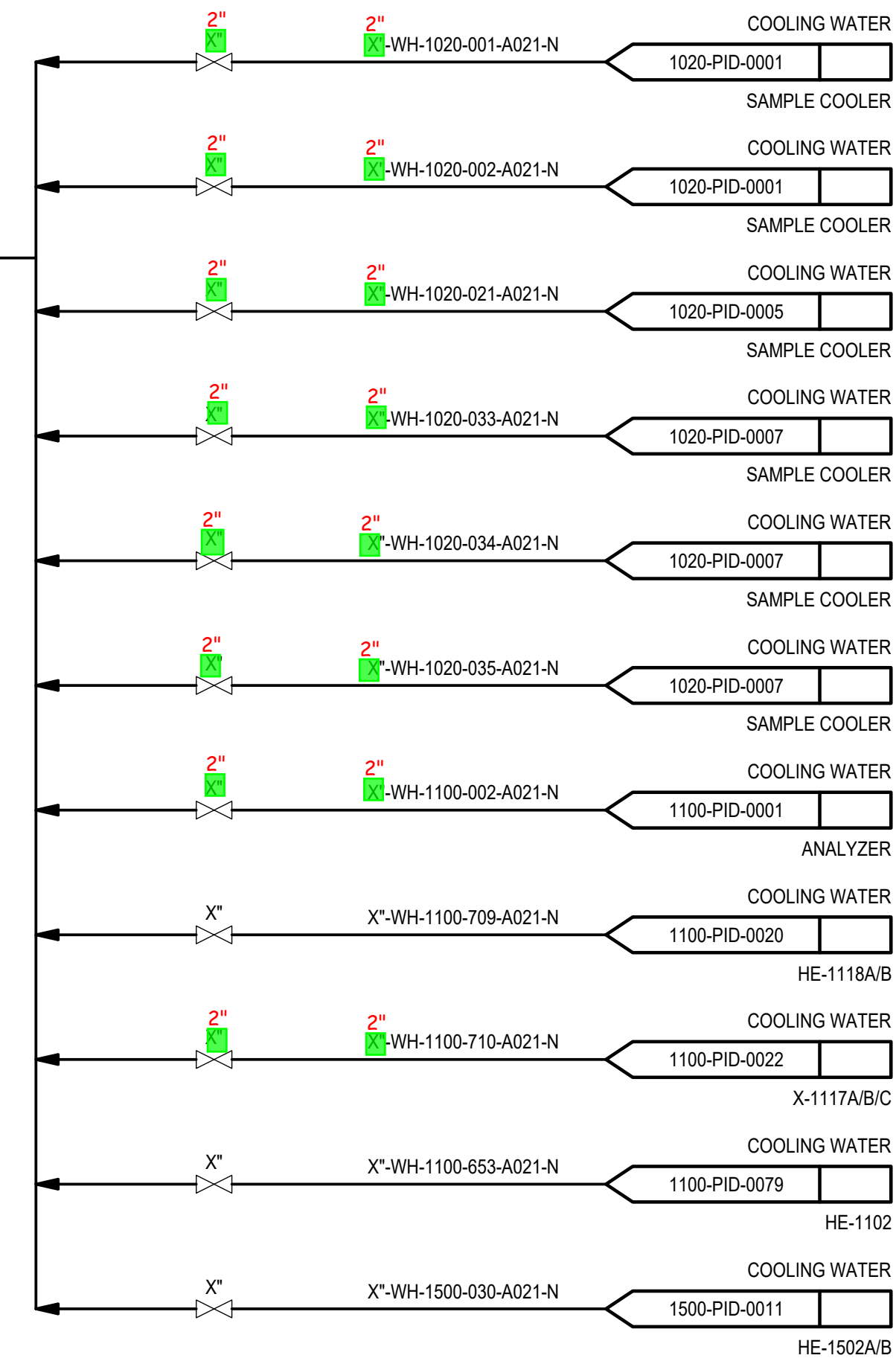
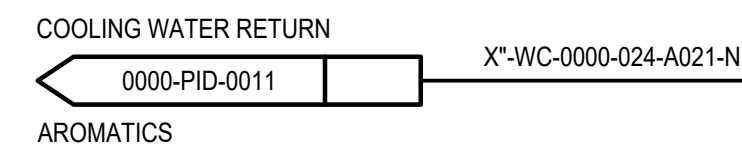
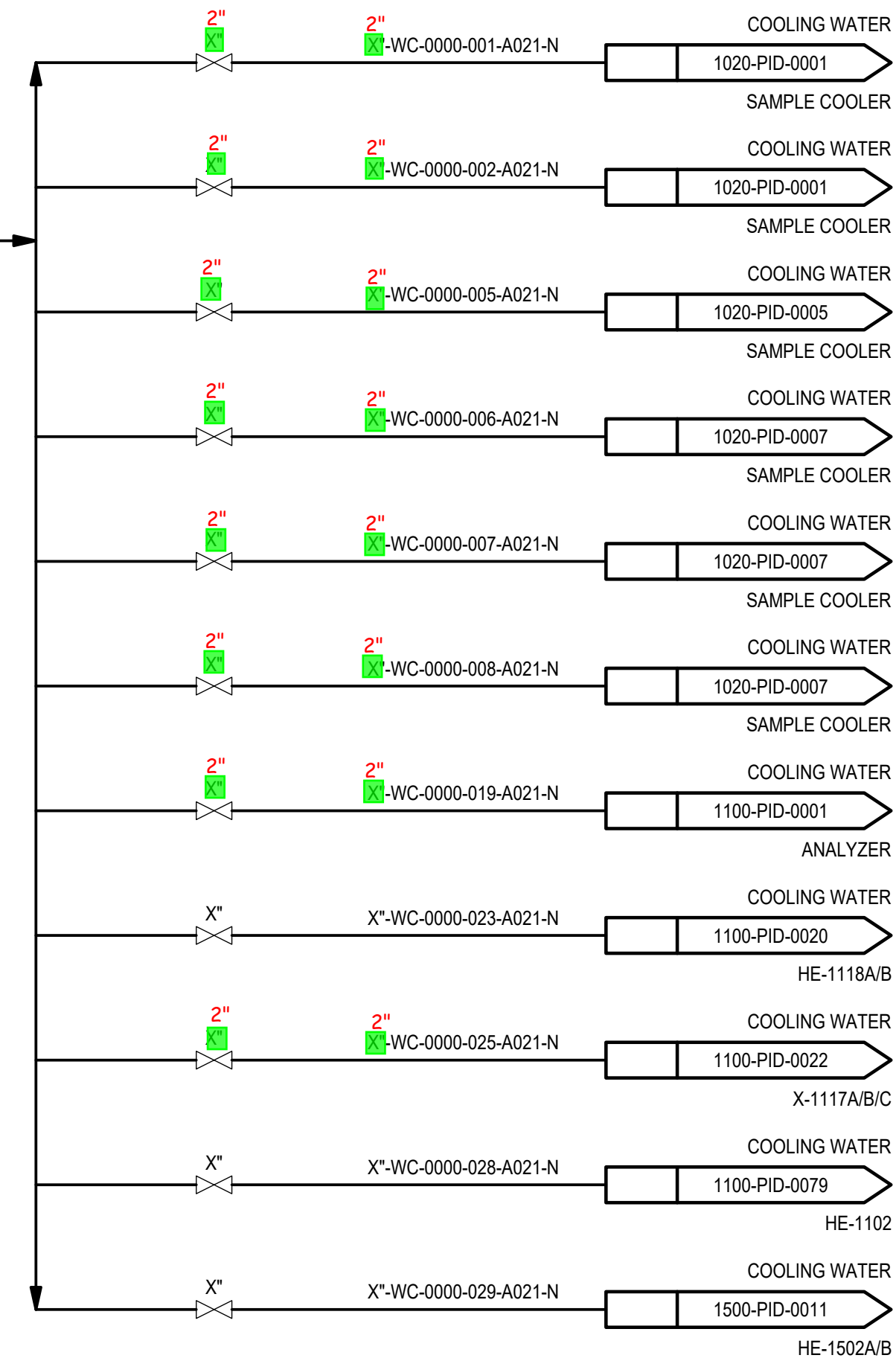
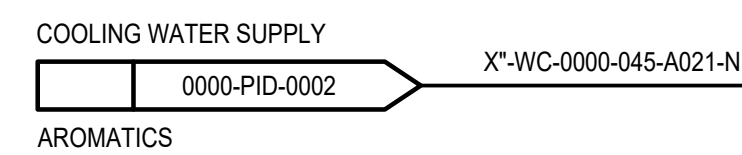
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	A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN													
																		PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	REV A



GENERAL NOTES

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[illegible]

REFERENCE DRAWINGS

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

IOR NO:

PIPING & INSTRUMENTATION DIAGRAM

COOLING WATER DISTRIBUTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

	SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619
	13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040		
	PROJECT NO: 25MA02	SCALE: NONE	
	DRAWN BY: GS	CHECKED: APB	
DESIGN BY:		ENG APVL: PDN	
DRAWING NO:		0000-PID-0020	

DRAWING LOCATION: c:\users\jason\one\active - eng\ssps files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids - (Border Rev1 5 16 25).
PLOT DATE: ---, ---, ---
PLOT BY: jason

- GENERAL NOTES
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CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

NITROGEN DISTRIBUTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

0000-PID-0030

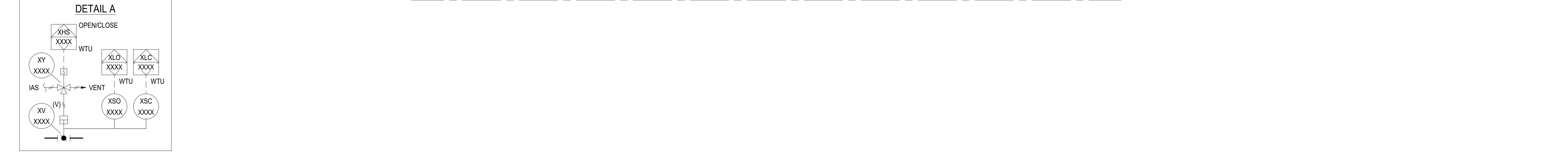
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TING NO:		1000-PID-0001	

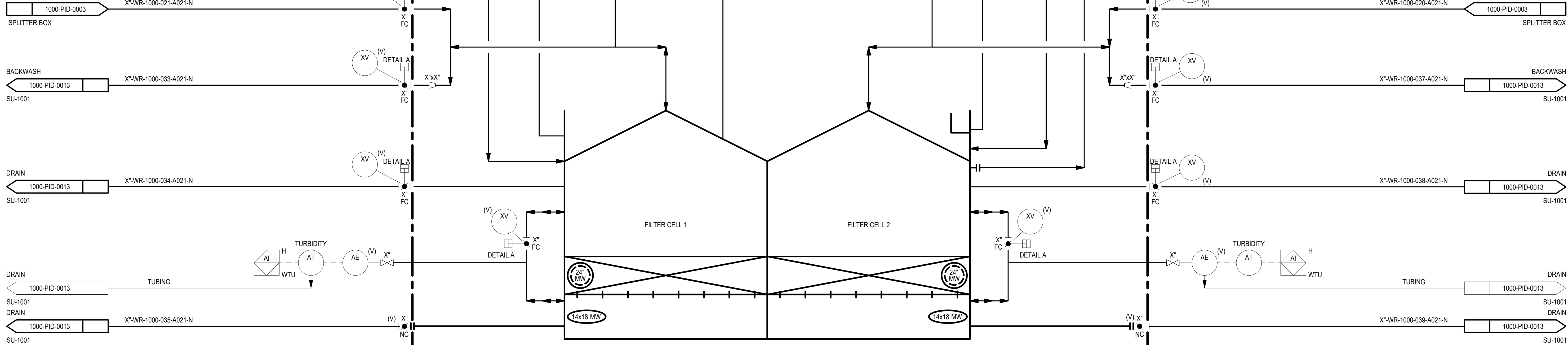
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						PLANT: PORT ARTHUR REFINERY																																			
						JOB NO:																																			

SAND FILTER
DESIGN CAPACITY :
RATED POWER :
MOC :
TRIM : VT-1000-042-A021-N
INSULATION :

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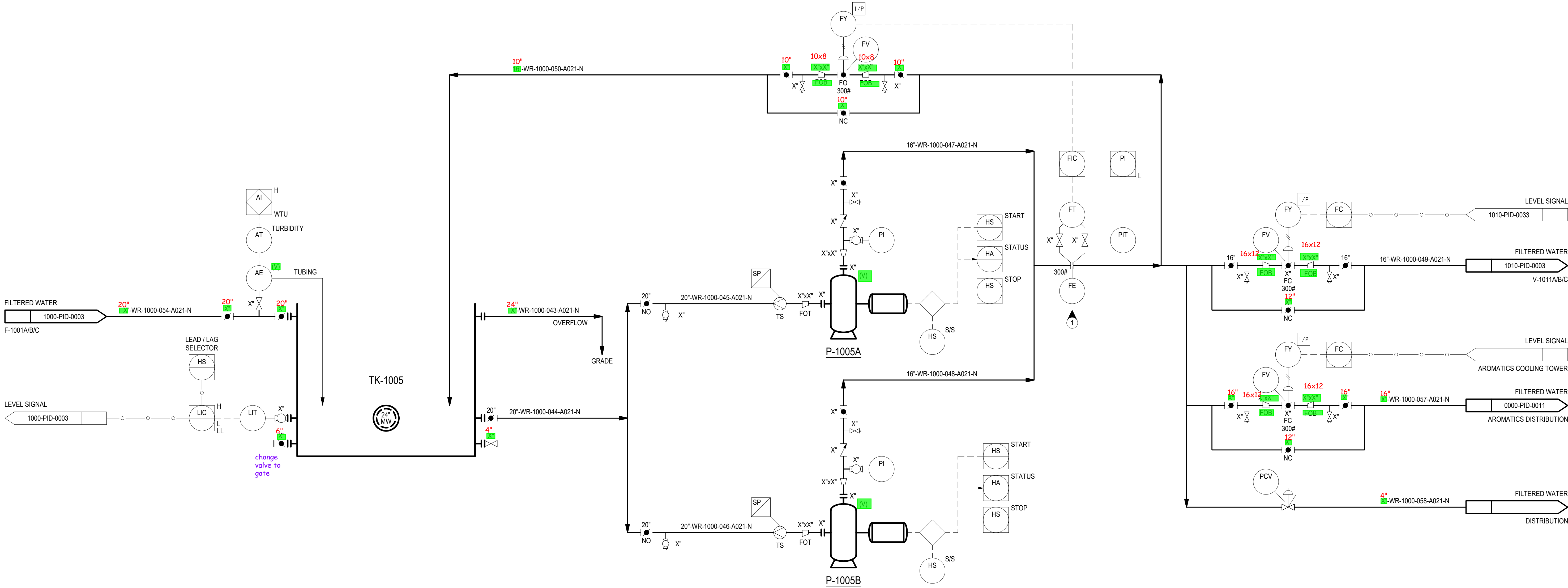
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TK-1005
REPRESSURIZATION TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1000-051-A021-N
INSULATION / TRACING :

P-1005A/B
FILTERED WATER PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1000-052-A021-N
SEAL PLAN :

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NOTES
1. ORIFICE PLATE IS FLOW CONDITIONING ORIFICE, WHICH REQUIRES 2 DIAMETERS STRAIGHT RUN UPSTREAM AND DOWNSTREAM.



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CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM
PIPING & INSTRUMENTATION DIAGRAM
SUPERNATANT SUMP

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

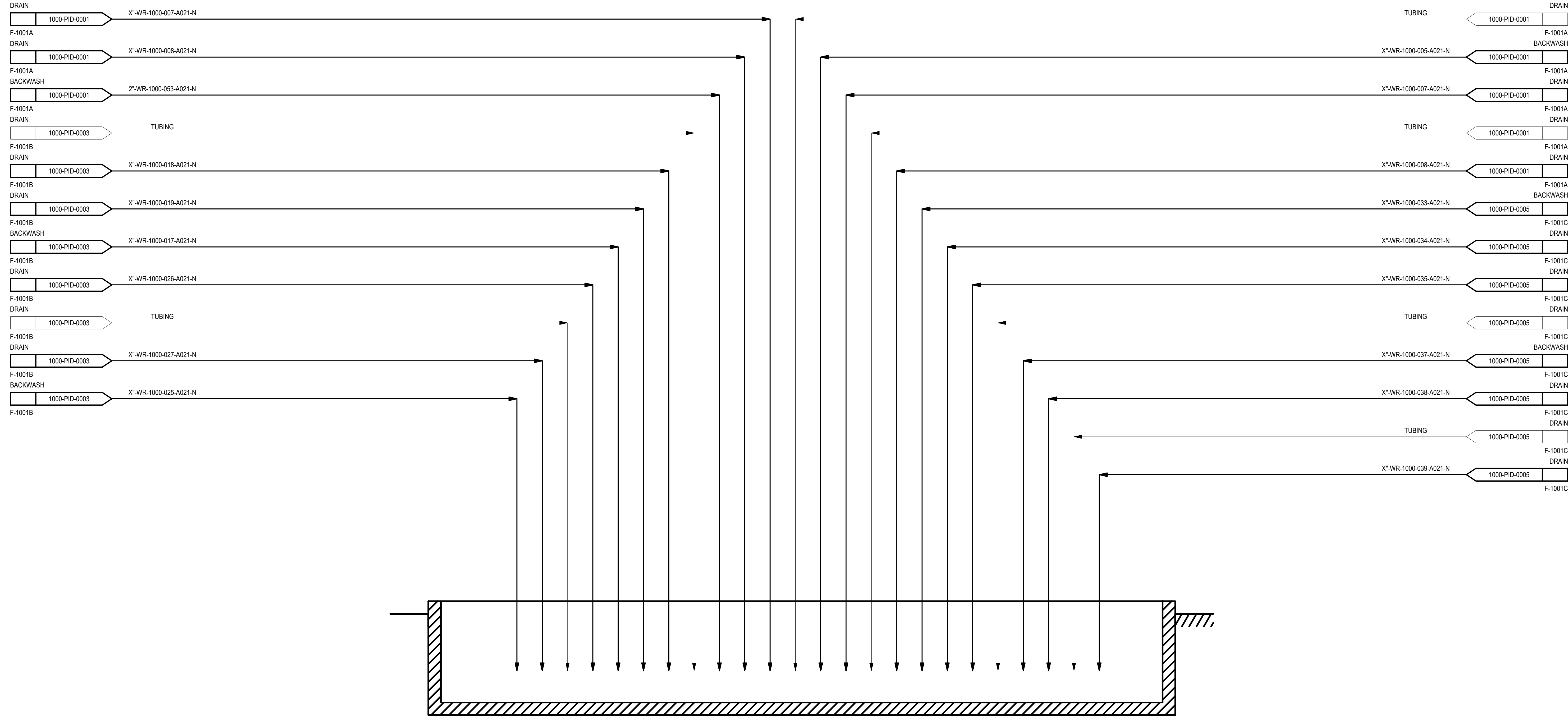
FIRM NUMBER
#13619

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:
1000-PID-0011

REV
A

SU-1001
SUPERNATANT SUMP
SIZE :
WORKING CAPACITY :
INTERNALS :
MOC :
TRIM : VT-1000-056-A021-N

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-000X.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM
PIPING & INSTRUMENTATION DIAGRAM
SUPERNATANT SUMP

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

1000-PID-0013

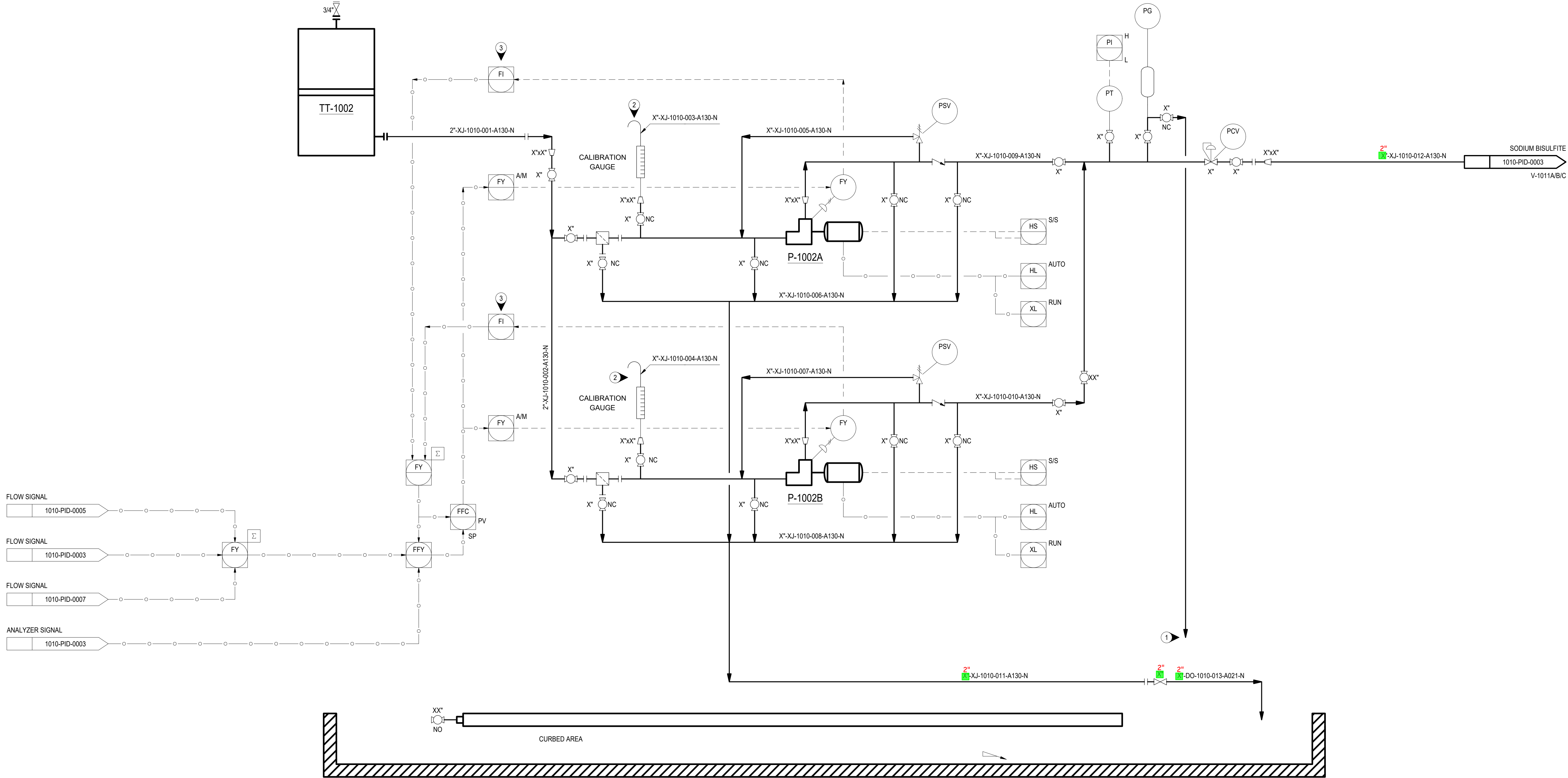
REV

A

TT-1002
SODIUM BISULFITE STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-014-A130-N
INSULATION / TRACING :

P-1002A/B
SODIUM BISULFITE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-015-A130-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. OPEN END DRAIN TO CURB AREA.
2. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.
3. STROKE FEEDBACK (FI-XXXX) AND STROKE CONTROL (FO-XXXX).



FLOW SIGNAL
1010-PID-0005

FLOW SIGNAL
1010-PID-0003

FLOW SIGNAL
1010-PID-0007

ANALYZER SIGNAL
1010-PID-0003

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MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PROCESS FLOW DIAGRAM

SODIUM BISULFITE INJECTION SYSTEM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

FIRM NUMBER #13619

SCALE: NONE

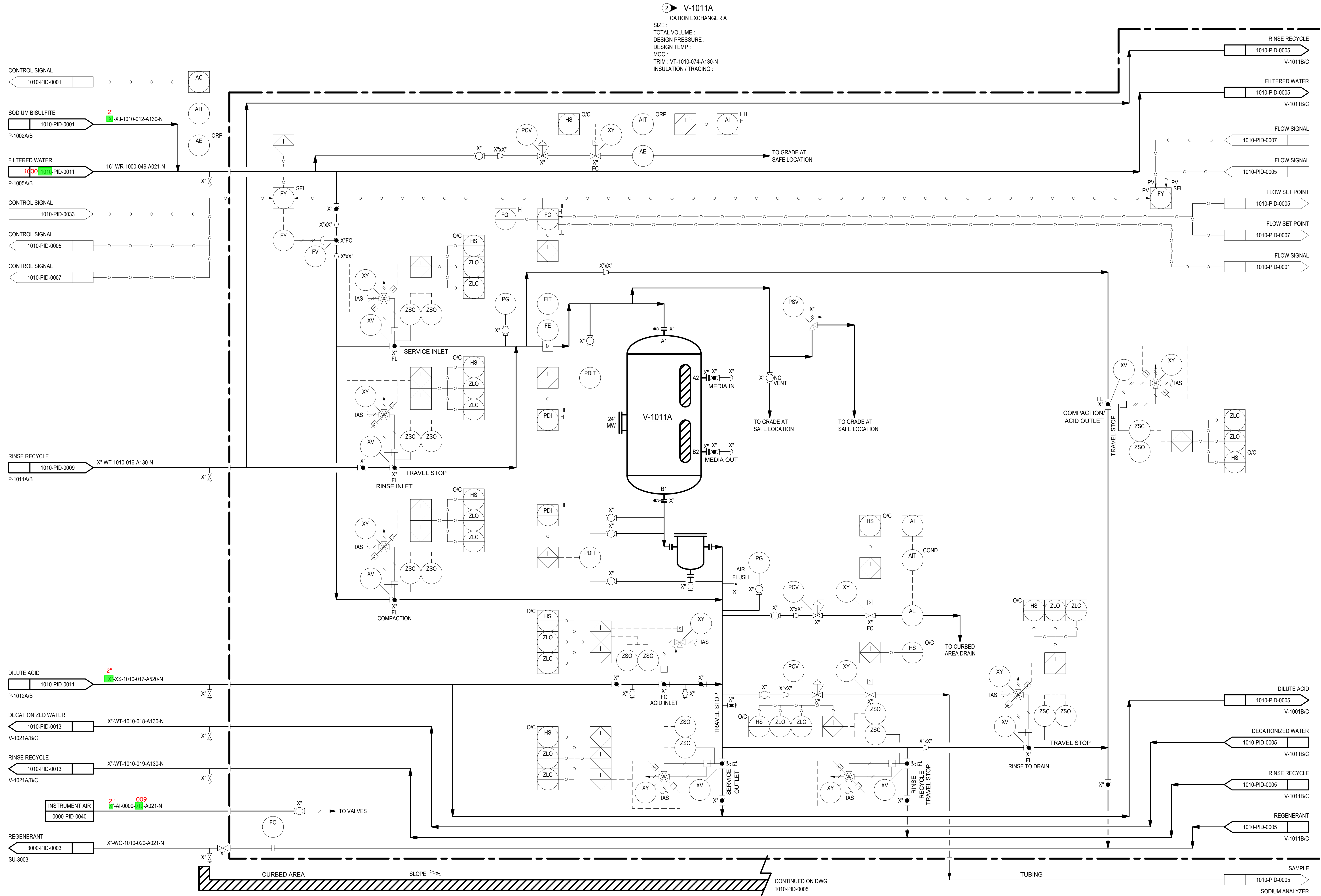
CHECKED: APB

ENG APVL: PDN

1010-PID-0001

REV A

CHECK PRINT



② V-1011A
CATION EXCHANGER A
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-074-A130-N
INSULATION / TRACING :

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- NOTES
1. ACID RESISTANT LINING.
2. CATION EXCHANGER SHOWN IS TYPICAL EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

DRAWING LOCATION: c:\users\jason\one-drive - eng\ssps\files - 25ma02 - aromatics utility scope old eyes review\04 - design\04 - process\04a -

PLOT DATE: ---

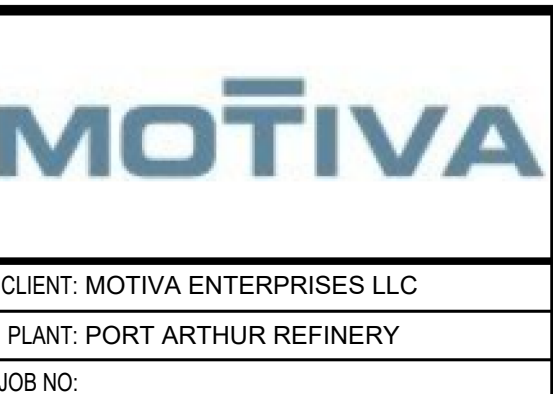
PLOTTED BY: jason

DRAWING: 1010-pid-0003.dwg

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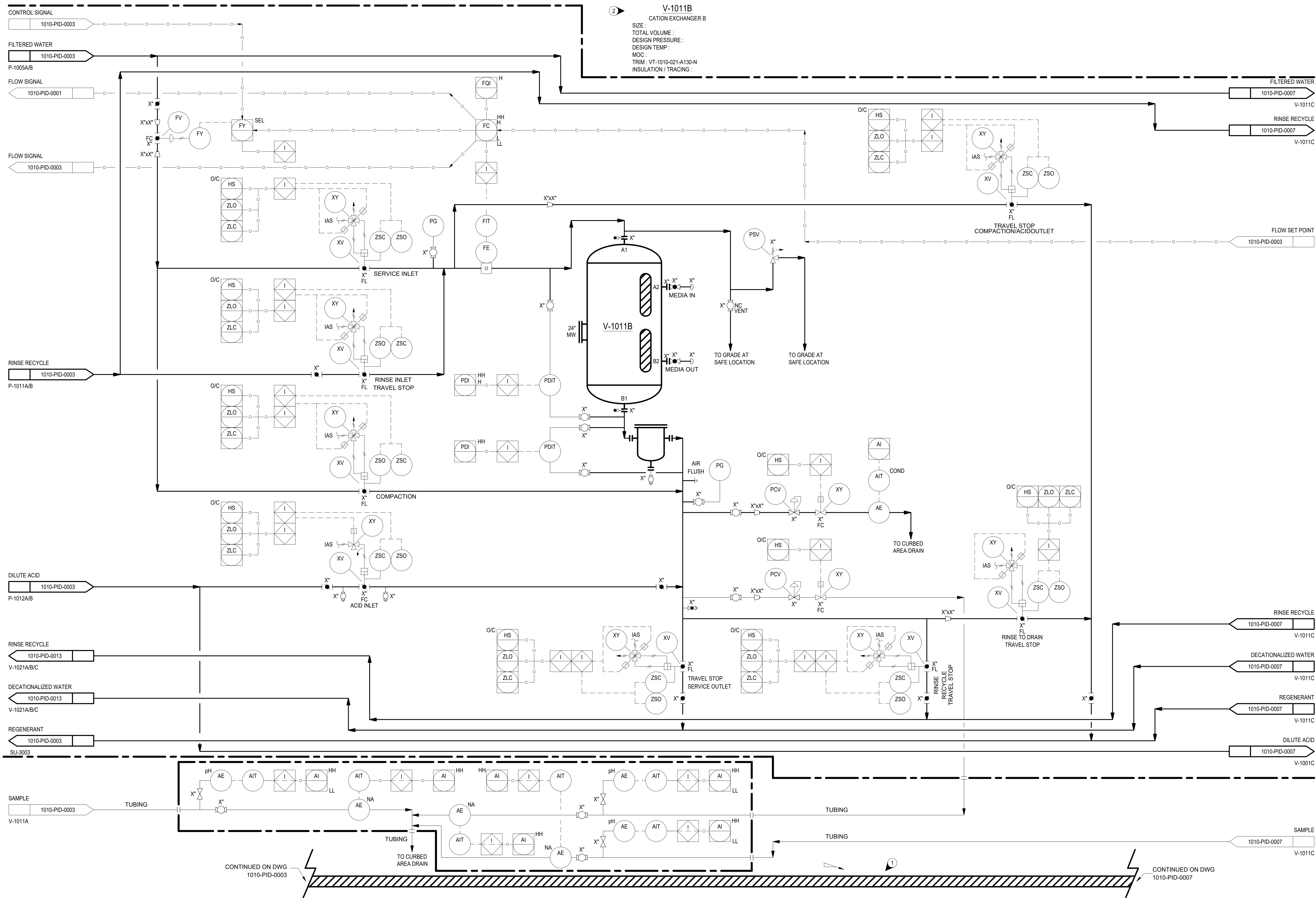


PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZED WATER SYSTEM
CATION EXCHANGER A

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040			
PROJECT NO: 25MA02	SCALE: NONE	DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN		
DRAWING NO:	1010-PID-0003	REV	A

CHECK PRINT



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION
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REV #	DATE	DESCRIPTION
A	12-05-25	ISSUED FOR APPROVAL

REV #	DATE	DESCRIPTION
A	12-05-25	ISSUED FOR APPROVAL

REV #	DATE	DESCRIPTION
A	12-05-25	ISSUED FOR APPROVAL

REV #	DATE	DESCRIPTION
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REV #	DATE	DESCRIPTION
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REV #	DATE	DESCRIPTION
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REV #	DATE	DESCRIPTION
A	12-05-25	ISSUED FOR APPROVAL

REV #	DATE	DESCRIPTION
A	12-05-25	ISSUED FOR APPROVAL



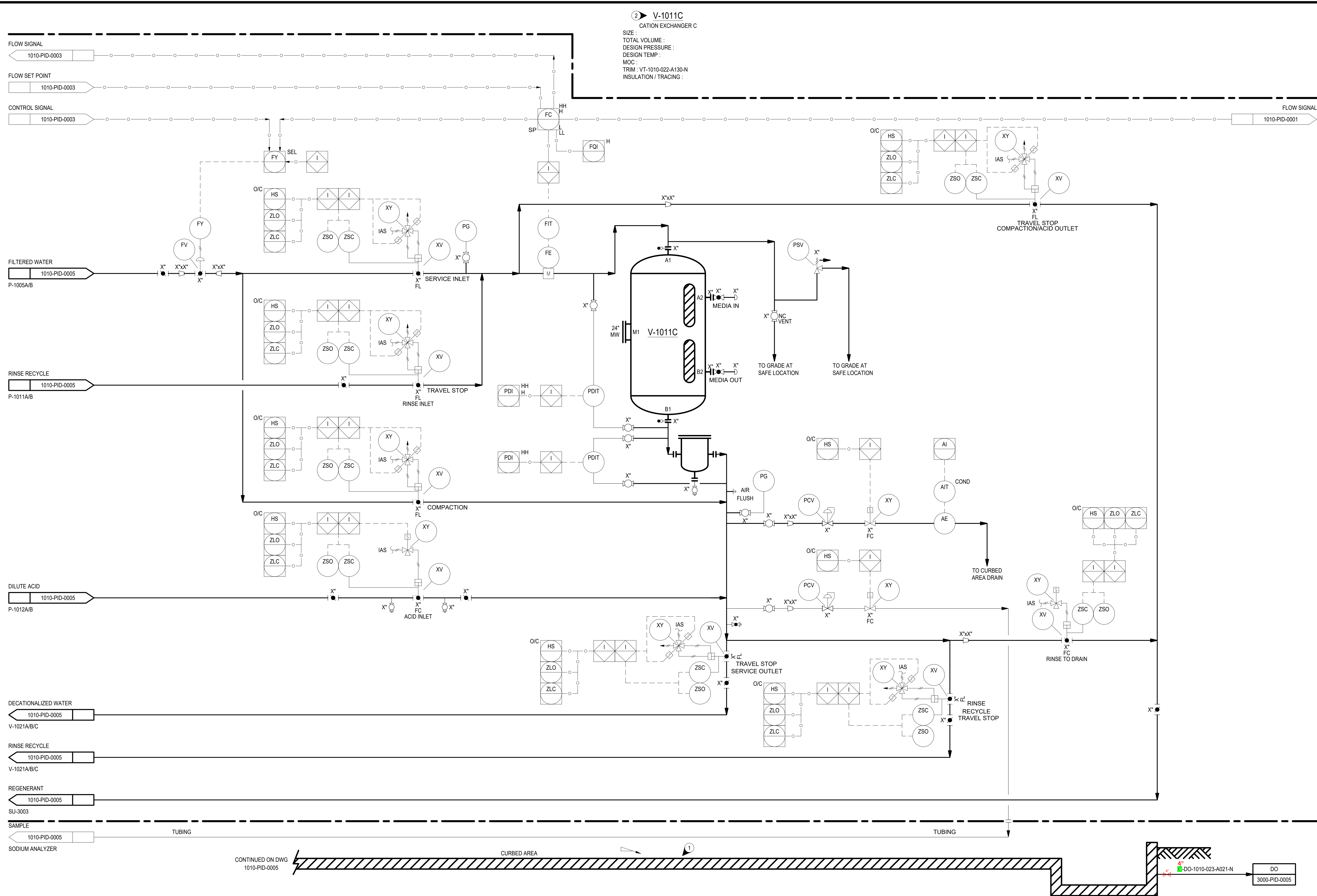
PIPING & INSTRUMENTATION DIAGRAM	
DEMINERALIZED WATER PACKAGE	
CATION EXCHANGER B	
CLIENT: MOTIVA ENTERPRISES LLC	PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW
PLANT: PORT ARTHUR REFINERY	
JOB NO:	

PROJECT NO: 25MA02	SCALE: NONE
DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN
DRAWING NO:	

SSP ENGINEERING & CONSULTING	FIRM NUMBER #13619
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	
1010-PID-0005	REV A

CHECK PRINT

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes rev\1010 - process\pids - 1010-pid-0007.dwg, PLOT DATE: ---, PLOTTED BY: jason, DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes rev\1010 - process\pids - 1010-pid-0007.dwg, (Border Rev 1 5/18/25)



- GENERAL NOTES
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- NOTES
1. ACID RESISTANT LINING.
2. CATION EXCHANGER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

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REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

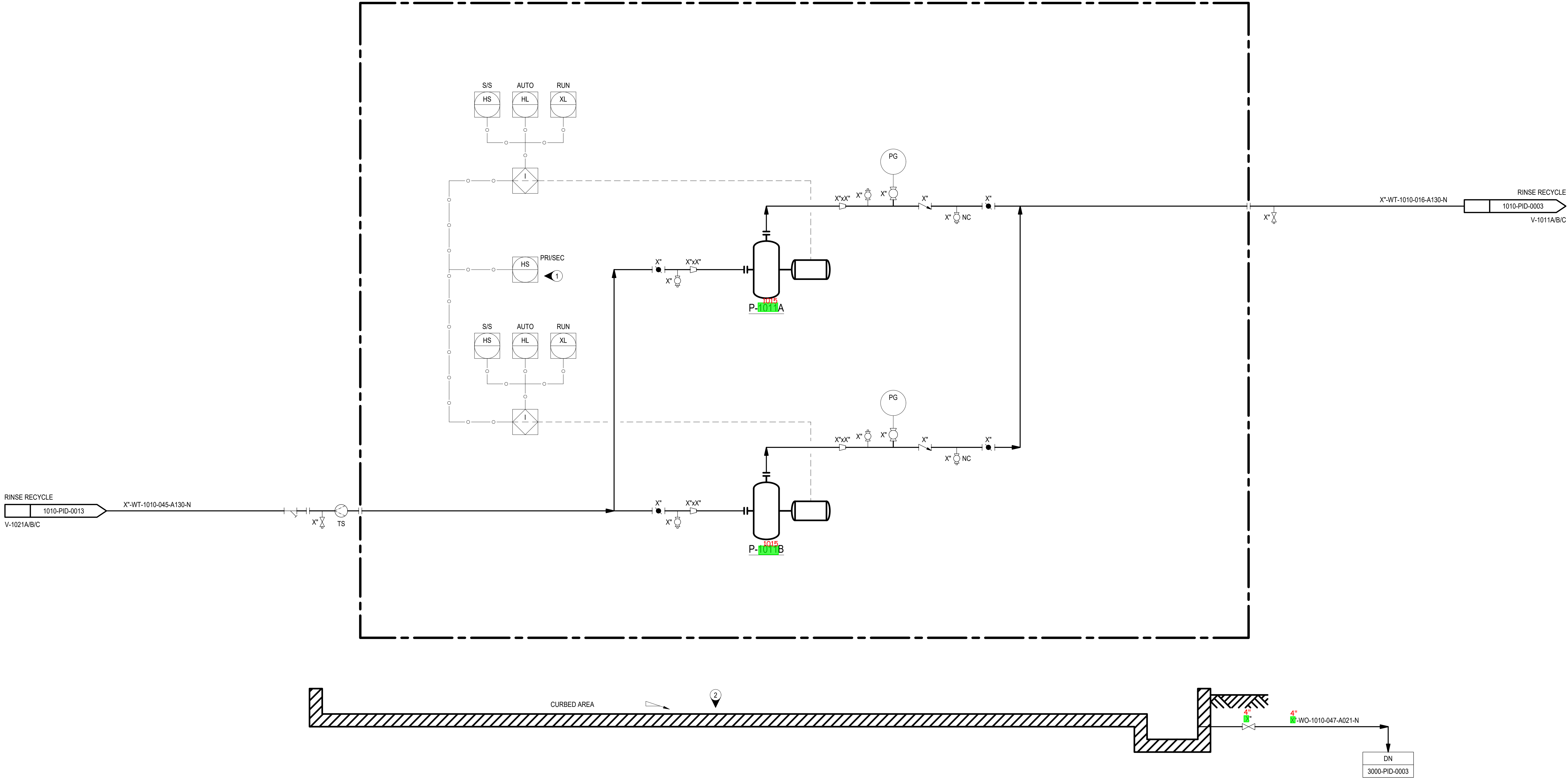
MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM		FIRM NUMBER	
DEMINERALIZED WATER PACKAGE		#13619	
CATION EXCHANGER C			
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		REV	
DRAWING NO:		1010-PID-0007	
PROJECT NO: 25MA02		SCALE: NONE	
DRAWN BY: GS		CHECKED: APB	
DESIGN BY:		ENG APVL: PDN	
DRAWING NO:		1010-PID-0007	

SSP ENGINEERING & CONSULTING	
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	
PROJECT NO: 25MA02	
SCALE: NONE	
DRAWN BY: GS	
CHECKED: APB	
DESIGN BY:	
ENG APVL: PDN	
DRAWING NO:	

1015
P-1011A/B
DEMIN RINSE RECYCLE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-048-A130-N
SEAL PLAN :

- GENERAL NOTES
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- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. RINSE RECYCLE PUMPS ARE USED BY CATION/ ANION. ONLY ONE PUMP CAN BE IN RINSE RECYCLE AT A TIME. PUMPS ALTERNATE AT EVERY START.
2. ACID/ CAUSTIC RESISTANT LINING.



CHECK PRINT

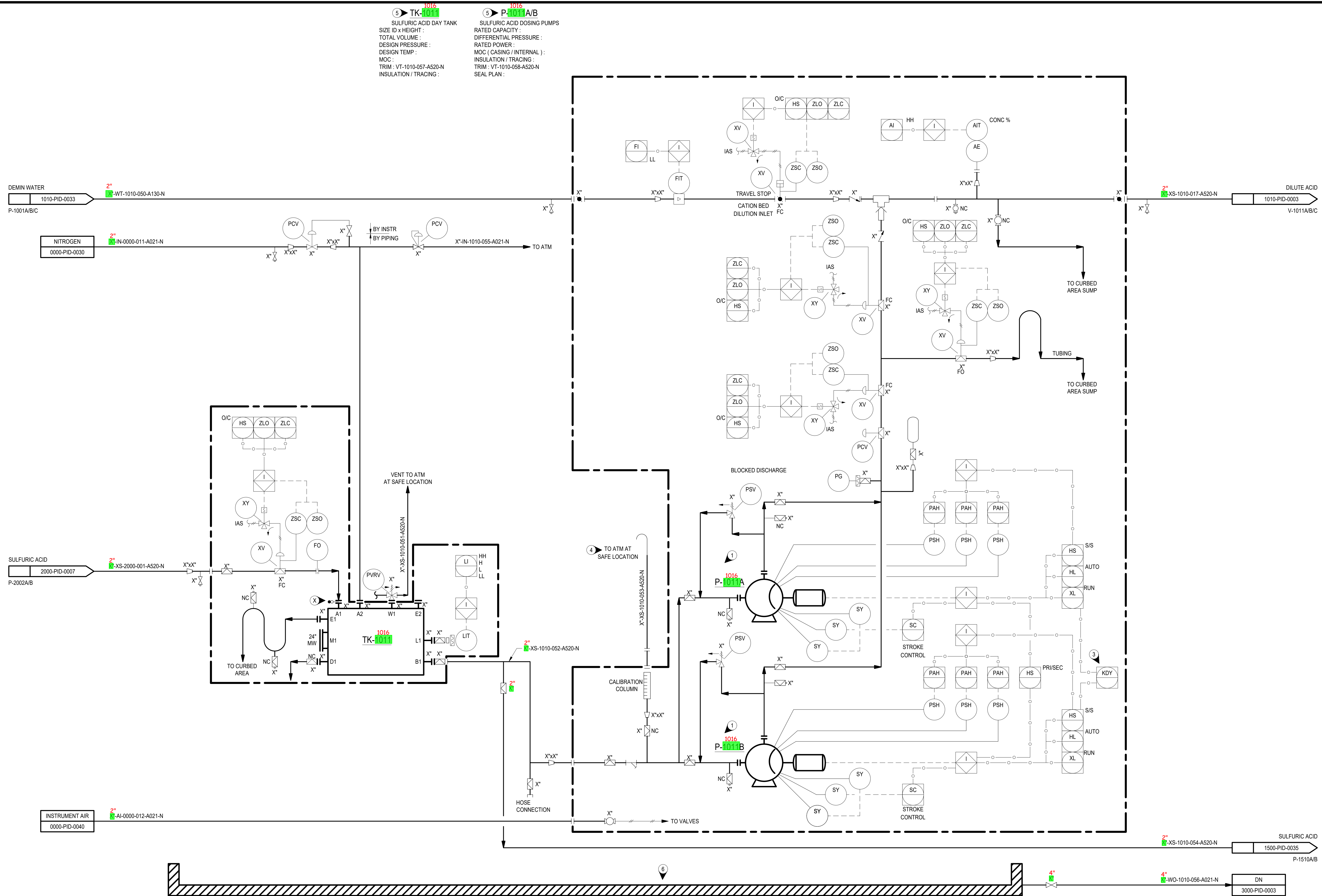
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A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA			PIPING & INSTRUMENTATION DIAGRAM		SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
CLIENT: MOTIVA ENTERPRISES LLC			DEMINERALIZED WATER PACKAGE		13100 NORTHWEST FREEWAY		SCALE: NONE	
PLANT: PORT ARTHUR REFINERY			DEMIN RINSE RECYCLE PUMPS		SUITE 625		CHECKED: APB	
JOB NO:			PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		HOUSTON, TEXAS 77040		ENG APVL: PDN	
					DRAWING NO:		REV	
					1010-PID-0009		A	

DRAWING LOCATION: c:\users\jason\one-drive - eng\ssps files - 25ma02 - aromatics utility scope add eyes trining\04 - design\04 - process\pids, (Border Rev 1 5.18.25),
PLOT DATE: ---, PLOT BY: jason,
DRAWING: 1010-pid-0011.dwg,



- GENERAL NOTES
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 - CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
- CHECK VALVES ARE INTERNAL TO THE PUMP.
 - VALVE STEM IS ACCESSIBLE FROM GRADE.
 - DOSING PUMPS ALTERNATE DUTY AFTER EACH REGEN.
 - CALIBRATION COLUMN VENT IS EXTENDED ABOVE THE MAXIMUM TANK ELEVATION.
 - SULFURIC ACID DILUTION SYSTEM SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.
 - ACID RESISTANT LINING.

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CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZER WATER PACKAGE
ACID DILUTION SYSTEM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

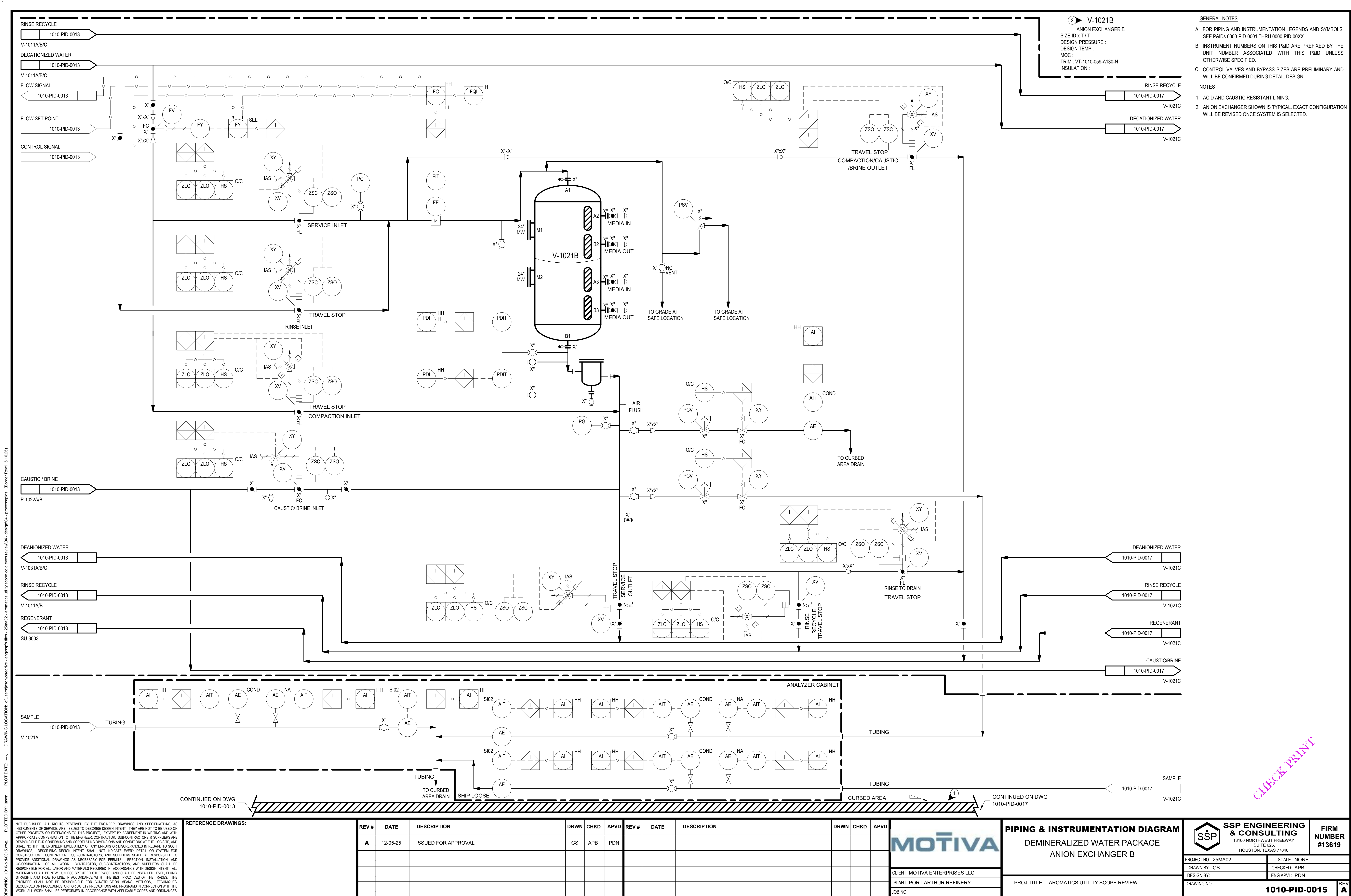
SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

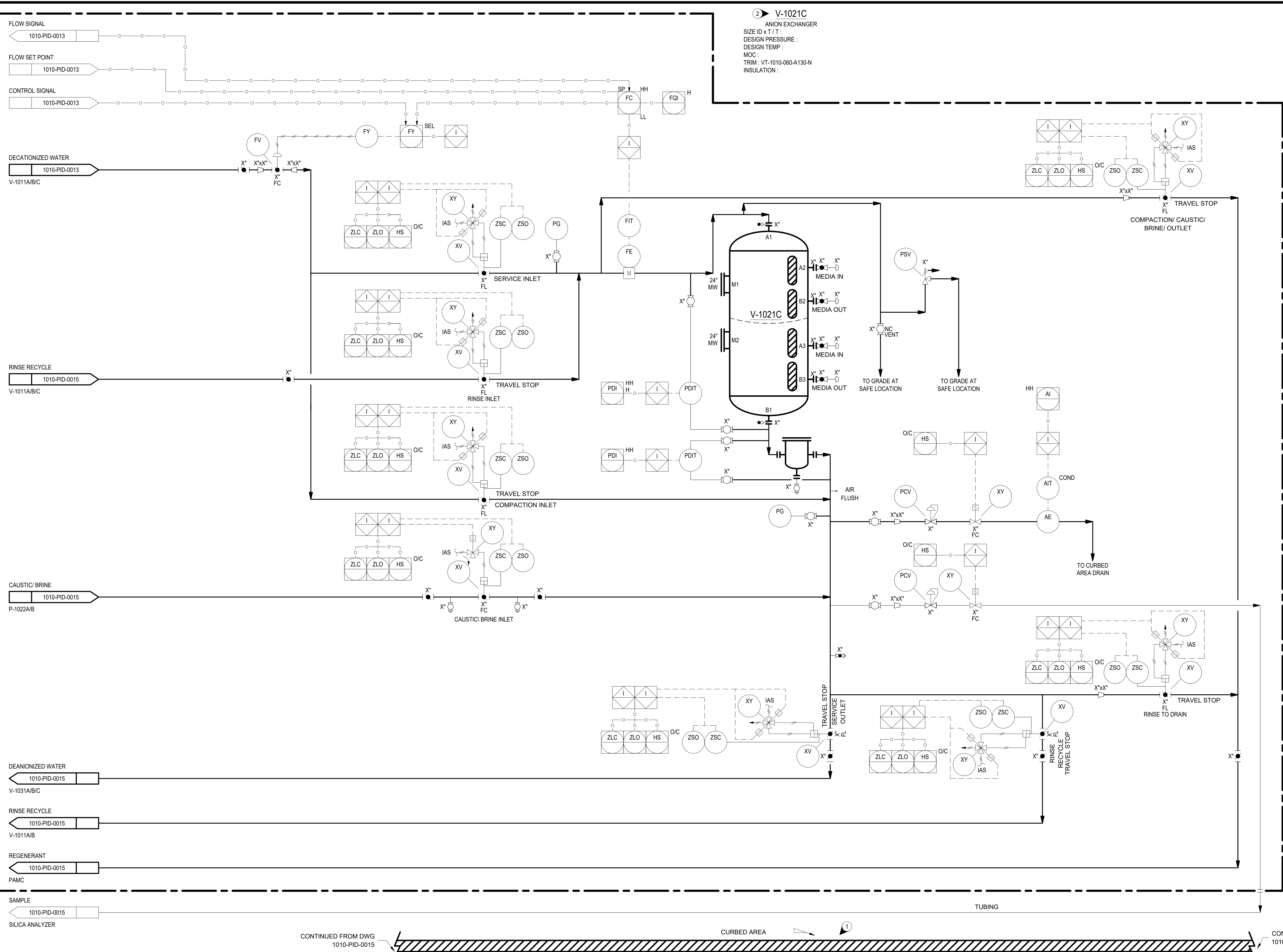
FIRM NUMBER
#13619

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

1010-PID-0011
A

CHECK PRINT





GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. ACID AND CAUSTIC RESISTANT LINING.

2. ANION EXCHANGER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:			
REV #	DATE	DESCRIPTION	DRWN
A	12-05-25	ISSUED FOR APPROVAL	GS

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN



CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZED WATER PACKAGE
ANION EXCHANGER C

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		FIRM NUMBER #13619
PROJECT NO: 25MA02	SCALE: NONE	
DRAWN BY: GS	CHECKED: APB	
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:	1010-PID-0017	REV A

HE-1021

CAUSTIC DILUTION HEATER

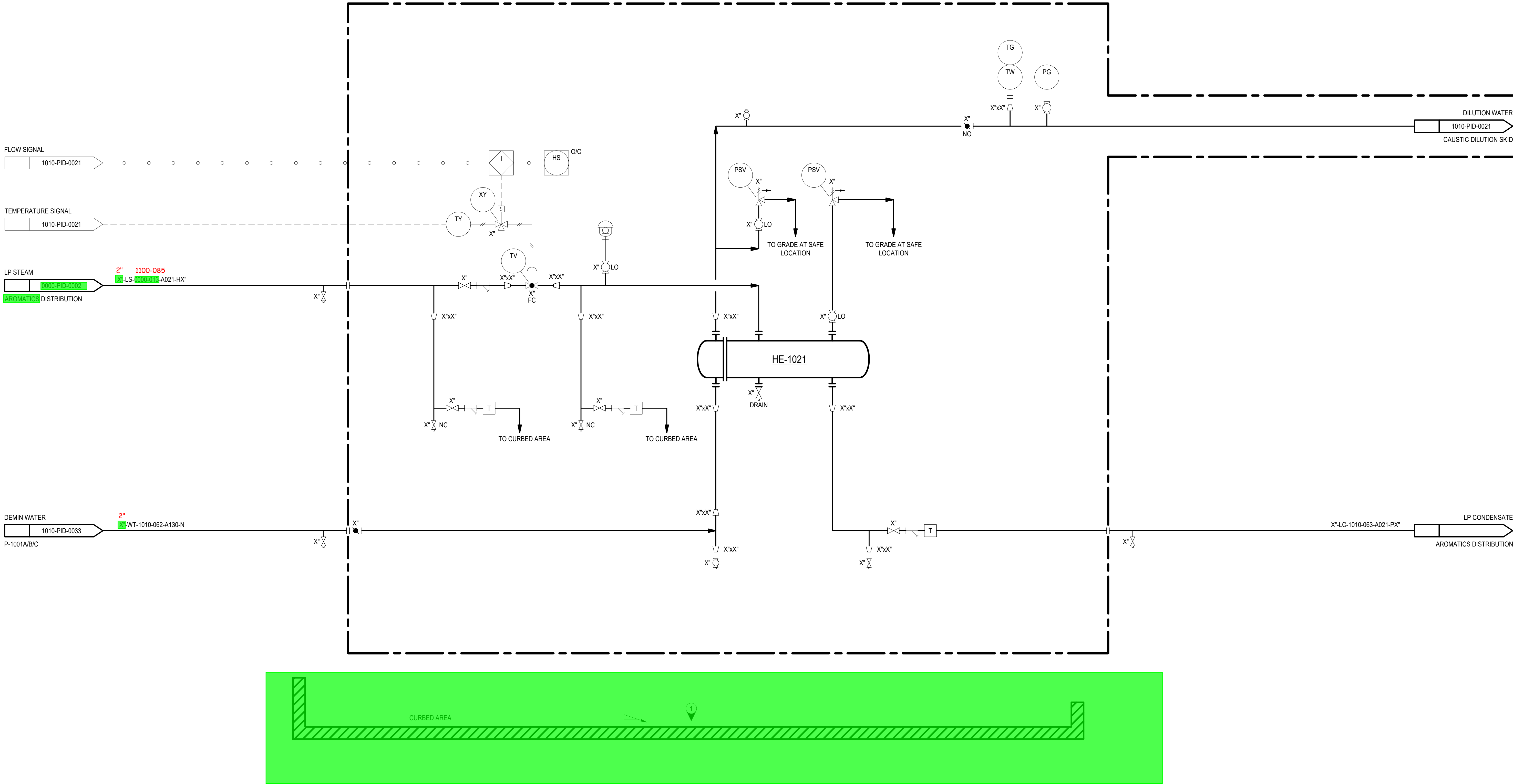
TYPE :
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1010-061-A130-N
INSULATION (SHELL / TUBES) :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. CAUSTIC-RESISTANT LINING.
2. CAUSTIC DILUTION HEATER CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER SYSTEM
CAUSTIC DILUTION HEATER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

1010-PID-0019

REV
A

4 TK-1021
CAUSTIC DAY TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-069-A130-N
INSULATION / TRACING :

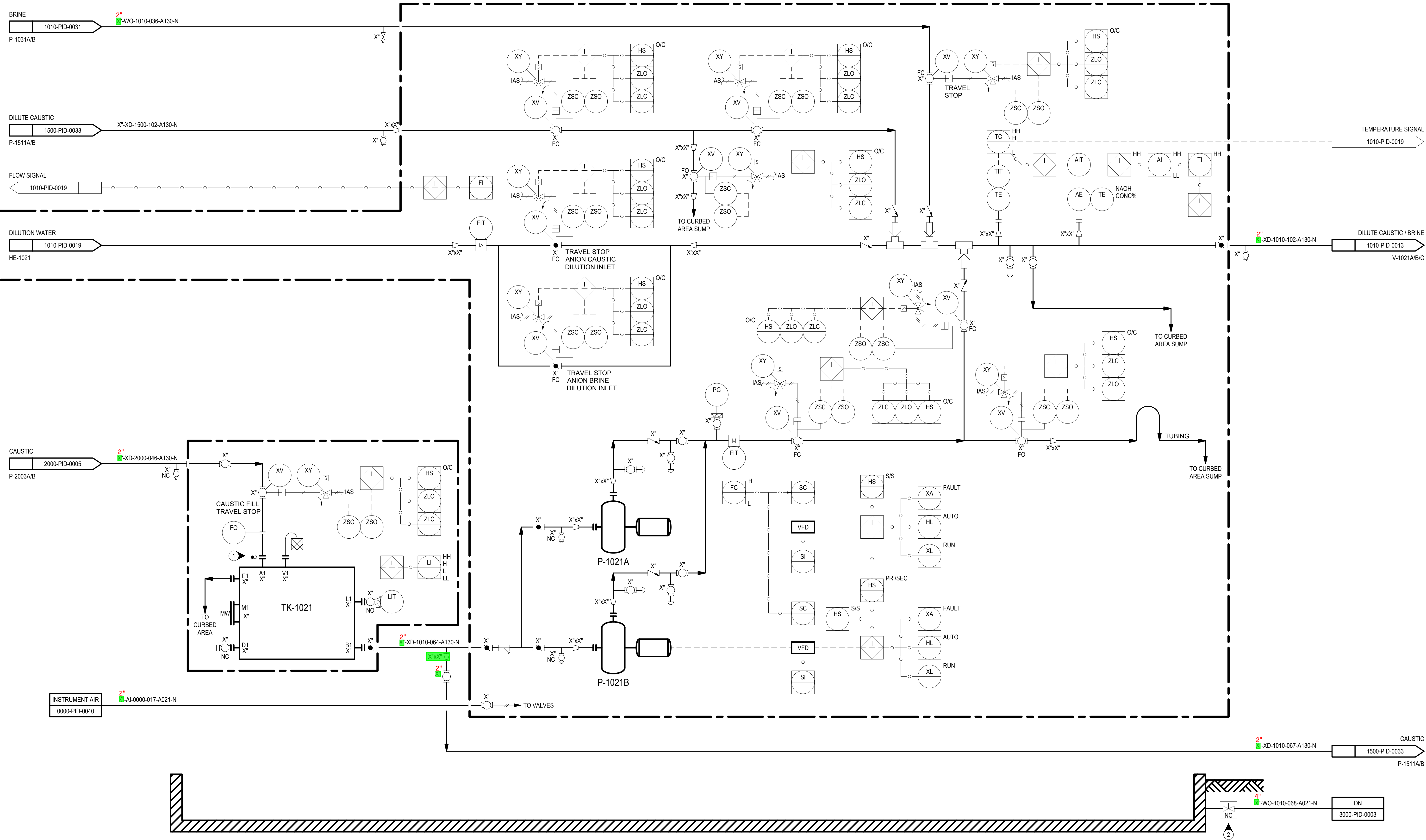
4 P-1021 A/B
CAUSTIC DOSING PUMP
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-070-A130-N
SEAL PLAN :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. PROVIDED BY CONTRACTOR.
2. VALVE STEM IS ACCESSIBLE FRON GRADE.
3. CAUSTIC RESISTANCE LINING.
4. CAUSTIC DILUTION SYSTEM SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER SYSTEM

CAUSTIC DILUTION SYSTEM

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1010-PID-0021

FIRM NUMBER #13619

REV A

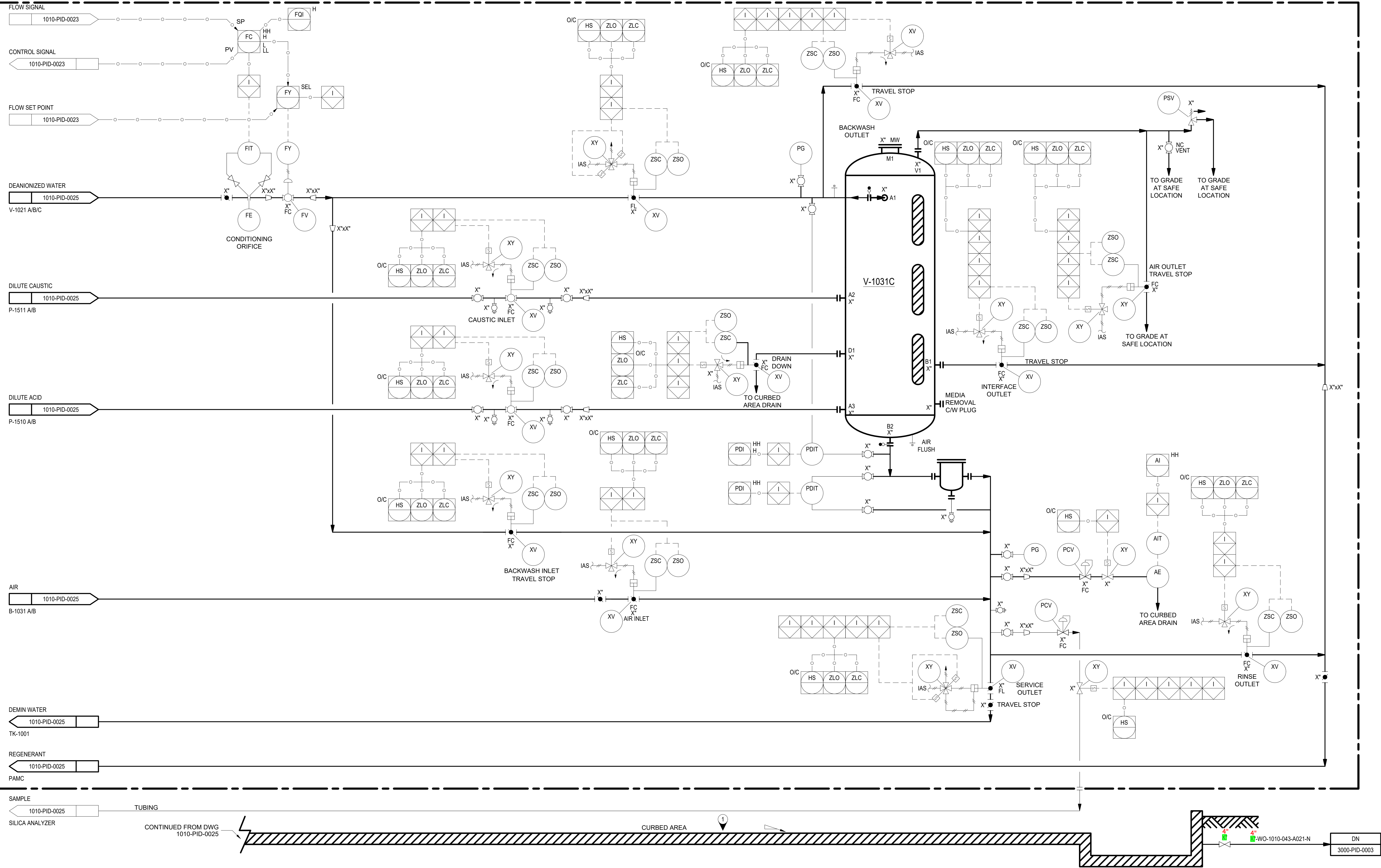
SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

② V-1031C
MIXED BED EXCHANGER C
SIZE 10 x 11 T
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-066-A130-N
INSULATION :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. ACID RESISTANT AND CAUSTIC RESISTANT LINING.
2. MIXED BED POLISHER SHOWN IS TYPICAL. EXACT CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

DEMINERALIZED WATER PACKAGE

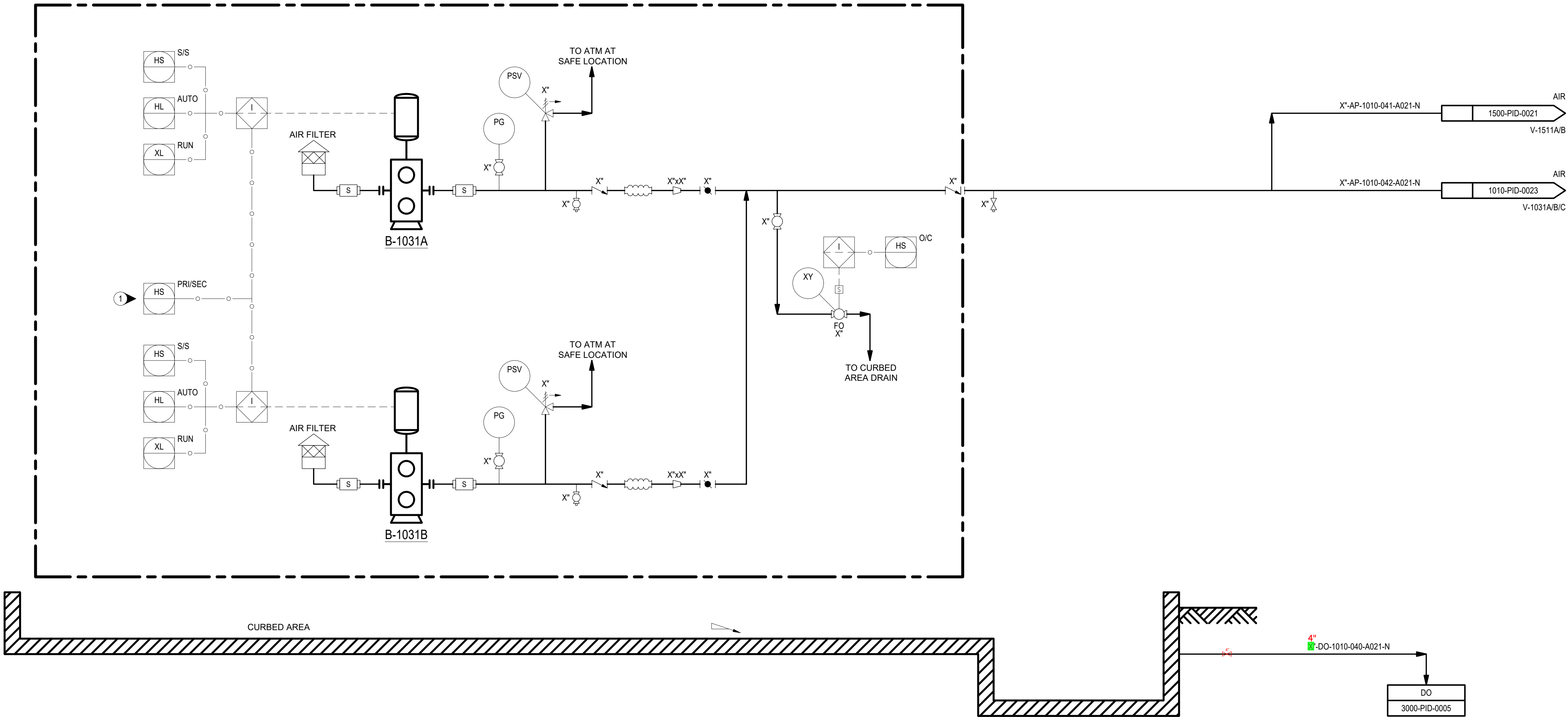
MIXED BED EXCHANGER C

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

PROJECT NO: 25MA02	SCALE: NONE	FIRM NUMBER #13619
DRAWN BY: GS	CHECKED: APB	
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:	1010-PID-0027	REV A

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. MIXED BED SCOUR BLOWERS ALTERNATE DUTY AFTER EACH USE.
2. MIXED BED SCOUR BLOWERS CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



B-1031A/B
MIXED BED AIR SCOUR BLOWERS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-065-A130-N
SEAL PLAN :

CHECK PRINT

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
DEMINERALIZED WATER PACKAGE
MIXED BED AIR SCOUR BLOWERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02 SCALE: NONE

DRAWN BY: GS CHECKED: APB

DESIGN BY: ENG APVL: PDN

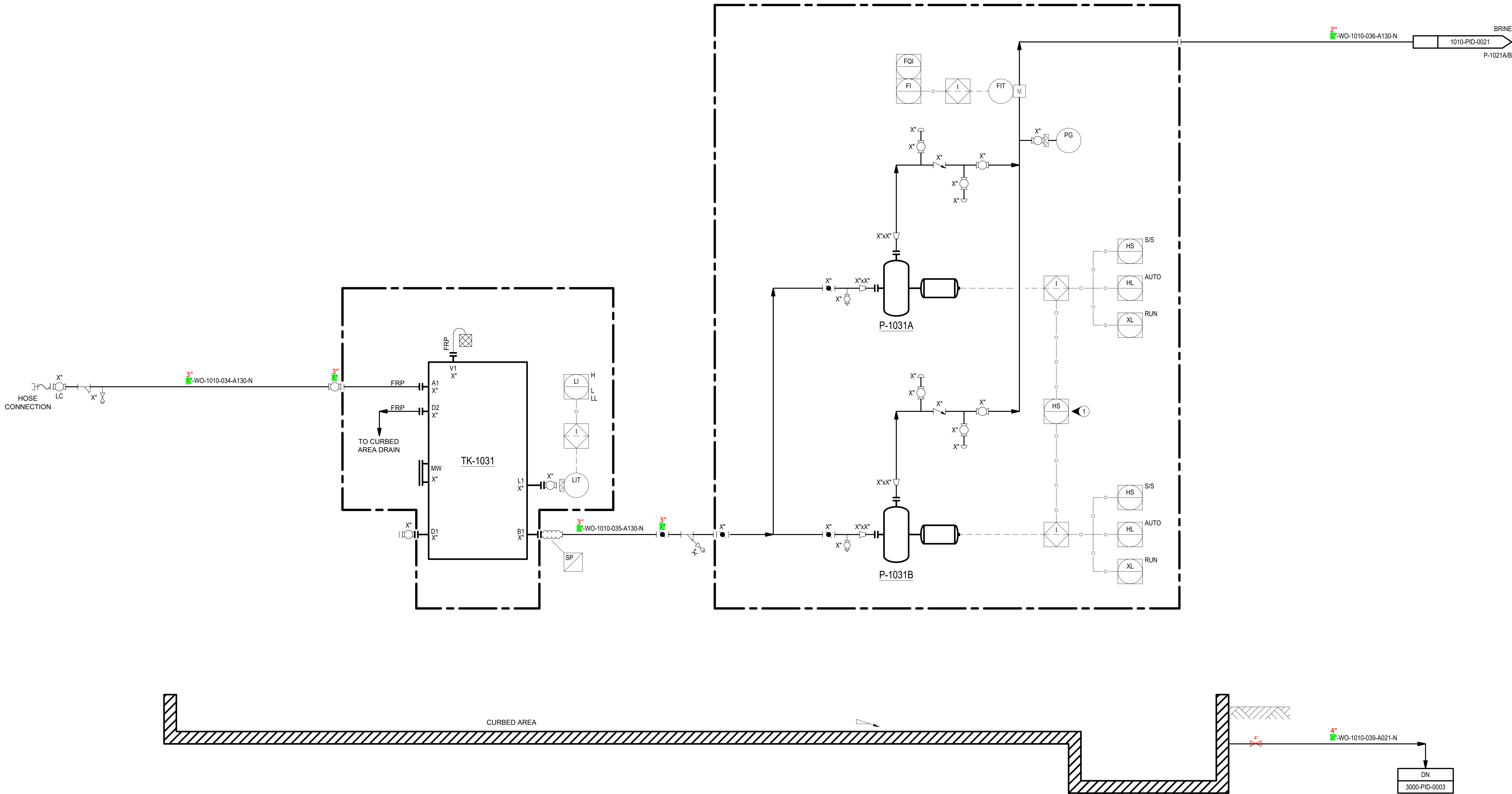
DRAWING NO: 1010-PID-0029

FIRM
NUMBER
#13619

REV

TK-1031
BRINE MEASURING TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-037-A130-N
INSULATION / TRACING :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. PUMPS ALTERNATE AFTER EVERY BRINE INJECTION.
2. BRINE SYSTEM CONFIGURATION IS TYPICAL. DRAWING TO BE REVISED WHEN SYSTEM IS SELECTED.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

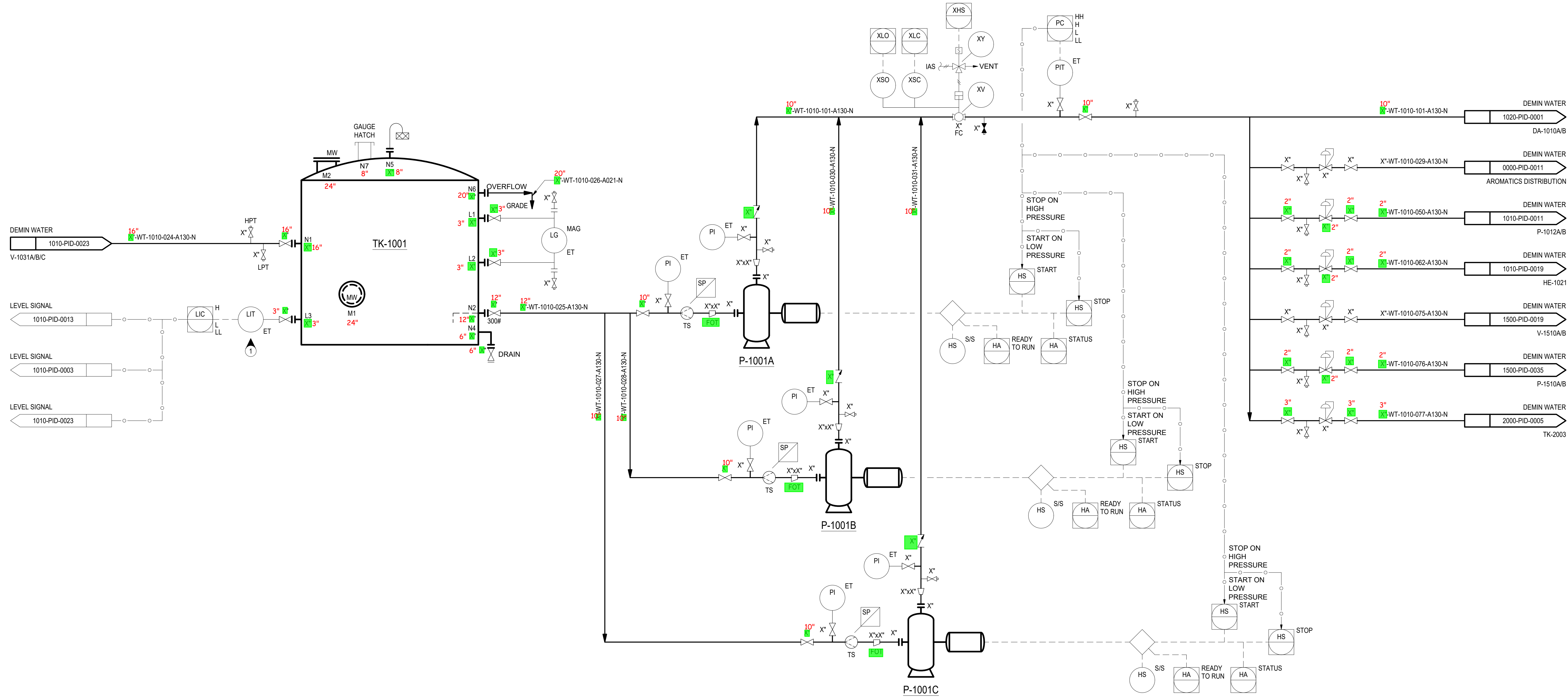
MOTIVA			PIPING & INSTRUMENTATION DIAGRAM		SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
CLIENT: MOTIVA ENTERPRISES LLC			DEMINERALIZER WATER PACKAGE		13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		PROJECT NO: 25MA02	
PLANT: PORT ARTHUR REFINERY			BRINE SYSTEM		SCALE: NONE		DRAWN BY: GS	
JOB NO:			PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		CHECKED: APB		DESIGN BY:	
					ENG APVL: PDN		DRAWING NO:	
							1010-PID-0031	
							A	

TK-1001
DEMIN WATER TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1010-032-A130-N
INSULATION / TRACING :

P-1001A/B/C
DEMIN WATER PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1010-033-A130-N
SEAL PLAN :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. THREADED BLEED RING ON WATER SERVICE.



CHECK PRINT

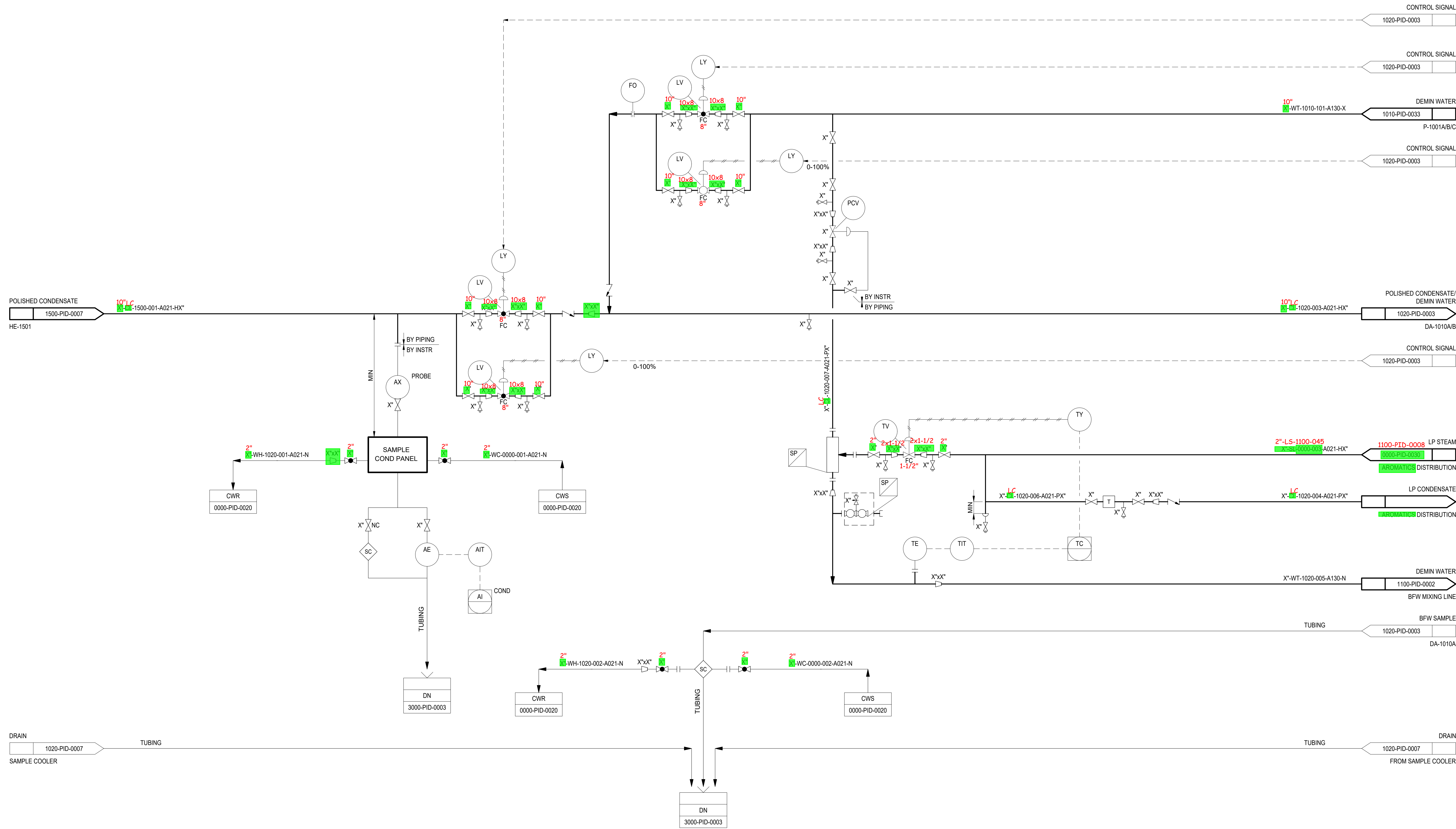
NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE, AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

	PIPING & INSTRUMENTATION DIAGRAM		 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	FIRM NUMBER #13619	
	DEMIN WATER TANK				
CLIENT: MOTIVA ENTERPRISES LLC			PROJECT NO: 25MA02	SCALE: NONE	
PLANT: PORT ARTHUR REFINERY	PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	DRAWN BY: GS		CHECKED: APB	
		DESIGN BY:		ENG APVL: PDN	
JOB NO:	DRAWING NO:		1010-PID-0033		REV A

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM	
STEAM UNIT DEAERATOR SAMPLE SYSTEM	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	
PROJECT NO: 25MA02	SCALE: NONE
DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN
DRAWING NO:	

FIRM NUMBER #13619
REV A

1020-PID-0001

DEAERATOR

POLISHED CONDENSATE/
DEMIN WATER

10" LG
X 10" ID

1020-PID-0001

1020-003-A021-HX

HE-1501A/B

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

3. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



REFERENCE DRAWINGS

	
CLIENT: MOTIVA ENTERPRISES LLC	
PLANT: PORT ARTHUR REFINERY	
JOB NO:	

PIPING & INSTRUMENT DIAGRAM

STEAM UNIT

DEAERATOR A

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO.

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

RE

DEAERATOR B

The diagram illustrates the process flow for two units, DA-1010A and DA-1010B, which are part of a larger industrial system. The units are represented by large horizontal vessels with internal components like trays and reboilers.

Inputs and Outputs:

- DA-1010A:** Receives feed from the top left (1020-PID-0003) and has a relief discharge (1020-PID-0003). It also has a control signal (1020-PID-0003) and a BFW input (1020-PID-0007).
- DA-1010B:** Receives feed from the bottom left (1020-PID-0003) and has a relief discharge (1020-PID-0003). It also has a control signal (1020-PID-0003) and a BFW input (1020-PID-0007).

Internal Components and Piping:

- DA-1010A:** Features a reboiler (R1) and a condenser (C1). The reboiler has a steam input (1020-PID-0003) and a steam output (1020-PID-0003). The condenser has a cooling water input (1020-PID-0003) and a cooling water output (1020-PID-0003).
- DA-1010B:** Features a reboiler (R2) and a condenser (C2). The reboiler has a steam input (1020-PID-0003) and a steam output (1020-PID-0003). The condenser has a cooling water input (1020-PID-0003) and a cooling water output (1020-PID-0003).

Control Systems:

- DA-1010A:** Includes a control signal (1020-PID-0003) and a BFW input (1020-PID-0007).
- DA-1010B:** Includes a control signal (1020-PID-0003) and a BFW input (1020-PID-0007).

Other Components:

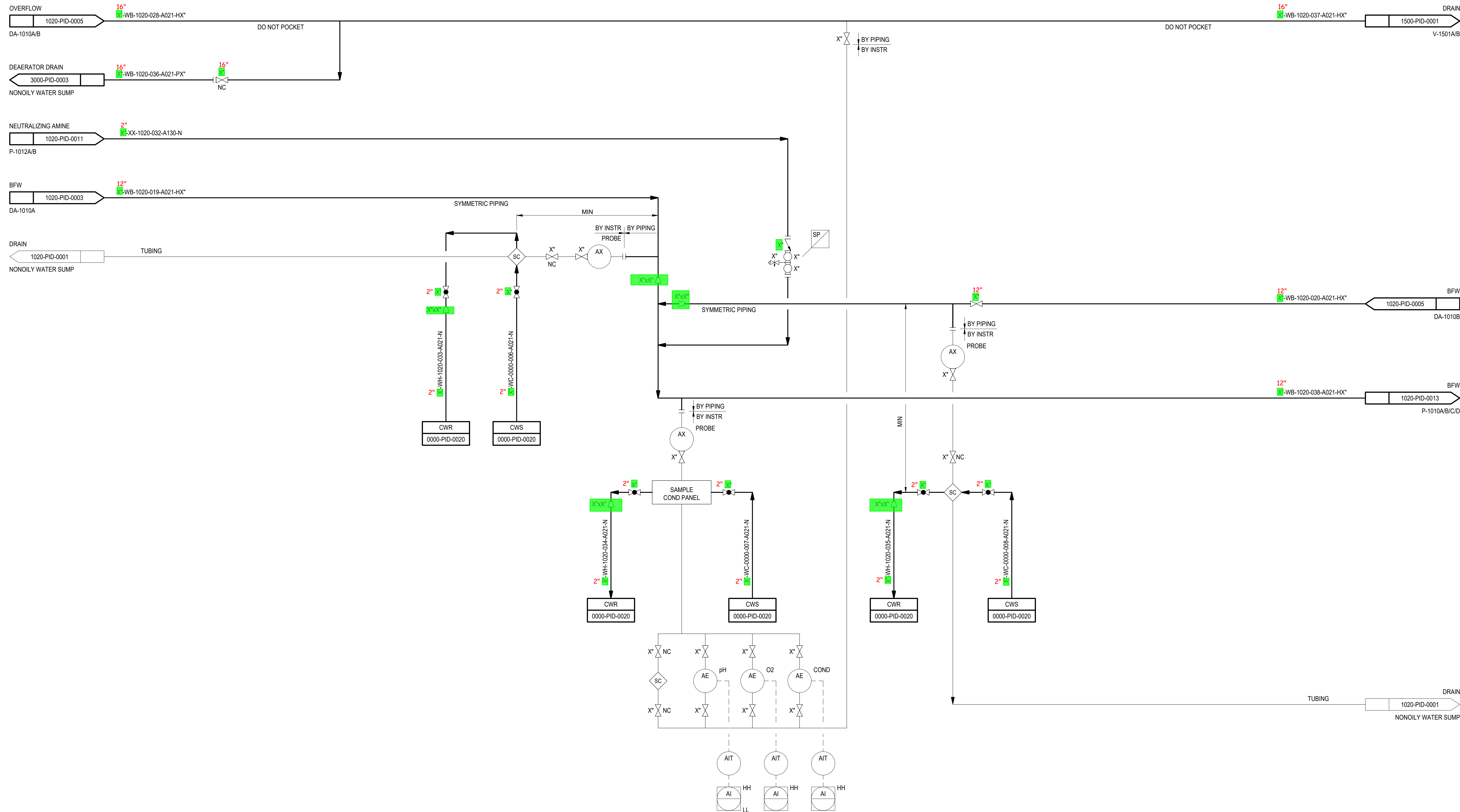
- Relief Discharge:** A line from the top of DA-1010A to a relief tank (1020-PID-0003).
- Control Signal:** A line from the bottom of DA-1010A to a control tank (1020-PID-0003).
- BFW Input:** A line from the bottom of DA-1010A to a BFW tank (1020-PID-0007).

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

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MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
BFW SAMPLE SYSTEM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING
& CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1020-PID-0007

REV

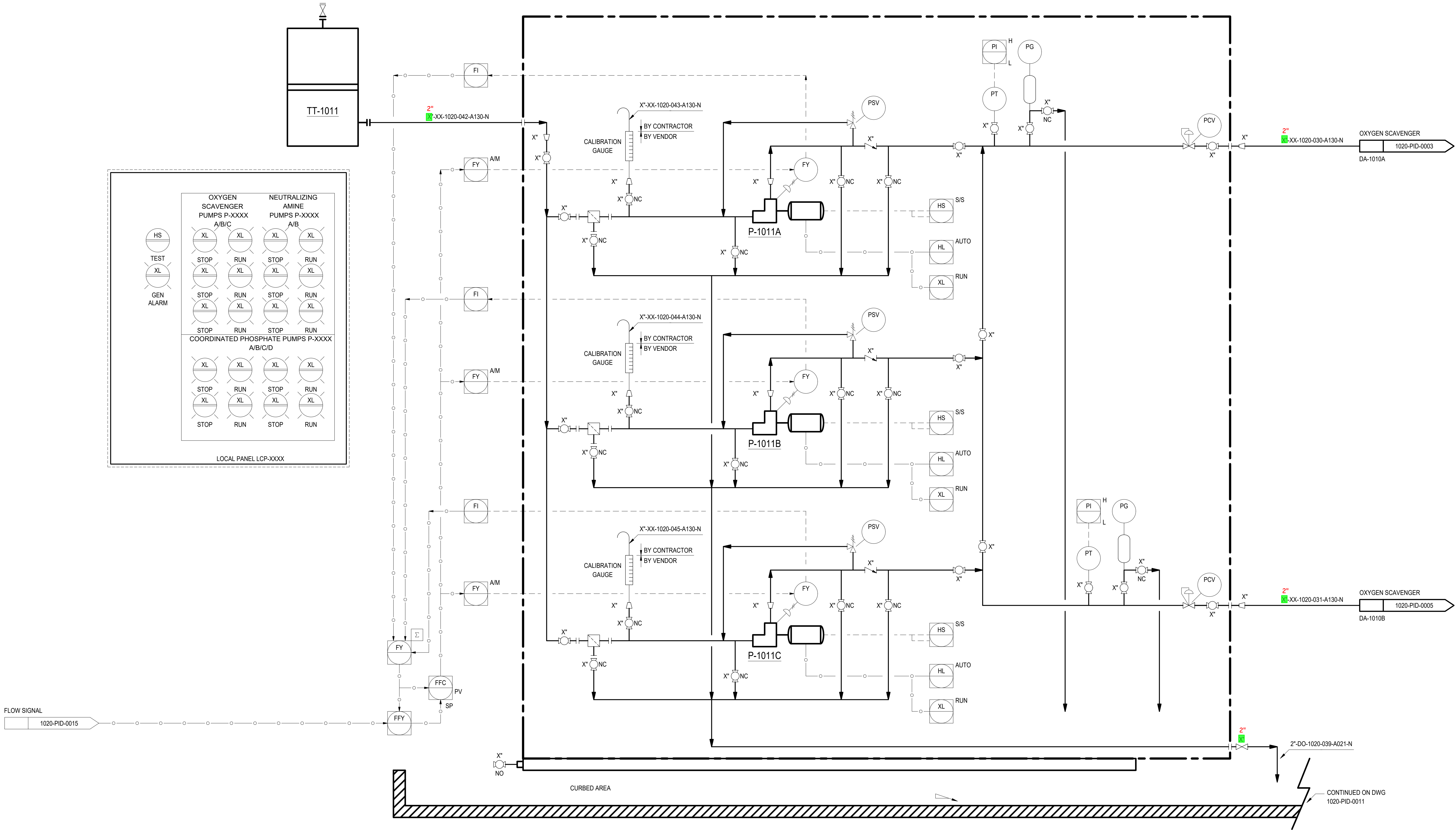
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TT-1011
OXYGEN SCAVENGER STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1020-040-A130-N
INSULATION / TRACING :

P-1011A/B/C
OXYGEN SCAVENGER PUMP
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-041-A130-N
SEAL PLAN :

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MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM	
OXYGEN SCAVENGER PACKAGE	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040	
PROJECT NO: 25MA02	SCALE: NONE
DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN
DRAWING NO:	

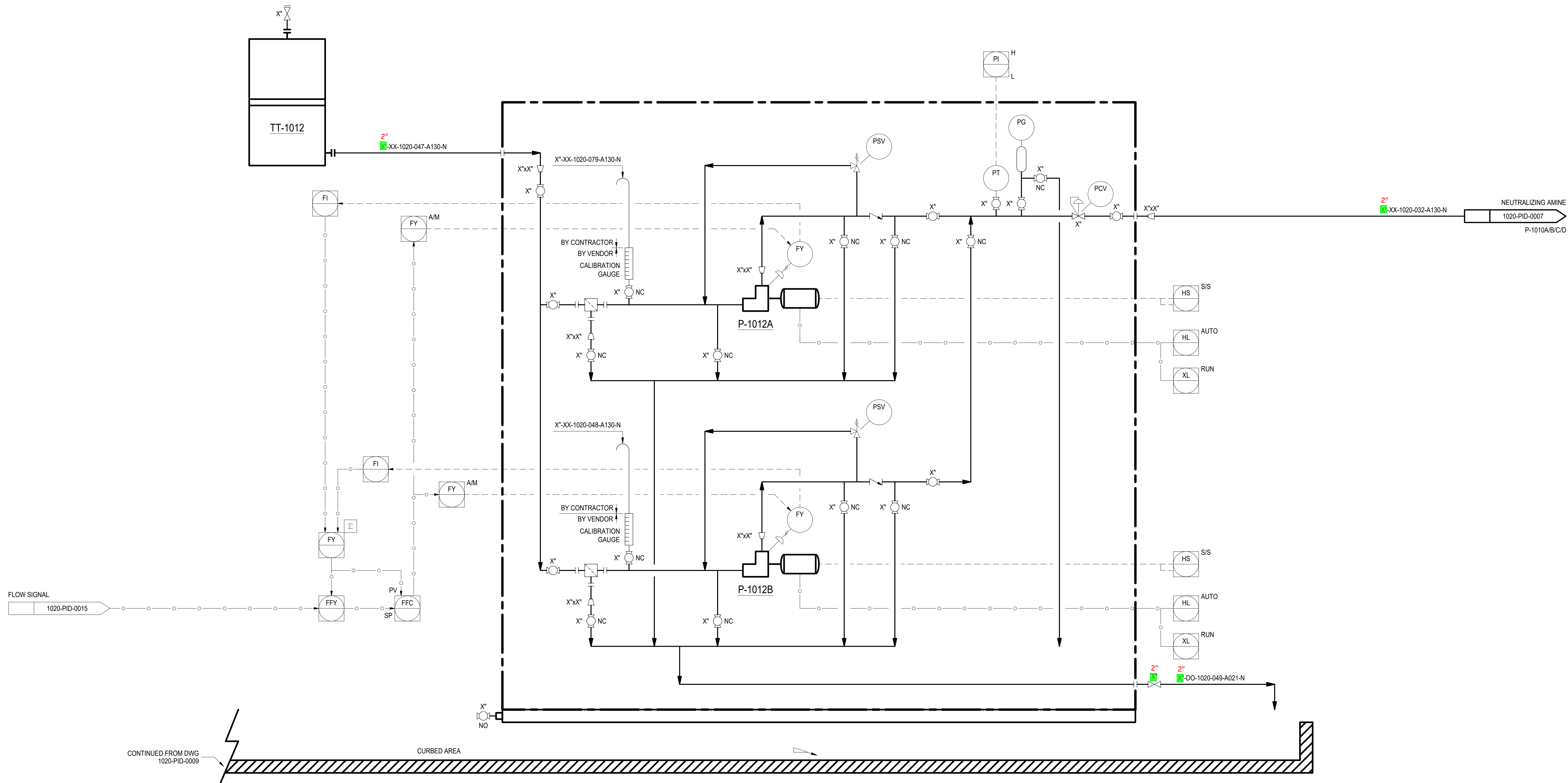
FIRM NUMBER	REV
#13619	A

CHECK PRINT

P-1012A/B
NEUTRALIZING AMINE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-051-A130-N
SEAL PLAN :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

REFERENCE DRAWINGS:

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

NEUTRALIZING AMINE PACKAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

FIRM
NUMBER
#13619

SCALE: NONE

CHECKED: APB

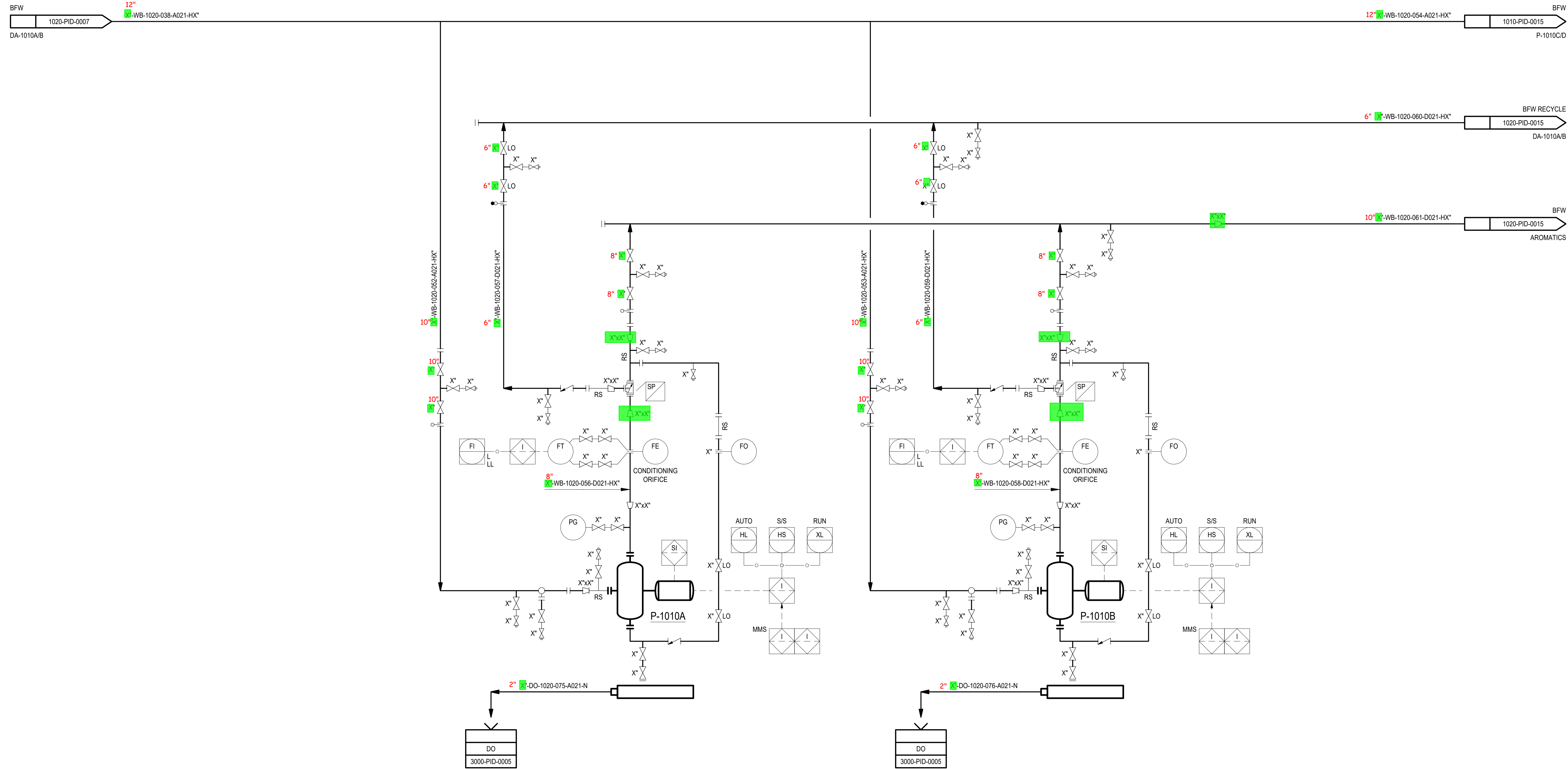
ENG APVL: PDN

1020-PID-0011

REV
A

P-1010A/B
BFW PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-062-D021-HX
SEAL PLAN :

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A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BFW PUMPS A/B

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

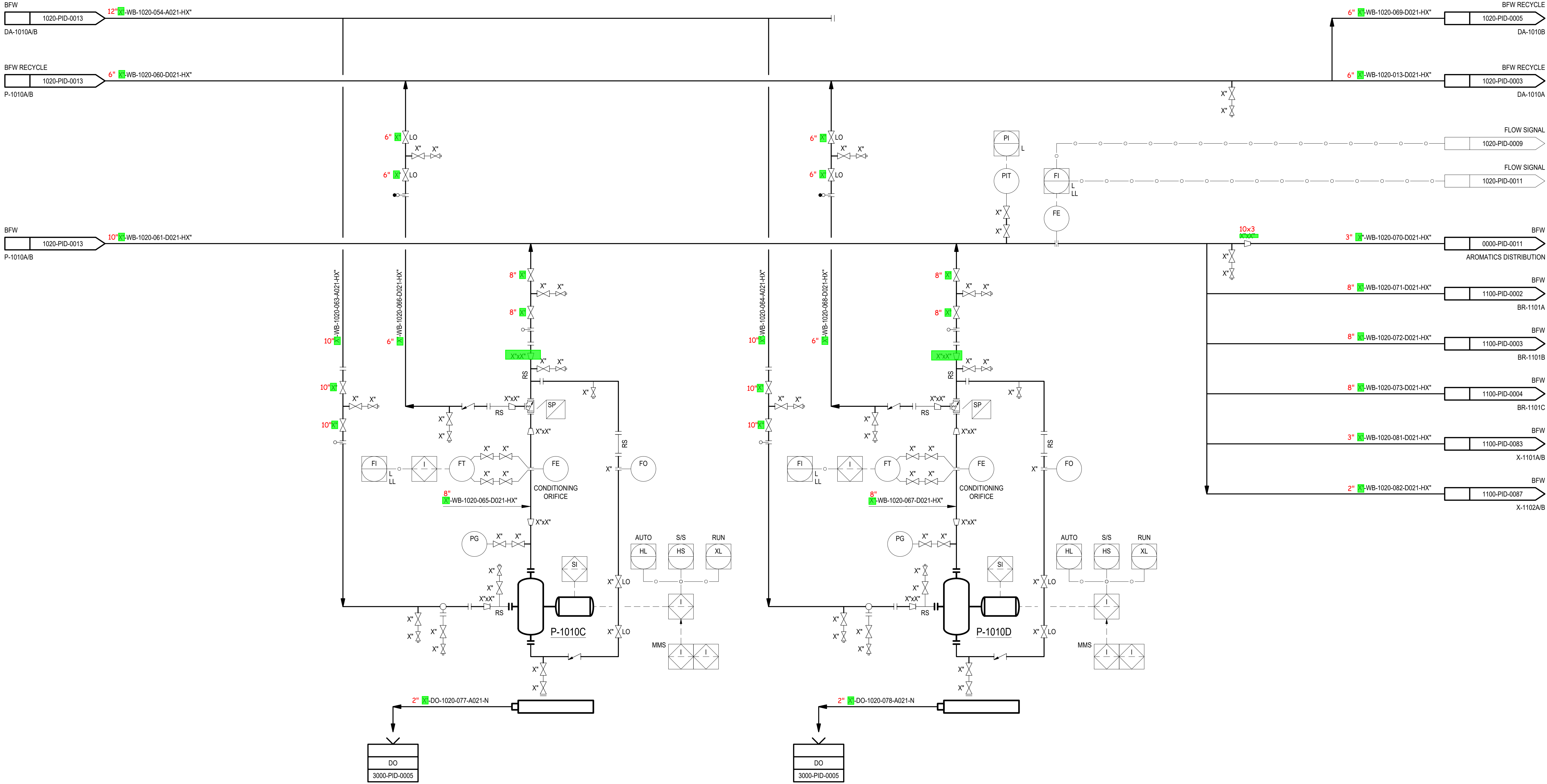
SCALE: NONE
CHECKED: APB
ENG APVL: PDN

1020-PID-0013

REV A

P-1010C/D
BFW PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1020-074-D021-HX"
SEAL PLAN :

GENERAL NOTES
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
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A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

STEAM UNIT
BFW PUMPS C/D

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

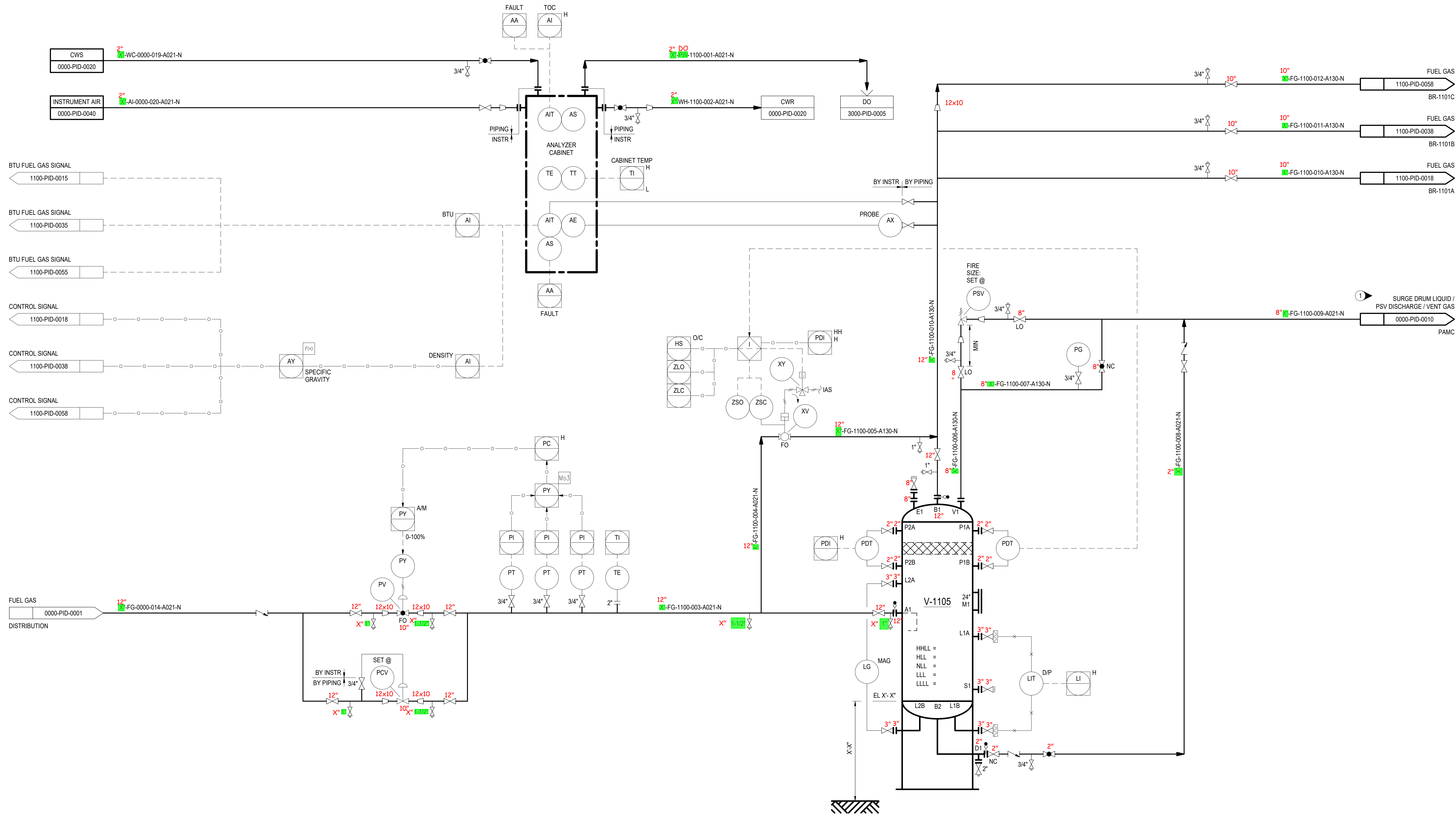
PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

1020-PID-0015
A

V-1105
FUEL GAS COALESCER
SIZE ID x T / I :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-013-A130-N
INSULATION :

- GENERAL NOTES
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- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. COALESCER PSV WILL BE ROUTED TO AROMATICS FLARE DEPENDING ON AVAILABILITY OF TIE-IN POINT.



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MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
FUEL GAS COALESCER

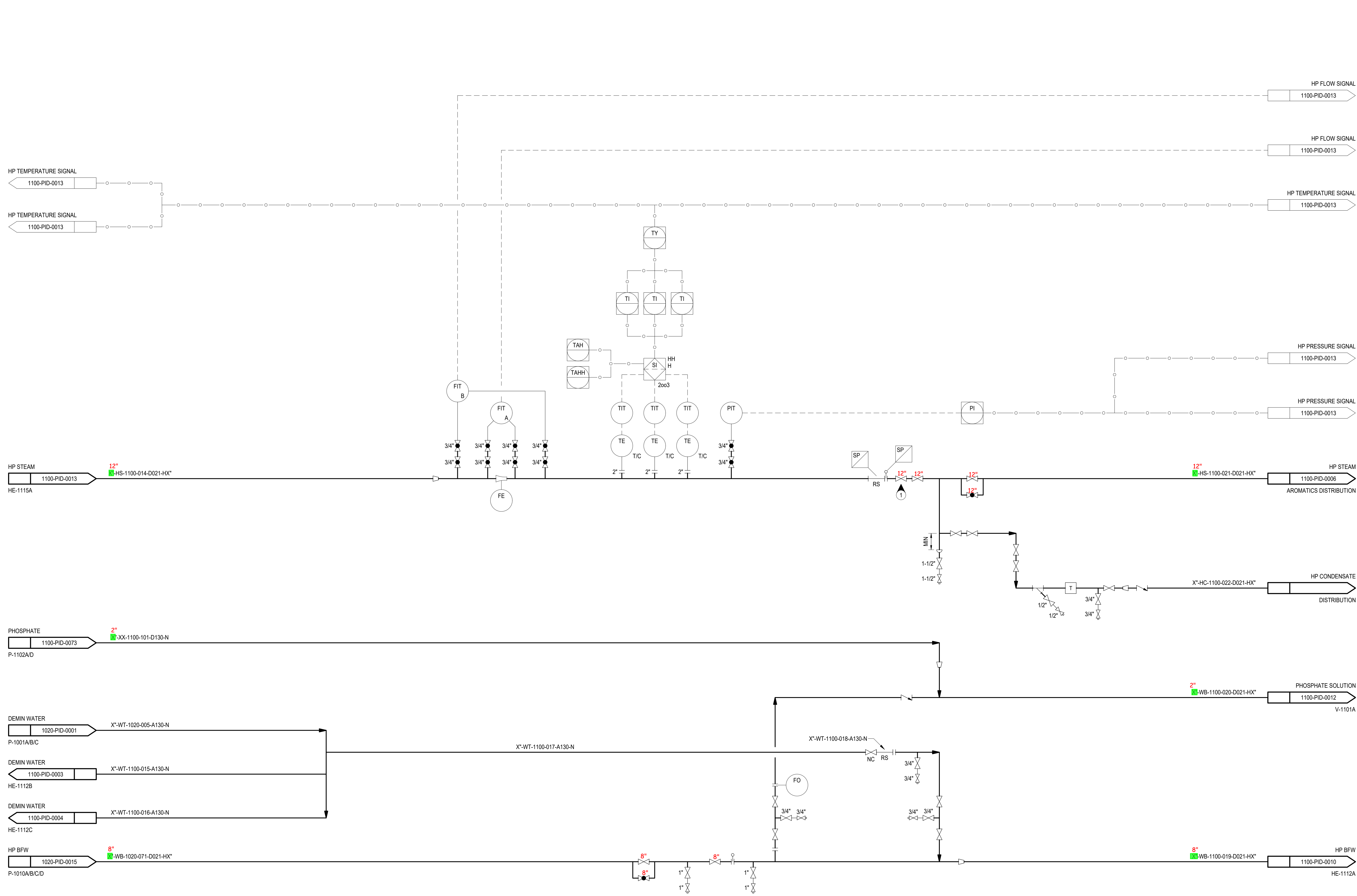
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:
1100-PID-0001
REV A

DRAWING: 1100-pid-0002.dwg, PLOT DATE: ---, PLOTTED BY: jason, DRAWING LOCATION: c:\users\jason\recent - engsage files - 25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pids, (Border Rev1 5.16.25)



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
 - B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
 - C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
- 1. VALVE FOR STEAM BLOWS OF HP SYSTEM.

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BFW AND HP STEAM BOILER A

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

1100-PID-0002

REV A

GENERAL NOTES

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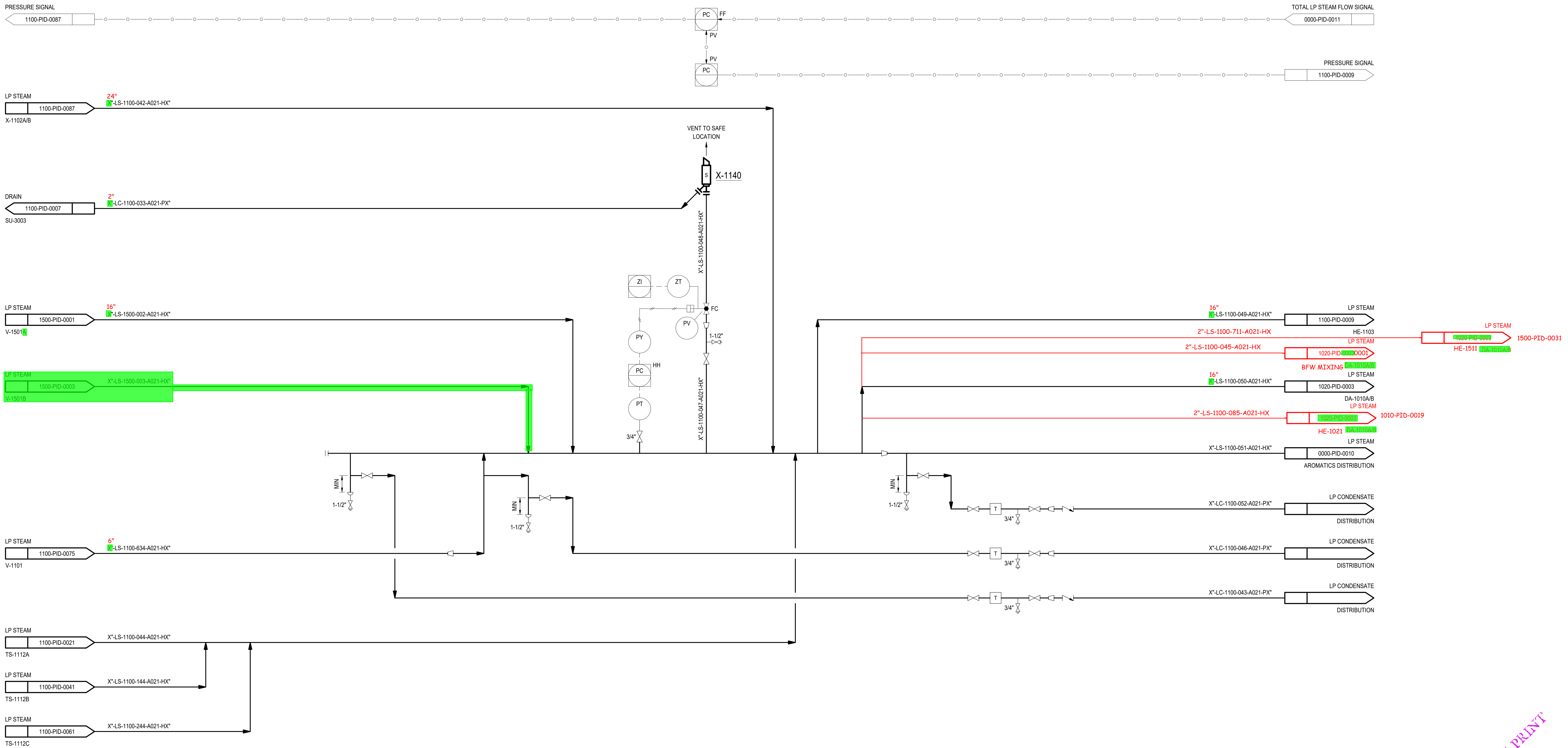


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X-1140
LP STEAM VENT SILENCER
RATED CAPACITY :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-704-A021-HX"
INSULATION :

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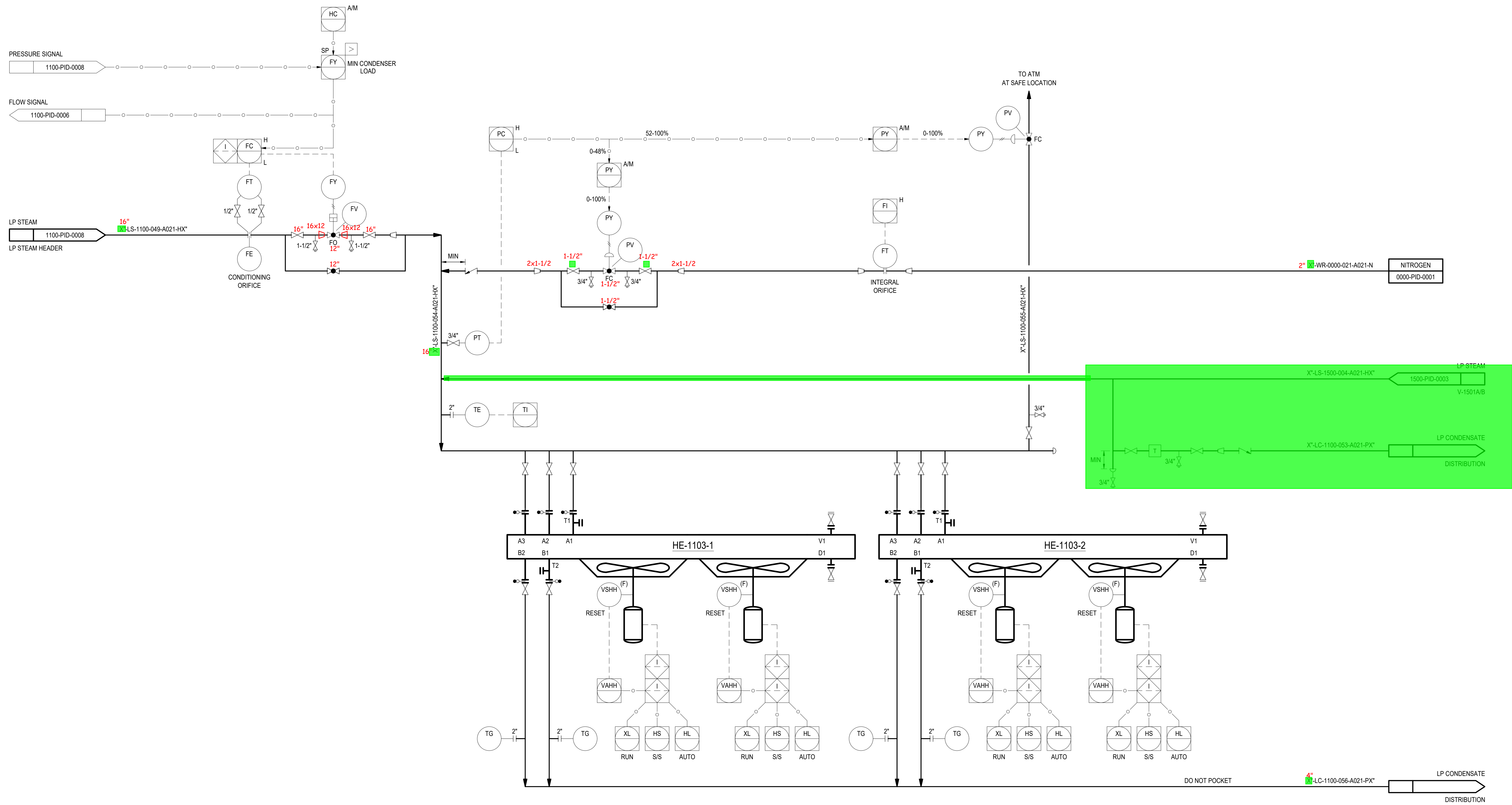


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A										12-05-25			ISSUED FOR APPROVAL			GS			DLH			PDN								
MOTIVA STEAM UNIT LP STEAM HEADER										CLIENT: MOTIVA ENTERPRISES LLC			PLANT: PORT ARTHUR REFINERY			PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW			SSP SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040			FIRM NUMBER #13619								
PROJECT NO: 25MA02										SCALE: NONE			DRAWN BY: GS			CHECKED: APB			DESIGN BY:			ENG APVL: PDN								
DRAWING NO:										1100-PID-0008			REVISION:																	

HE-1103
LP STEAM VENT CONDENSER
RATED DUTY :
NUMBER OF BAYS :
DESIGN TEMP :
RATED POWER :
MOC :
TRIM : VT-1100-057-A021-PX"
INSULATION :

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A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
LP STEAM VENT CONDENSER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

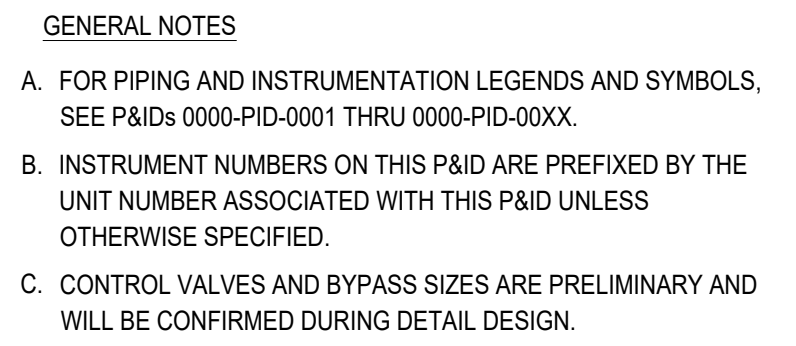
SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

1100-PID-0009

REV

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REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
HP BOILER FEED WATER HEADER

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

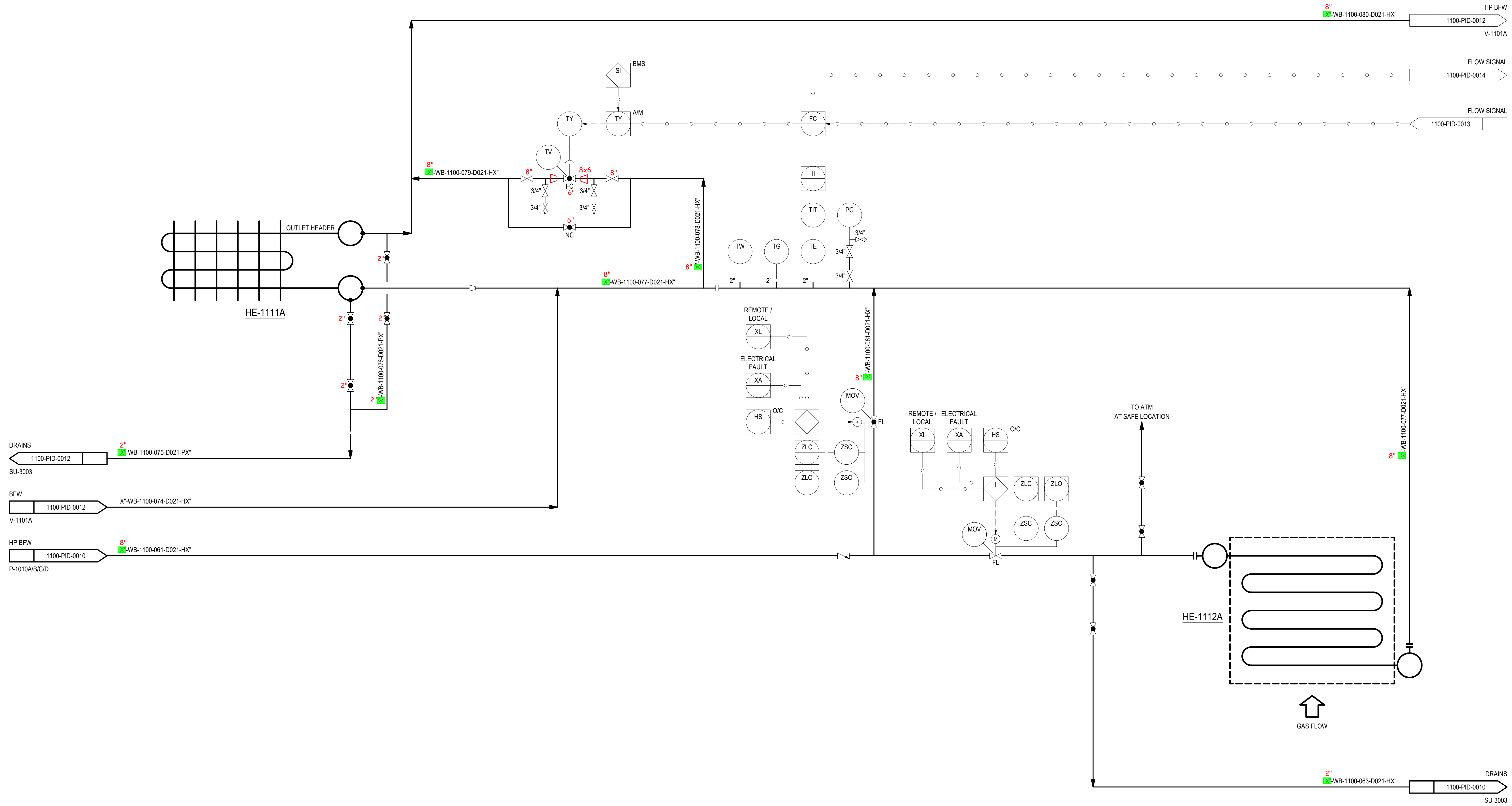
1100-PID-0010

REV A

HE-1111A
INTERNAL ECONOMIZER
RATED DUTY :
SURFACE AREA :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-082-D021-HX"
INSULATION :

HE-1112A
EXTERNAL ECONOMIZER
RATED DUTY :
SURFACE AREA :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-083-D021-HX"
INSULATION :

GENERAL NOTES
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CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
EXTERNAL AND INTERNAL ECONOMIZER

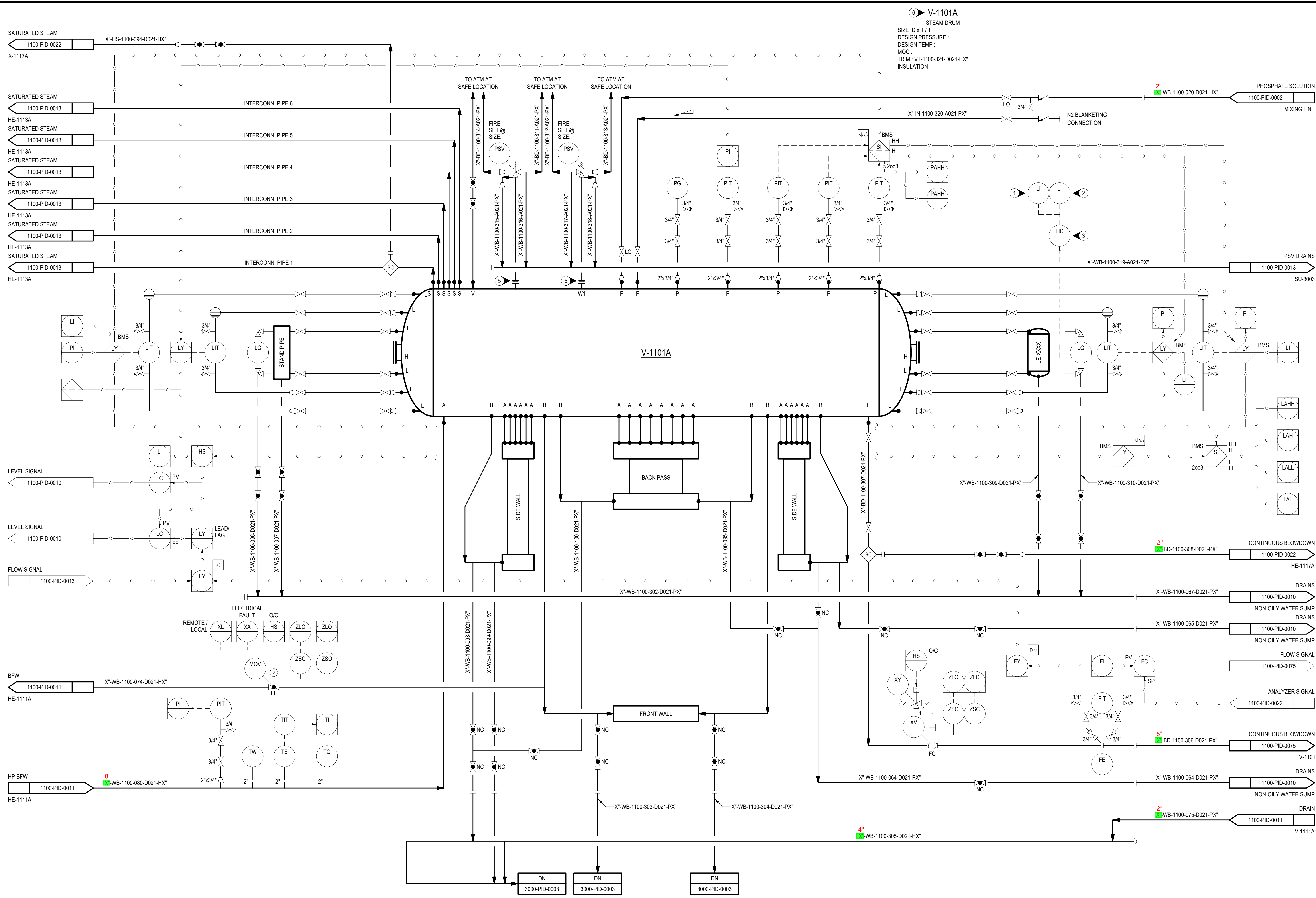
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

FIRM NUMBER #13619
REV A



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- NOTES
1. LEVEL INDICATOR MOUNTED LOCALLY AT GRADE ELEVATION.
2. LEVEL INDICATOR MOUNTED IN CONTROL ROOM.
3. WIRING AND JUNCTION BOX INTEGRAL WITH WATER COLUMN.
4. DRUM CENTER LINE (DCL).
5. THESE CONNECTIONS ARE LOCATED ON STEAM DRUM HEAD.
6. BOILER CONFIGURATION IS TYPICAL. DRAWING WILL BE MODIFIED ONCE BOILER IS SELECTED.

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CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
STEAM DRUM

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

DRAWING NO:

FIRM NUMBER
#13619

REV
1100-PID-0012

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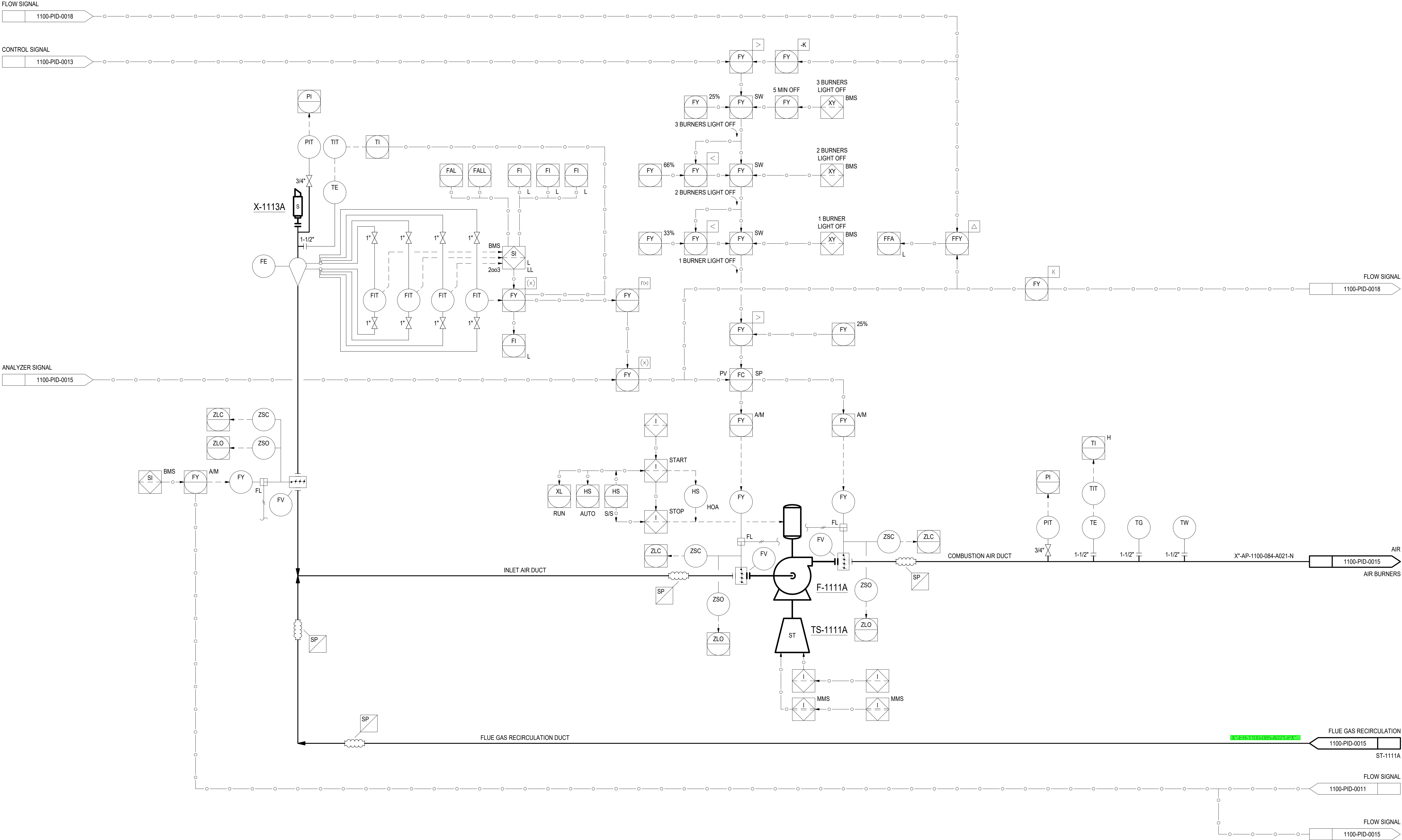
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	A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN										

X-1113A
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RATED CAPACITY :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-086-A021-N
INSULATION :

F-1111A
FORCED DRAFT FAN
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC :
TRIM : VT-1100-087-A021-N
INSULATION :

TS-1111A
STEAM TURBINE
RATED POWER :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-086-D021-HX"
INSULATION :

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- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM
STEAM UNIT
BOILER A
FORCED DRAFT FAN

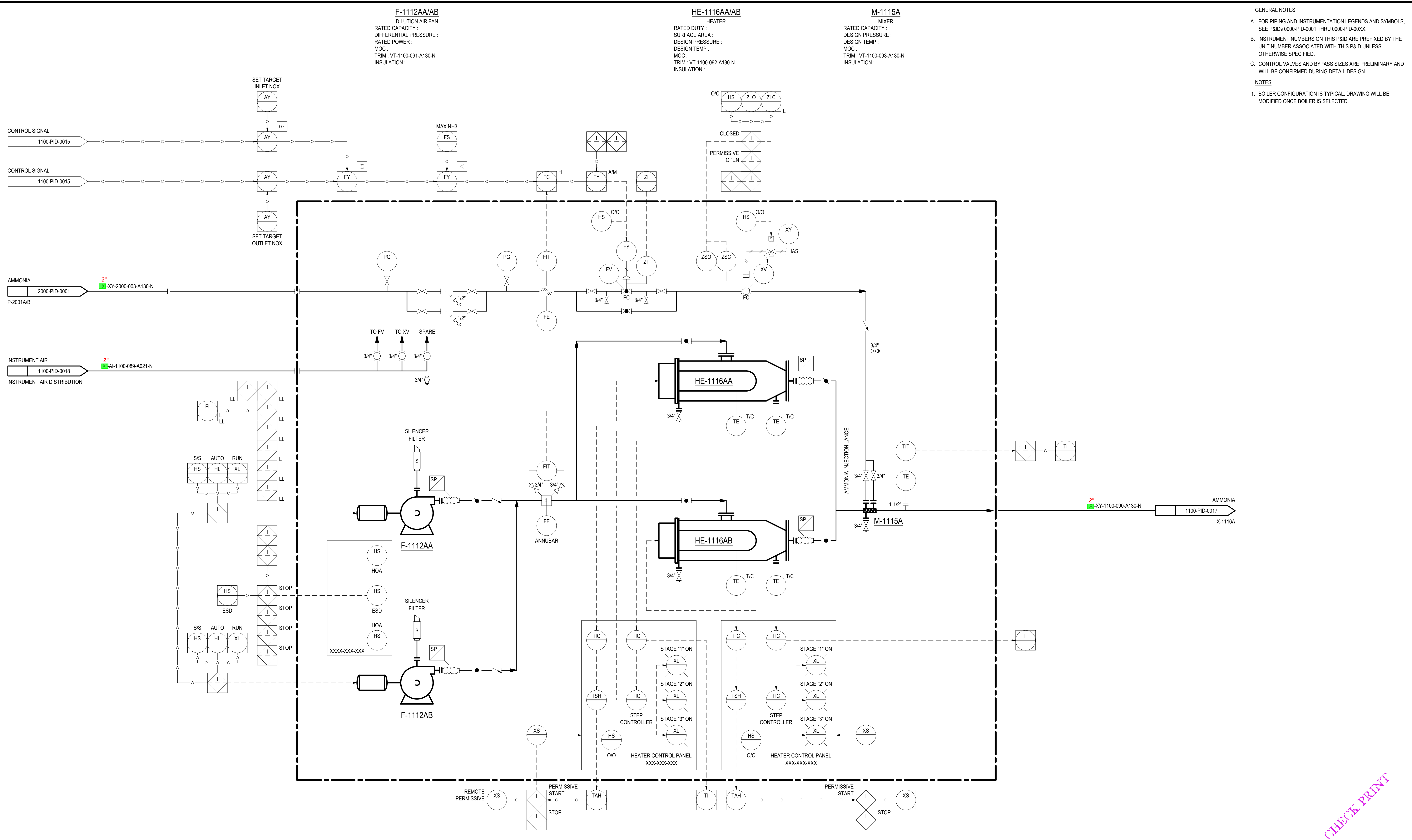
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

1100-PID-0014

REV



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- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
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REFERENCE DRAWINGS:			
1100-PID-0016	1100-PID-0017	1100-PID-0018	1100-PID-0019

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
BOILER A
AFCU

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
DRAWN BY: GS
DESIGN BY:
DRAWING NO:

SCALE: NONE
CHECKED: APB
ENG APVL: PDN

FIRM NUMBER
#13619

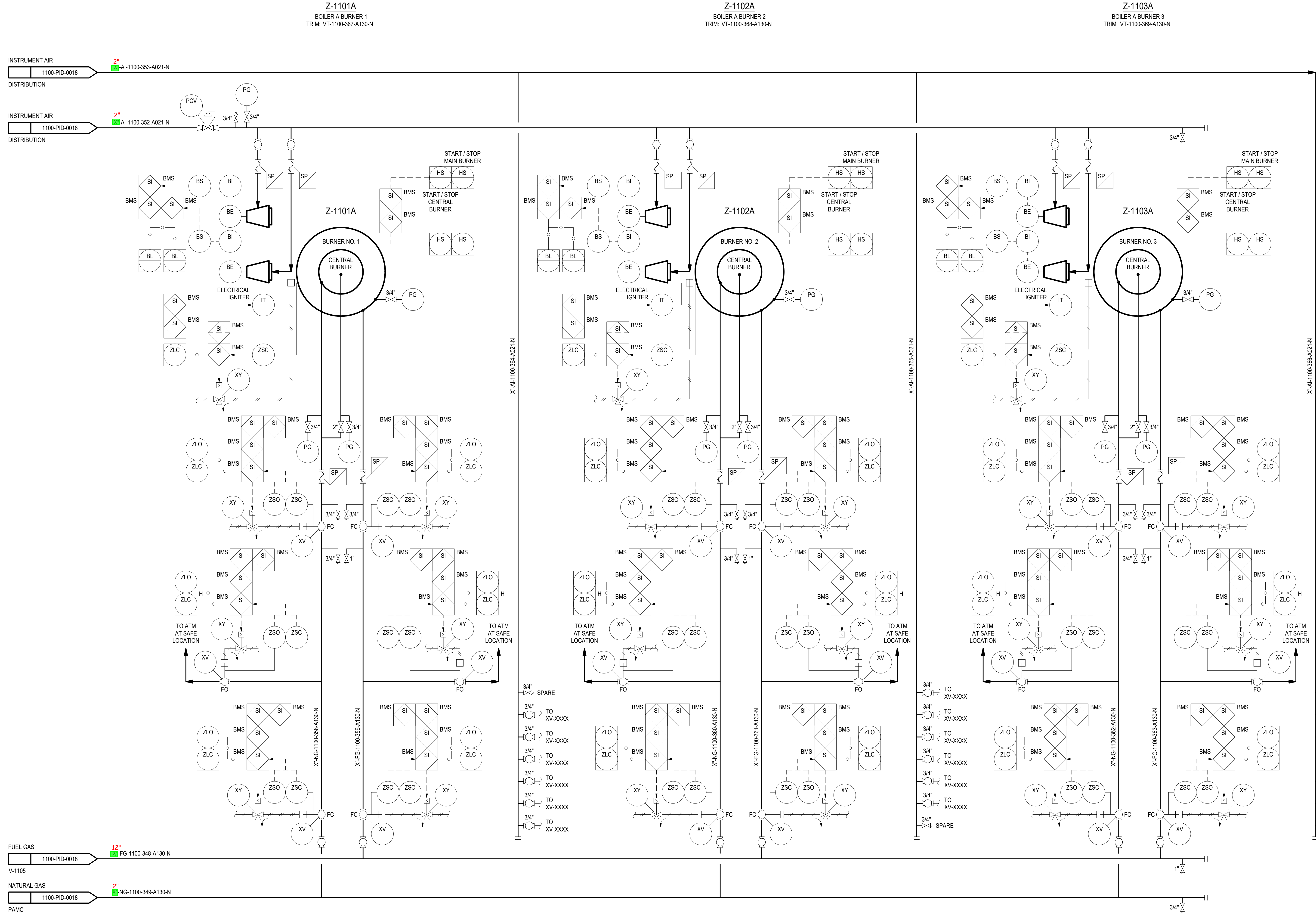
REV
A

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REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD																																																																				
A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN																																																																				
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A	12-05-25	ISSUED FOR APPROVAL	GS	DLH	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT

BOILER A BURNERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

1100-PID-0019

FIRM NUMBER **#13619**

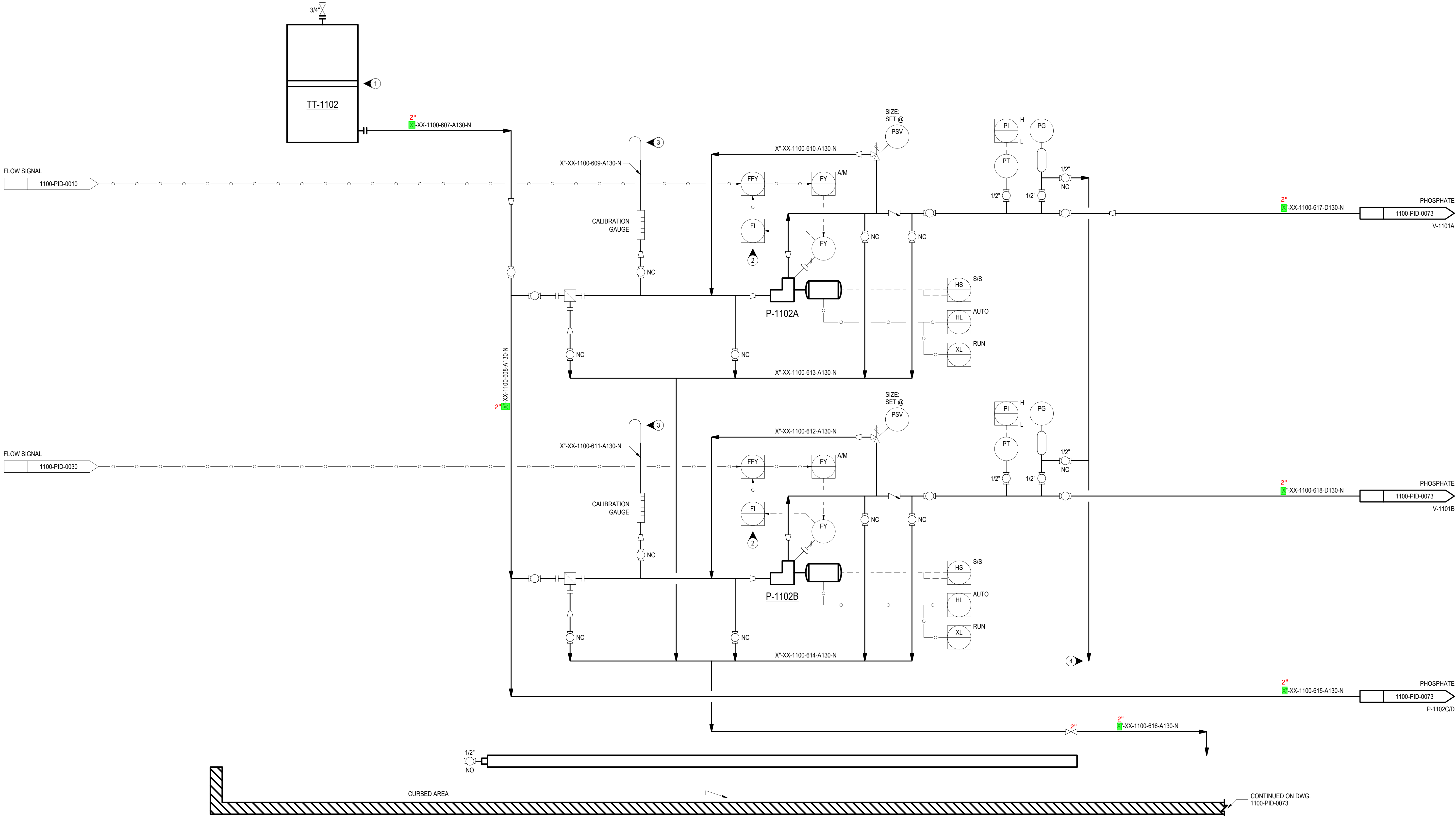
REV **A**

TT-1102
PHOSPHATE STACKED TOTE
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-631-A130-N
INSULATION / TRACING :

P-1102A/B
PHOSPHATE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1100-632-D130-N
SEAL PLAN :

GENERAL NOTES
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C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. STACKED TOTE.
 2. STROKE FEEDBACK.
 3. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.
 4. OPEN END DRAIN TO CURB AREA.
 5. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START / RUN / STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAINED STOP.
 6. ALL CHEMICAL INJECTIONS WILL BE IN COMMON CURBED AREA WITH ONE SUMP AND ONE 2" STAND PIPE.
 7. THE SELECTION SWITCH IS USED TO SELECT THE SPARE INJECTION DEVICE TO REPLACE ANY ONE OF THE THREE INJECTION DEVICES WHICH IS OFFLINE FOR MAINTENANCE.



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REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STEAM UNIT
PHOSPHATE PACKAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE
DRAWN BY: GS
CHECKED: APB
DESIGN BY:
ENG APVL: PDN
DRAWING NO:

FIRM NUMBER #13619

1100-PID-0071 **A**

SEAL PLAN :

NOTES

1. TOP OF VENT IS LOCATED HIGHER THAN TOP OF TANK.
2. OPEN END DRAIN TO CURB AREA.
3. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START / RUN / STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAINED STOP.
4. ALL CHEMICAL INJECTIONS WILL BE IN COMMON CURBED AREA WITH ONE SUMP AND ONE 2" STAND PIPE.
5. THE SELECTION SWITCH IS USED TO SELECT THE SPARE INJECTION DEVICE TO REPLACE ANY ONE OF THE THREE INJECTION DEVICES WHICH IS OFFLINE FOR MAINTENANCE.
3. STROKE FEEDBACK.



	SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040		FIRM NUMBER #13619
	PROJECT NO: 25MA02	SCALE: NONE	
	DRAWN BY: GS	CHECKED: APB	
	DESIGN BY:	ENG APVL: PDN	
DRAWING NO:		1100-PID-0073	REV A

V-1101
CONTINUOUS BLOWDOWN DRUM
SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1100-639-D021-PX"
INSULATION / TRACING :

GENERAL NOTES

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

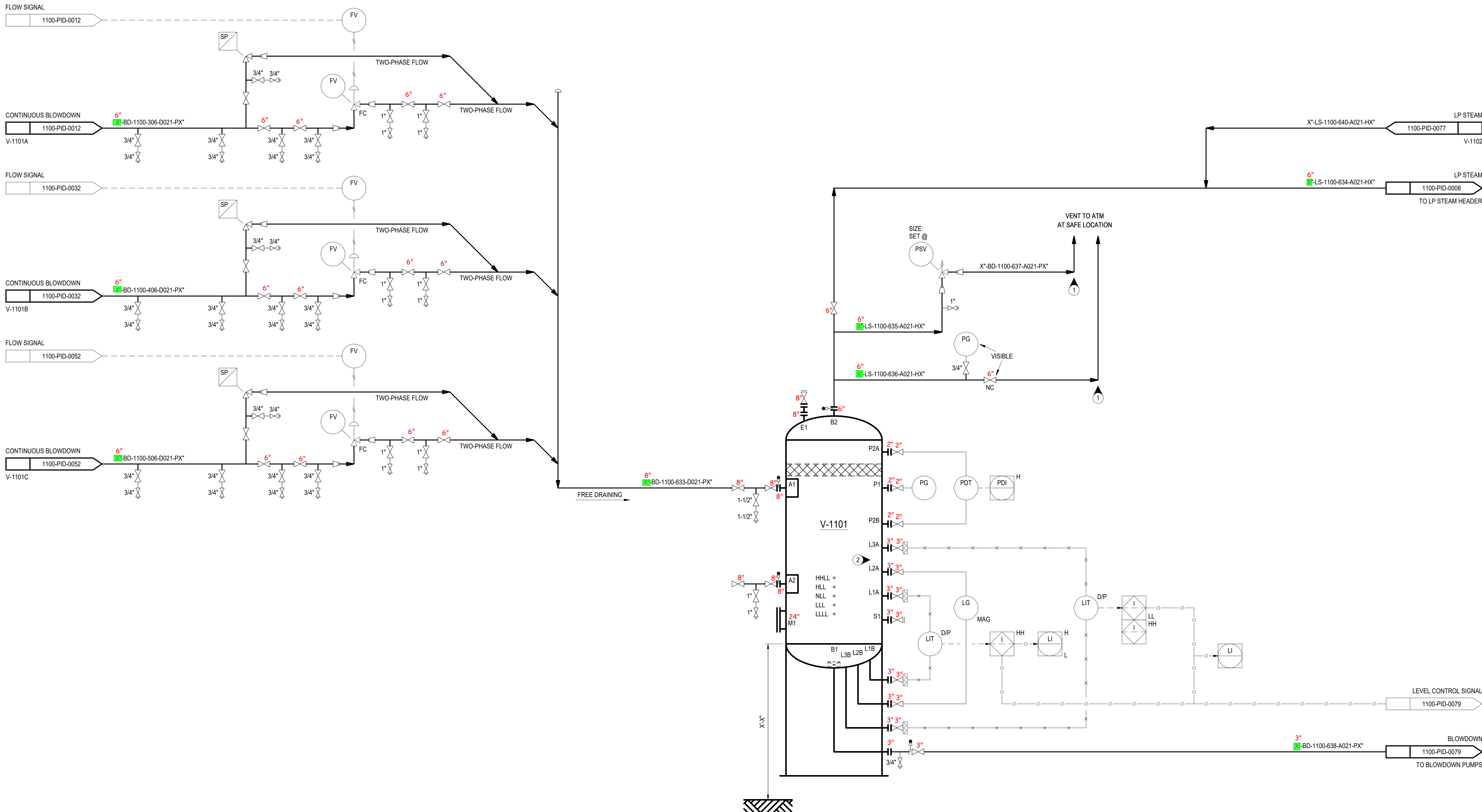
8. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

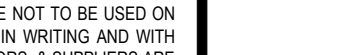
C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

NOTES

1. 3/8" WEEP HOLE AT LOW POINT.

2. ALL UPPER L1A, L2A AND L3A NOZZLES ARE AT SAME ELEVATION. ALL LOWER L1B, L2B AND L3B NOZZLES ARE AT SAME ELEVATION.



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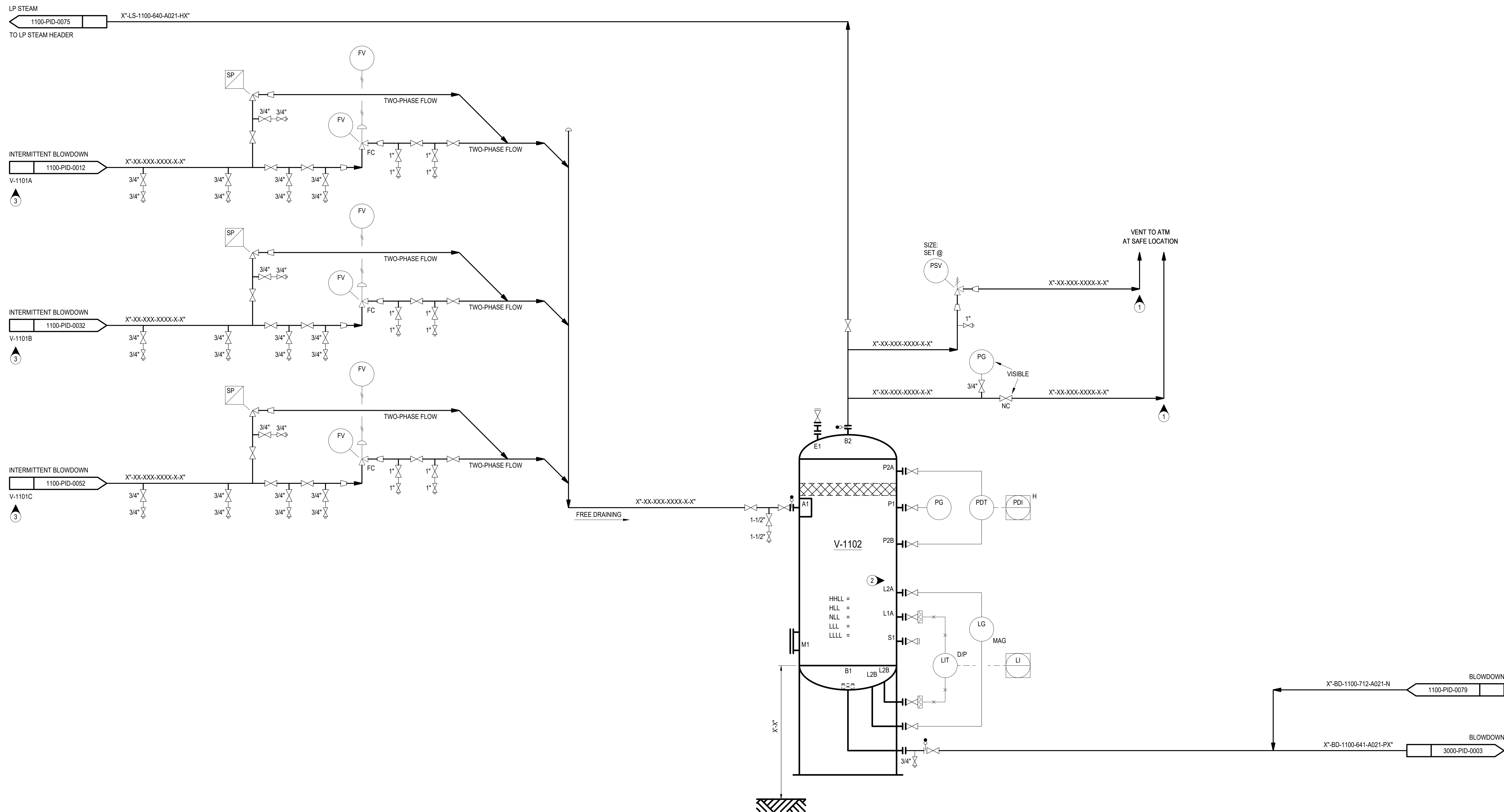
③ **V-1102**
 INTERMITTENT BLOWDOWN DRUM
 SIZE :
 TOTAL VOLUME :
 DESIGN PRESSURE :
 DESIGN TEMP :
 MOC :
 TRIM :
 INSULATION / TRACING :

GENERAL NOTES

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NOTES

1. 3/8" WEEP HOLE AT LOW POINT.
2. UPPER L1A AND L2A NOZZLES ARE AT SAME ELEVATION.
LOWER L1B AND L2B NOZZLES ARE AT SAME ELEVATION.
3. NEED FOR INTERMITTENT BLOWDOWN WILL BE DETERMINED
WHEN BOILER IS SELECTED.

DRAWING: 1100-pic-0077.dwg, PLOTTED BY: ssp_t, PLOT DATE: ---, DRAWING LOCATION: c:\users\ssp_t\eng\ssp - projects\motiva\25ma02 - aromatics utility scope cold eyes review\04 - design\04 - process\pics, (Border Rev1 5.16.25)

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										A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN	MOTIVA STEAM UNIT INTERMITTENT BLOWDOWN DRUM CLIENT: MOTIVA ENTERPRISES LLC PLANT: PORT ARTHUR REFINERY JOB NO: PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW DRAWINGS NO: 1100-PID-0077 REV A											

GENERAL NOTES

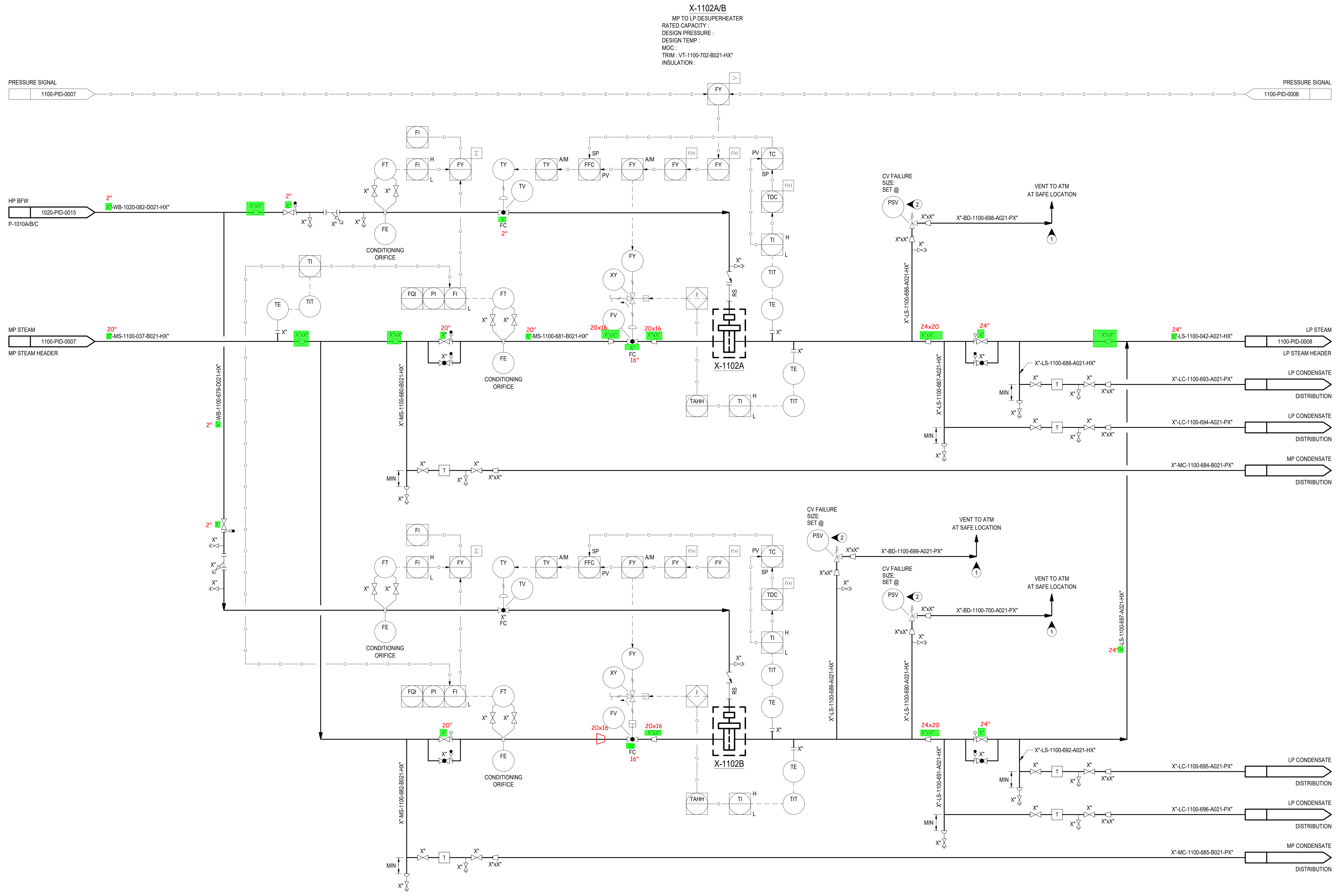
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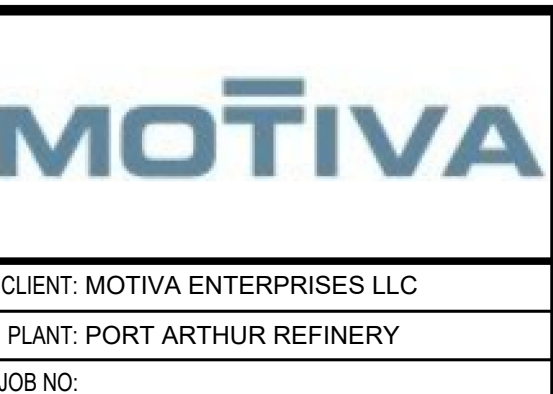
 SSP ENGINEERING & CONSULTING 13100 NORTHWEST FREEWAY SUITE 625, HOUSTON, TEXAS 77040	FIRM NUMBER	
	#13619	
	PROJECT NO: 25MA02	SCALE: NONE
	DRAWN BY: GS	CHECKED: APB
DESIGN BY:	ENG APVL: PDN	
DRAWING NO:	1100-PID-0079	REV: A



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REV #	DATE	DESCRIPTION	
A	12-05-25	ISSUED FOR APPROVAL	

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

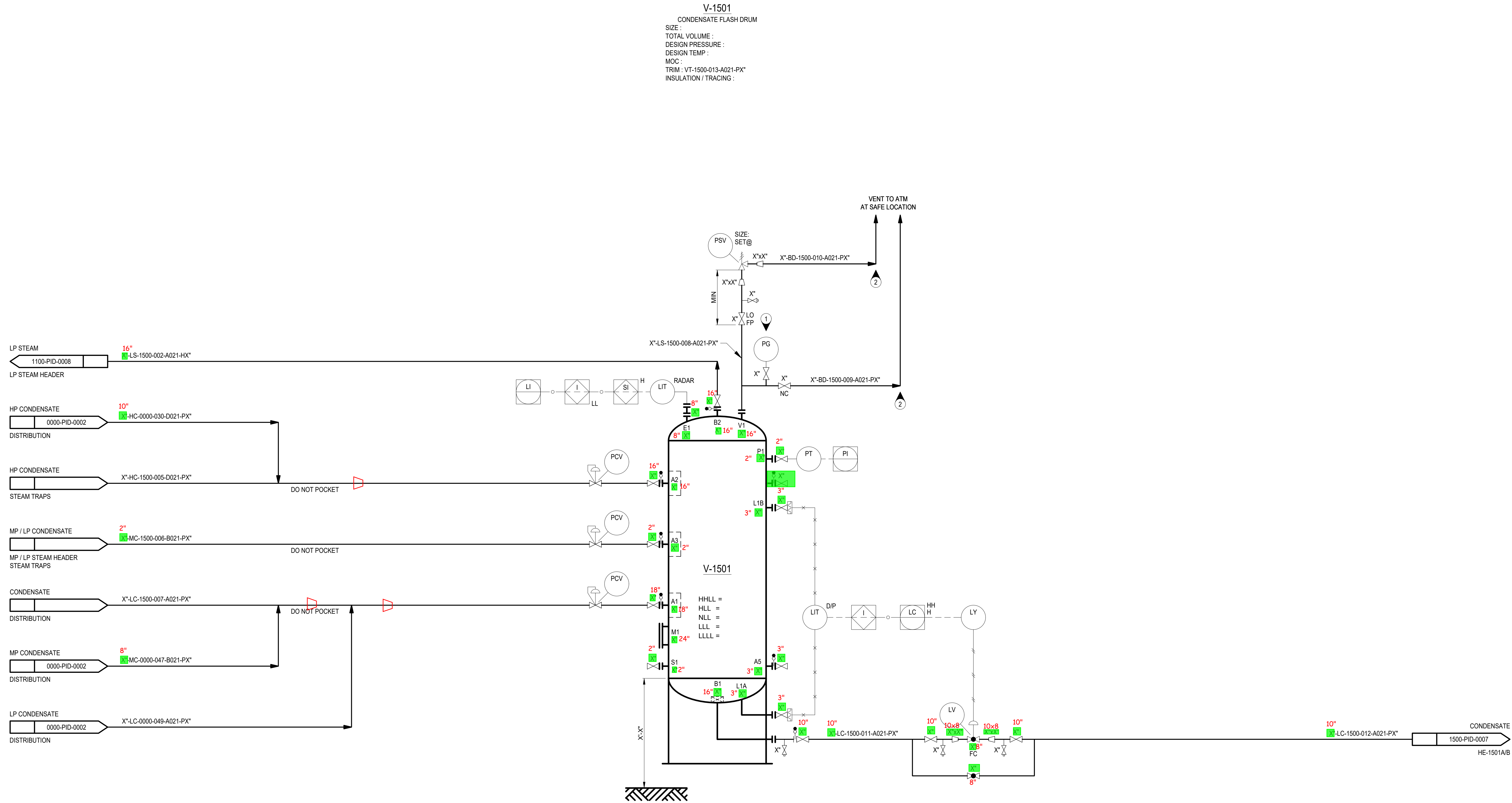


PIPING & INSTRUMENTATION DIAGRAM	
STEAM UNIT	
MP TO LP DESUPERHEATER	
CLIENT: MOTIVA ENTERPRISES LLC	
PLANT: PORT ARTHUR REFINERY	
JOB NO:	

SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040		SCALE: NONE	
PROJECT NO: 25MA02		DRAWN BY: GS	
DESIGN BY:		CHECKED: APB	
DRAWING NO:		ENG APVL: PDN	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW		REV	
		1100-PID-0087	

CHECK PRINT

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes review\04 - design\04 - process\pids, (Border Rev1 5 18 25), PLOT DATE: ---, PLOT BY: jason, DRAWING NO: 1500-pid-0001.dwg, NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT, SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-000X.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. PRESSURE INDICATOR TO BE READABLE FROM BYPASS VALVE.
2. PROVIDE 3/8" WEEP HOLE AT LOW POINT.

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM
PIPING & INSTRUMENTATION DIAGRAM
CONDENSATE FLASH DRUM A

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02
SCALE: NONE

DRAWN BY: GS
CHECKED: APB

DESIGN BY:
ENG APVL: PDN

DRAWING NO:
1500-PID-0001

FIRM
NUMBER
#13619

REV
A

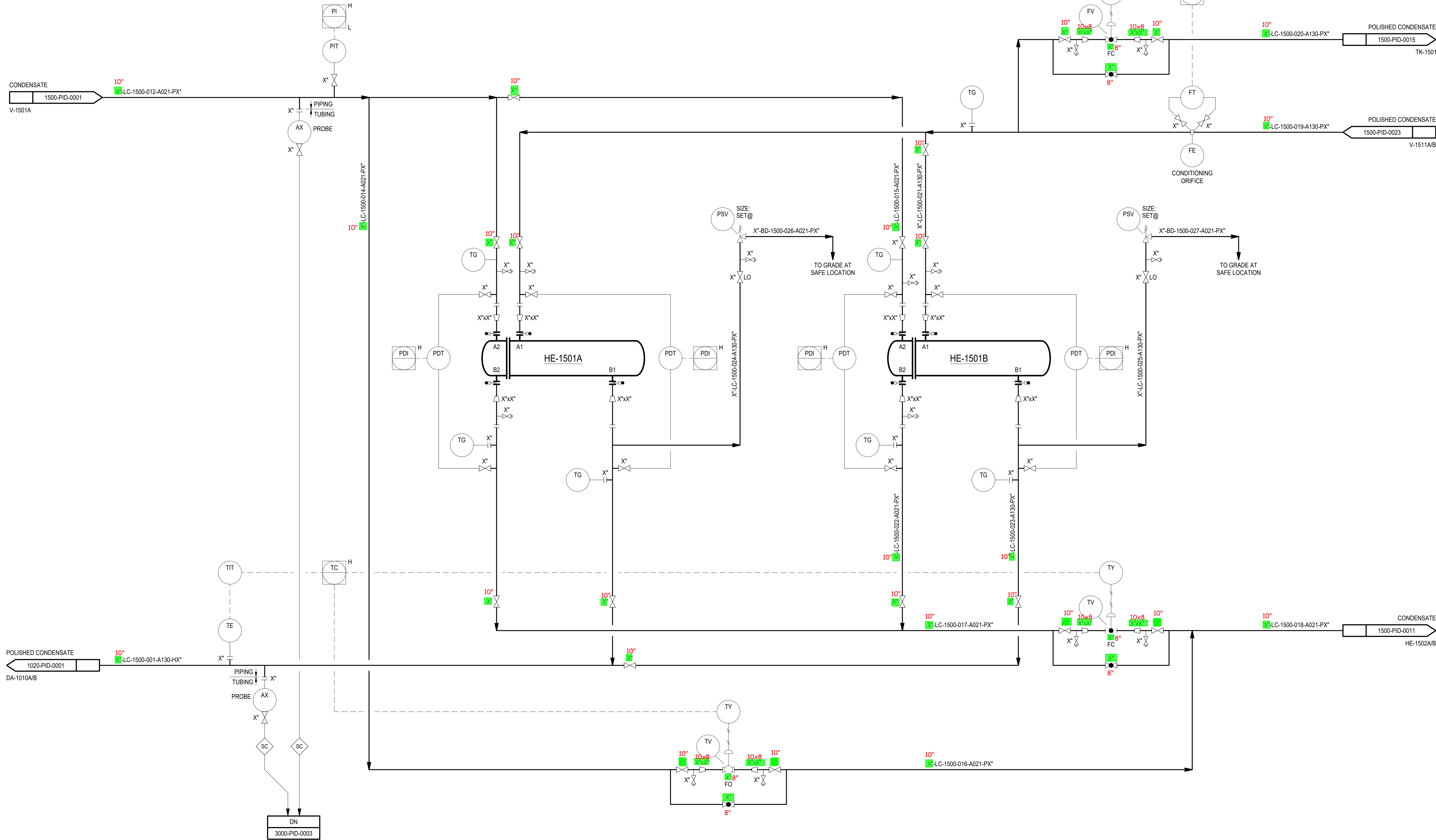
HE1501A/B

CONDENSATE EXCHANGER

TYPE :
RATED DUTY :
SURFACE AREA :
DESIGN PRESS @ TEMP (HOT SIDE) :
DESIGN PRESS @ TEMP (COLD SIDE) :
MOC :
TRIM (HOT / COLD) : VT-1500-028-A021-PX*
VT-1500-029-A130-PX*
INSULATION :

GENERAL NOTES

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



CHECK PRINT

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PIPING & INSTRUMENTATION DIAGRAM
CONDENSATE SYSTEM
CONDENSATE EXCHANGERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

REV

1500-PID-0007

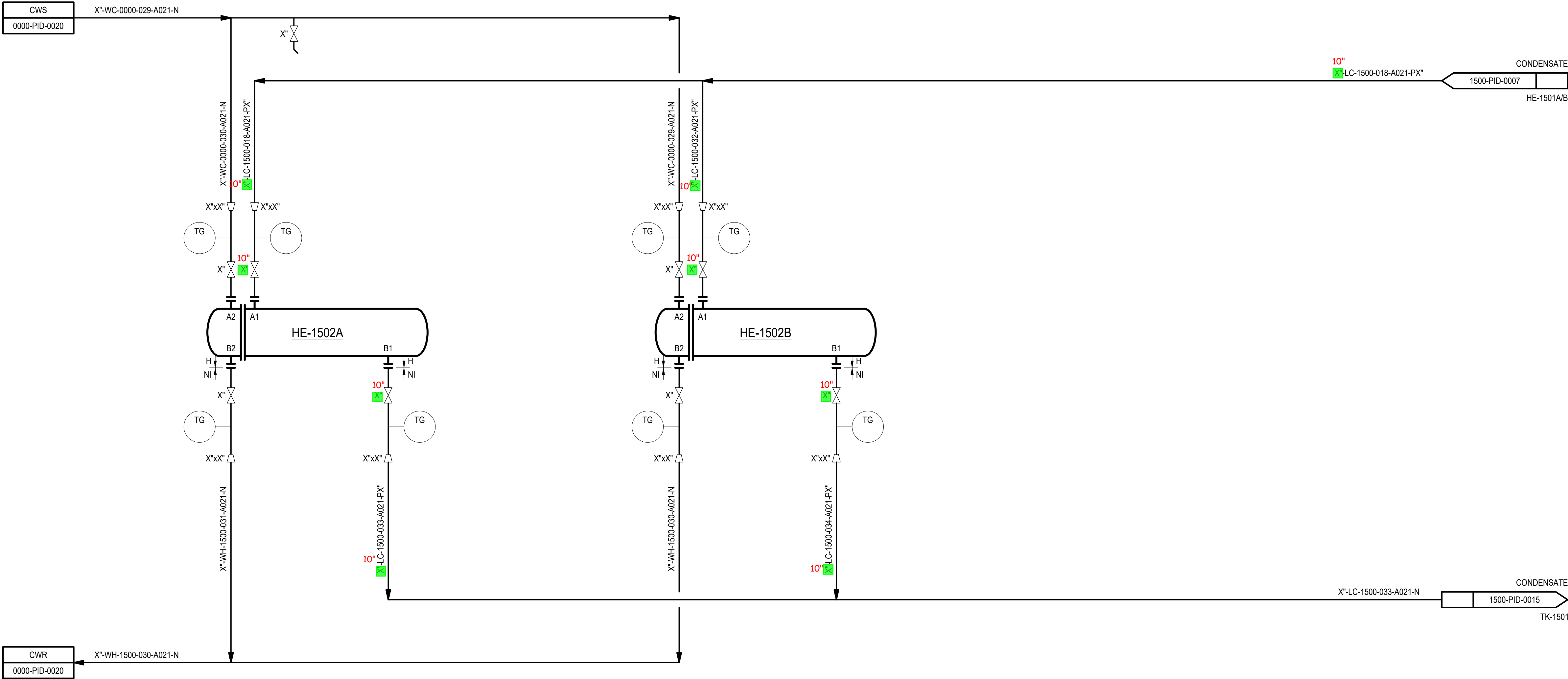
A

FIRM
NUMBER
#13619

HE-1502A/B
CONDENSATE TRIM COOLER

TYPE :
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1500-035 A021-N
VT-1500-036 A021-N
INSULATION (SHELL / TUBES) :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PROCESS FLOW DIAGRAM
PIPING & INSTRUMENTATION DIAGRAM
CONDENSATE SYSTEM
CONDENSATE TRIM COOLERS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



SSP ENGINEERING
& CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM
NUMBER
#13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

1500-PID-0011

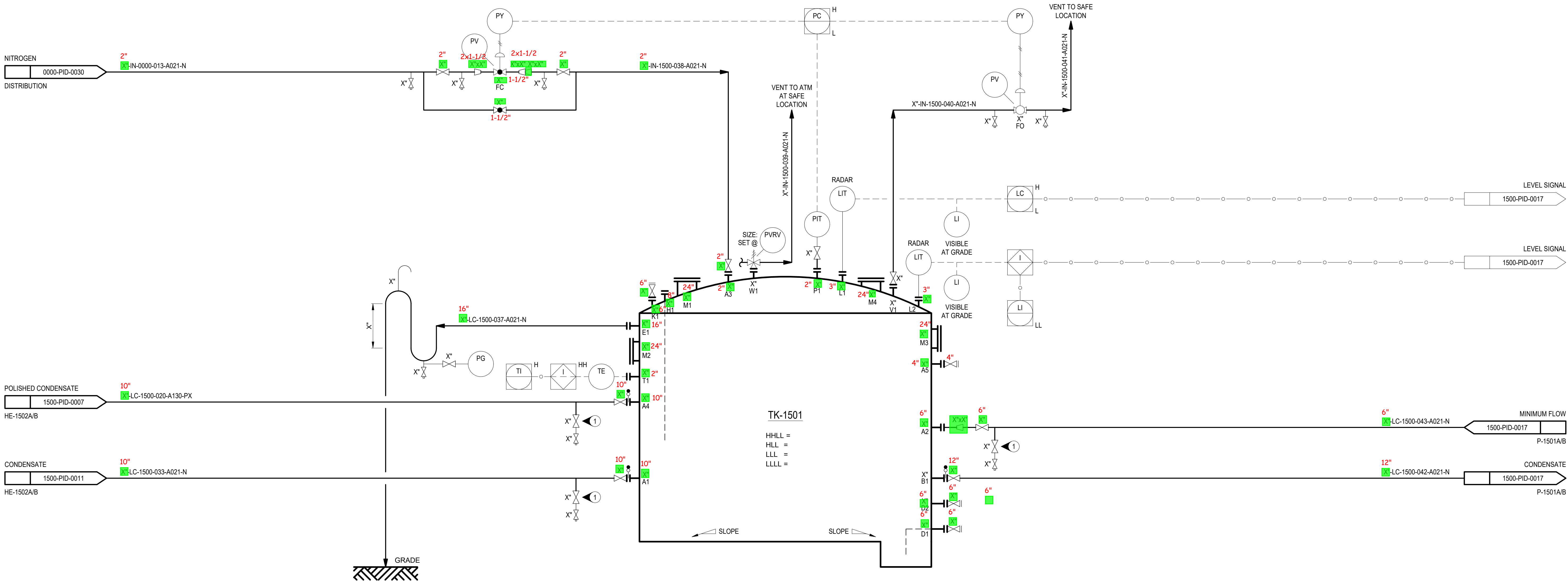
REV

A

TK-1501
CONDENSATE TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1500-044-A021-N
INSULATION / TRACING :

3. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.

1. CONNECTION TO BYPASS TANK DURING TANK INSPECTION AND MAINTENANCE.



CHECK PRINT

[illegible]

REFERENCE DRAWINGS

[illegible]**MOTIVA**

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE SYSTEM
CONDENSATE TANK

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DRAWN BY: GS

DESIGN BY:

**FIRM
NUMBER
#13619**

SCALE: NONE

CHECKED: ABB

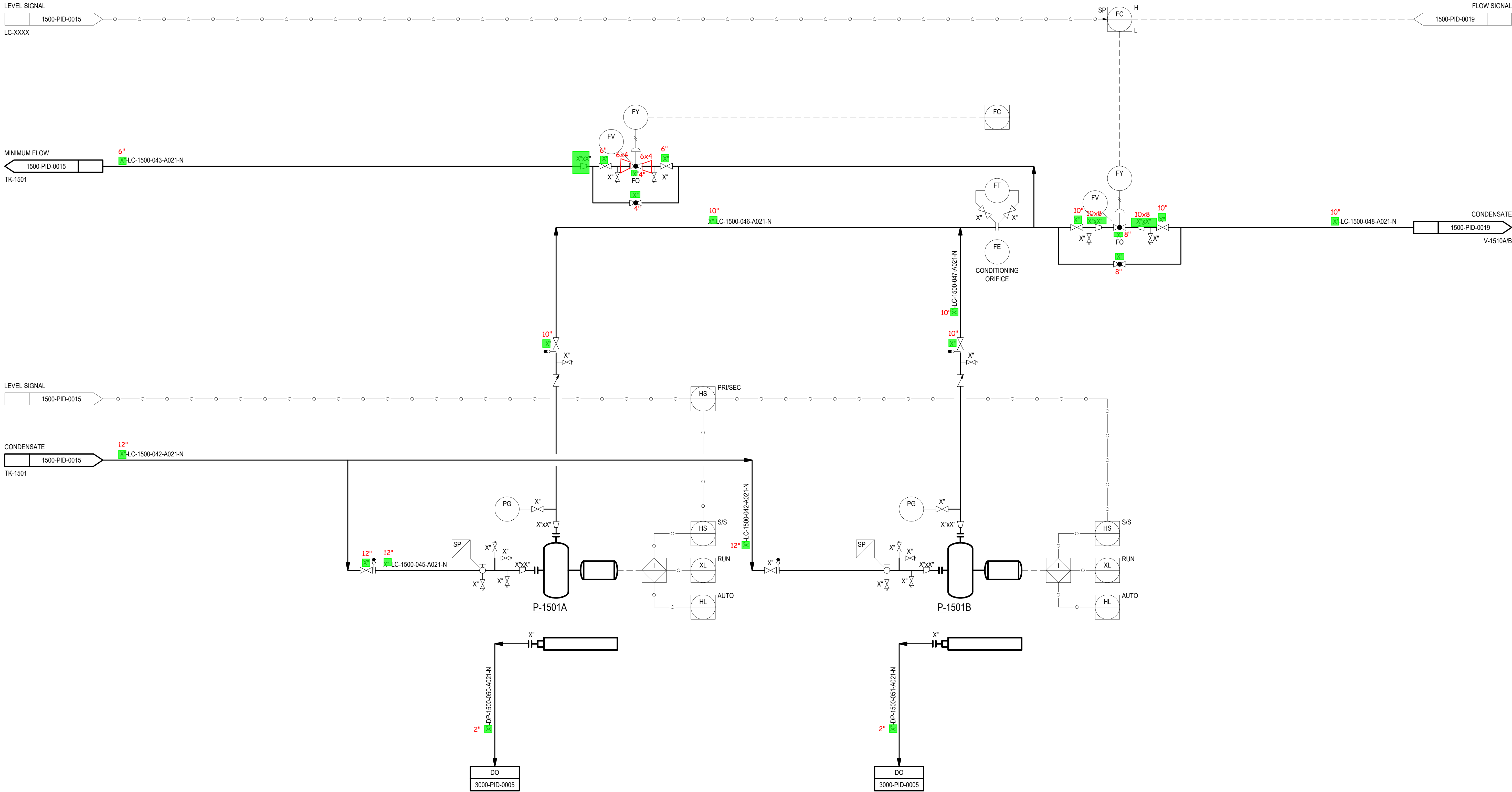
CHECKED. AFD
MAIL ROOM

ENG APVL: PDN

RE

P-1501A/B
CONDENSATE PUMPS
RATED CAPACITY :
DIFFERENTIAL PRESSURE :
RATED POWER :
MOC (CASING / INTERNAL) :
INSULATION / TRACING :
TRIM : VT-1500-048-A021-N
SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE SYSTEM

CONDENSATE PUMPS

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER #13619

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

1500-PID-0017

REV **A**

ACTIVATED CARBON FILTERS

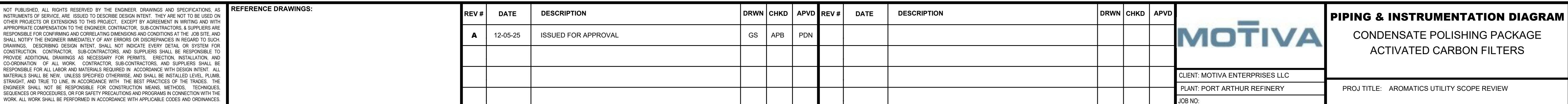
TRIM : VT-1500-055-A021-N
INSULATION / TRACING :

A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS,
SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

OTHERWISE SPECIFIED.

1. SELECTOR SWITCH FOR SELECTING BETWEEN ACF V-1510A AND V-1510B.

BE REVISED ONCE SYSTEM IS SELECTED.



CHECK PRINT

SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1500-058-A130-N
INSULATION / TRACING :

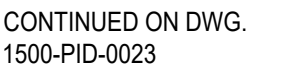
A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.

C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

1. FV-XXXX APPLY AS AN ON/OFF VALVE AND HC-XXXX IS MANUAL SET TO FULLY OPEN OR CLOSE THE VALVE.

2. COMMON CURBED AREA FOR MIXED BED POLISHER TRAIN 1 AND 2. ACID AND CAUSTIC RESISTANT LINING.

3. CONDENSATE POLISHER SHOWN IS TYPICAL. CONFIGURATION WILL BE REVISED ONCE SYSTEM IS SELECTED.



REFERENCE DRAWINGS:

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE POLISHING PACKAGE

CONDENSATE POLISHER A

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



PROJECT NO: 25MA02

DRAWN BY: G5

DESIGN BY:

DRAWING NO.

SCALE: NONE

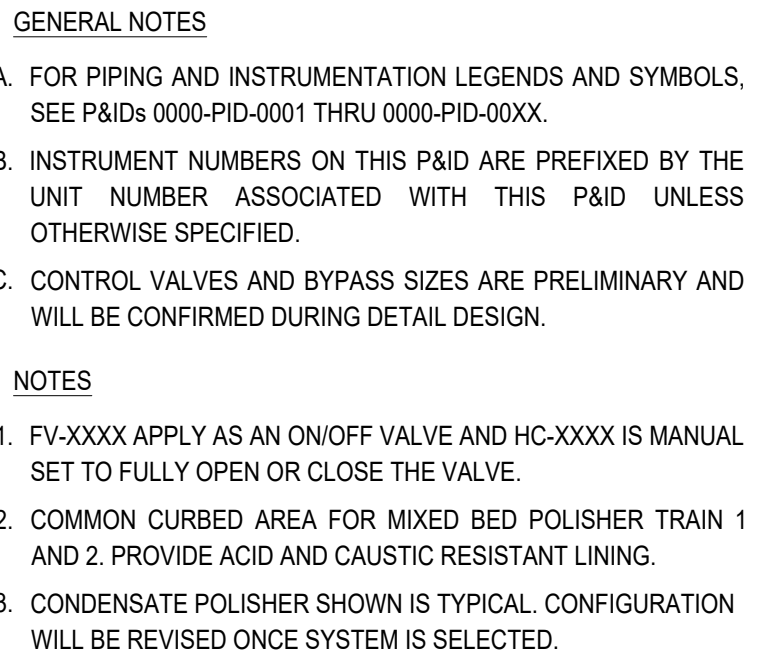
CHECKED: APB

ENGINEER: T. D. N.

1500-PID-0021

A

SIZE :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-1500-058-A130-N
INSULATION / TRACING :



CHECK PRINT

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

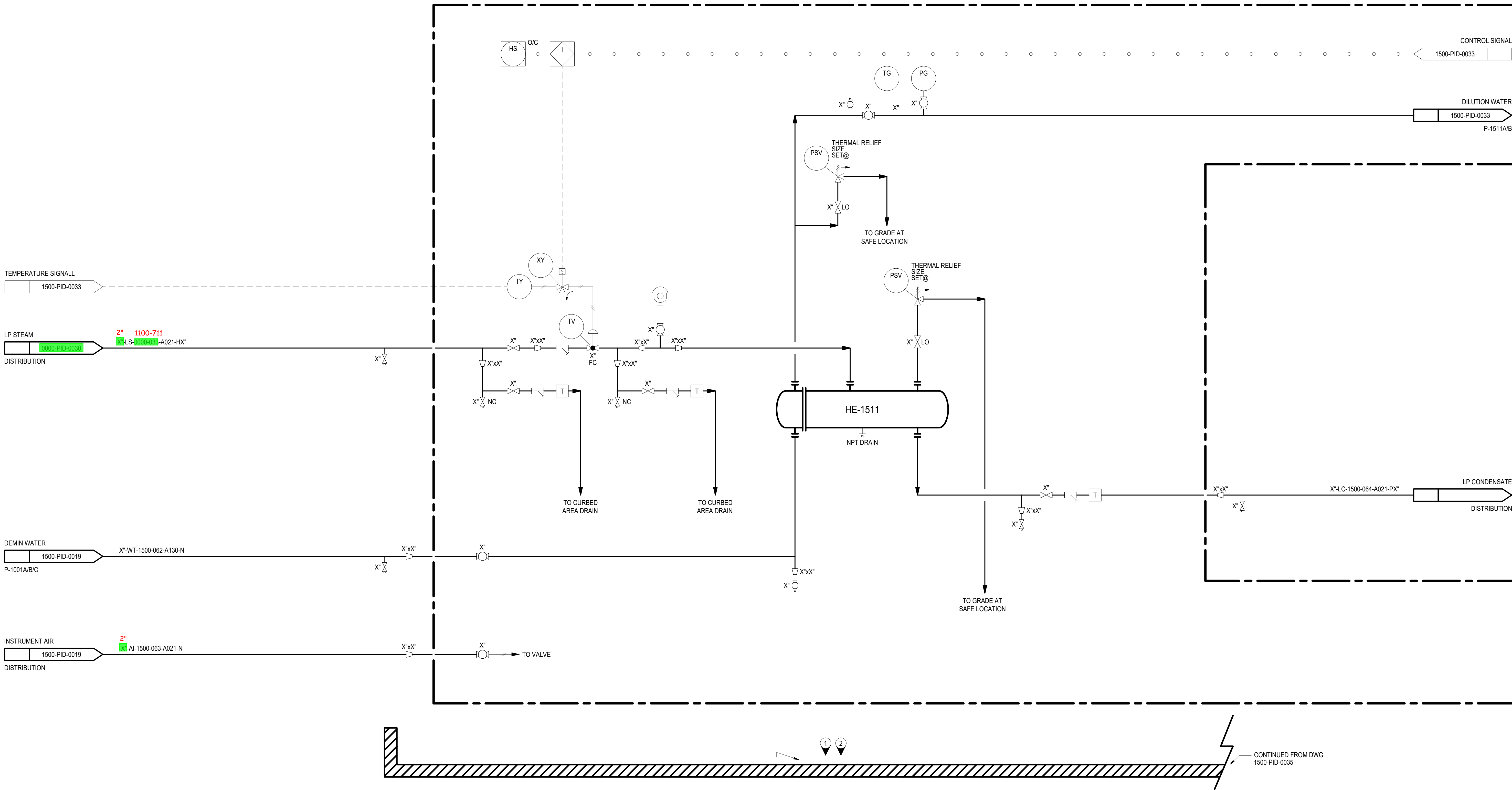
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



HE-1511
CONDENSATE CAUSTIC DILUTION HEATER
TYPE:
RATED DUTY :
SURFACE AREA :
SHELL DESIGN PRESS @ TEMP :
TUBE DESIGN PRESS @ TEMP :
MOC (SHELL / TUBES) :
TRIM (SHELL / TUBES) : VT-1500-065 A021-HX*
VT-1500-066 A130-PX*
INSULATION (SHELL / TUBES) :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. COMMON CURBED AREA FOR DEMIN CAUSTIC DILUTION SKID 1 (DRAWING XXXX-PID-XXXX), AND CONDENSATE POLISHING PACKAGE CAUSTIC DILUTION SKID.
2. CAUSTIC RESISTANT LINING.



CHECK PRINT

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REFERENCE DRAWINGS:

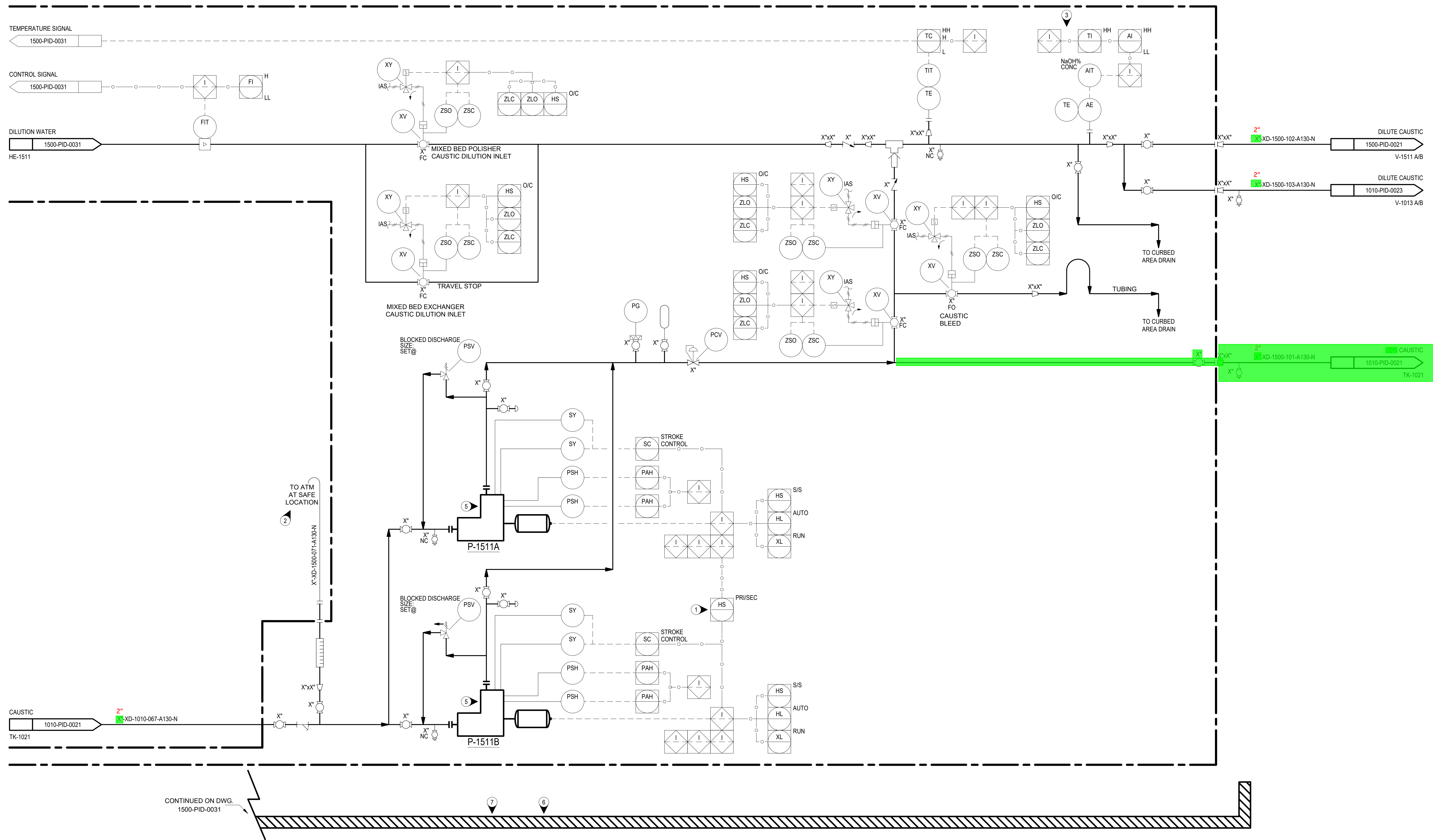
REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA		
CLIENT: MOTIVA ENTERPRISES LLC		
PLANT: PORT ARTHUR REFINERY		
JOB NO:		

PROCESS FLOW DIAGRAM	
PIPING & INSTRUMENTATION DIAGRAM	
CONDENSATE POLISHING PACKAGE	
CONDENSATE CAUSTIC DILUTION HEATER	
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW	

SSP ENGINEERING & CONSULTING		FIRM NUMBER #13619	
13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040			
PROJECT NO: 25MA02	SCALE: NONE		
DRAWN BY: GS	CHECKED: APB		
DESIGN BY:	ENG APVL: PDN		
DRAWING NO:	1500-PID-0031	REV	A

DRAWING LOCATION: c:\users\jason\one-drive - eng\ssps files - 25ma02 - aromatics utility scope old eyes\revin\04 - design\04 - process\04s - (Border Rev 1 5 18 25). PLOT DATE: ---. PLOT BY: jason. DRAWING: 1500-pid-0033.dwg.



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. DOSING PUMPS ALTERNATE DUTY AFTER EACH USE.
2. CALIBRATION COLUMN VENT IS EXTENDED TO A HEIGHT ABOVE THE MAXIMUM TANK ELEVATION.
3. INTEGRAL TEMPERATURE ANALOG SIGNAL IN ANALYZER.
4. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START/ RUN/ STOP SWITCH. SPRING RETURN FROM START TO RUN. MAINTAIN STOP.
5. CHECK VALVES INTERNAL TO THE PUMP.
6. COMMON CURBED AREA FOR DEMIN CAUSTIC DILUTION SYSTEM AND CONDENSATE POLISHING PACKAGE CAUSTIC DOSING PUMPS AND CONDENSATE CAUSTIC DILUTION HEATER.
7. CAUSTIC RESISTANT LINING.

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT, EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER. CONTRACTOR, SUB-CONTRACTORS & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW, UNLESS SPECIFIED OTHERWISE, AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

CONDENSATE POLISHING PACKAGE

CAUSTIC DOSING PUMPS

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

1500-PID-0033

FIRM NUMBER #13619

REV A

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY SUITE 625 HOUSTON, TEXAS 77040

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

CHECK PRINT

SULFURIC ACID DOSING PUMPS
 RATED CAPACITY :
 DIFFERENTIAL PRESSURE :
 RATED POWER :
 MOC (CASING / INTERNAL) :
 INSULATION / TRACING :
 TRIM : VT-1500-069-A520-N
 SEAL PLAN :

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

1. DOSE PUMPS ALTERNATE DUTY AFTER EACH USE.
2. CHECK VALVES INTERNAL TO THE PUMP.
3. ALL MOTOR DRIVEN EQUIPMENT PROVIDED WITH A 3 POSITION START/ RUN/ STOP SWITCH. SPRING RETURN FROM START TO RUN, MAINTAIN STOP.
4. COMMON CURBED AREA FOR DEMIN ACID DILUTION SKID AND CONDENSATE POLISHING PACKAGE SULFURIC ACID DOSING PUMPS.
5. CALIBRATION COLUMN VENT IS EXTENDED TO A HEIGHT ABOVE THE MAXIMUM TANK ELEVATION.
6. ACID RESISTANT LINING.



REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN

JOB NO.

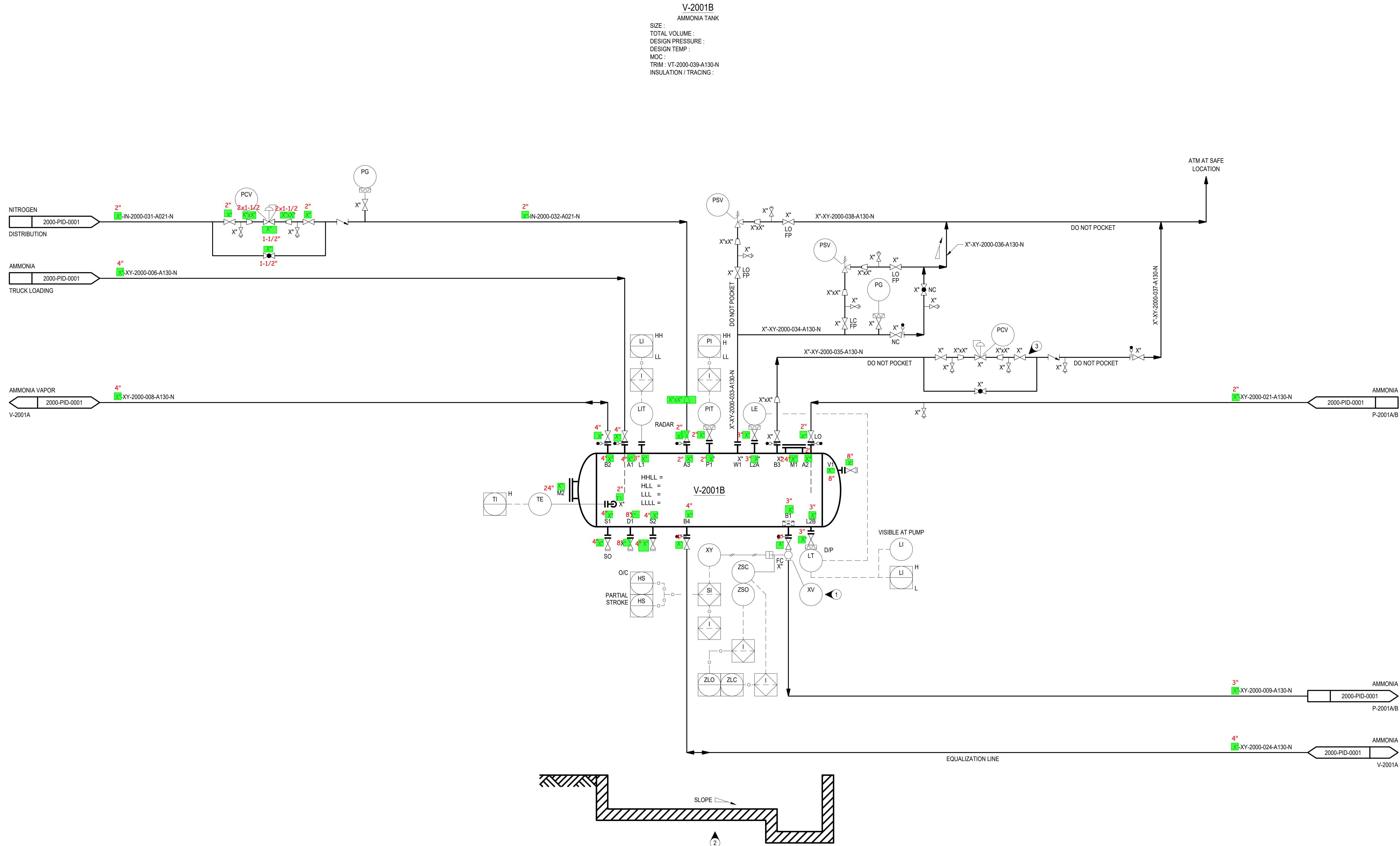
PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



1500 BID 0005

REV

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps\files - 25ma02 - aromatics utility scope old eyes\revin\04 - design\04 - process\04a -
PLOT DATE: ---, PLOT BY: jason, PLOTTED BY: jason, 2000-pid-0003.dwg, (Border Rev 1 5/18/25)



CHECK PRINT

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.:

PIPING & INSTRUMENTATION DIAGRAM

AMMONIA TANK B

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

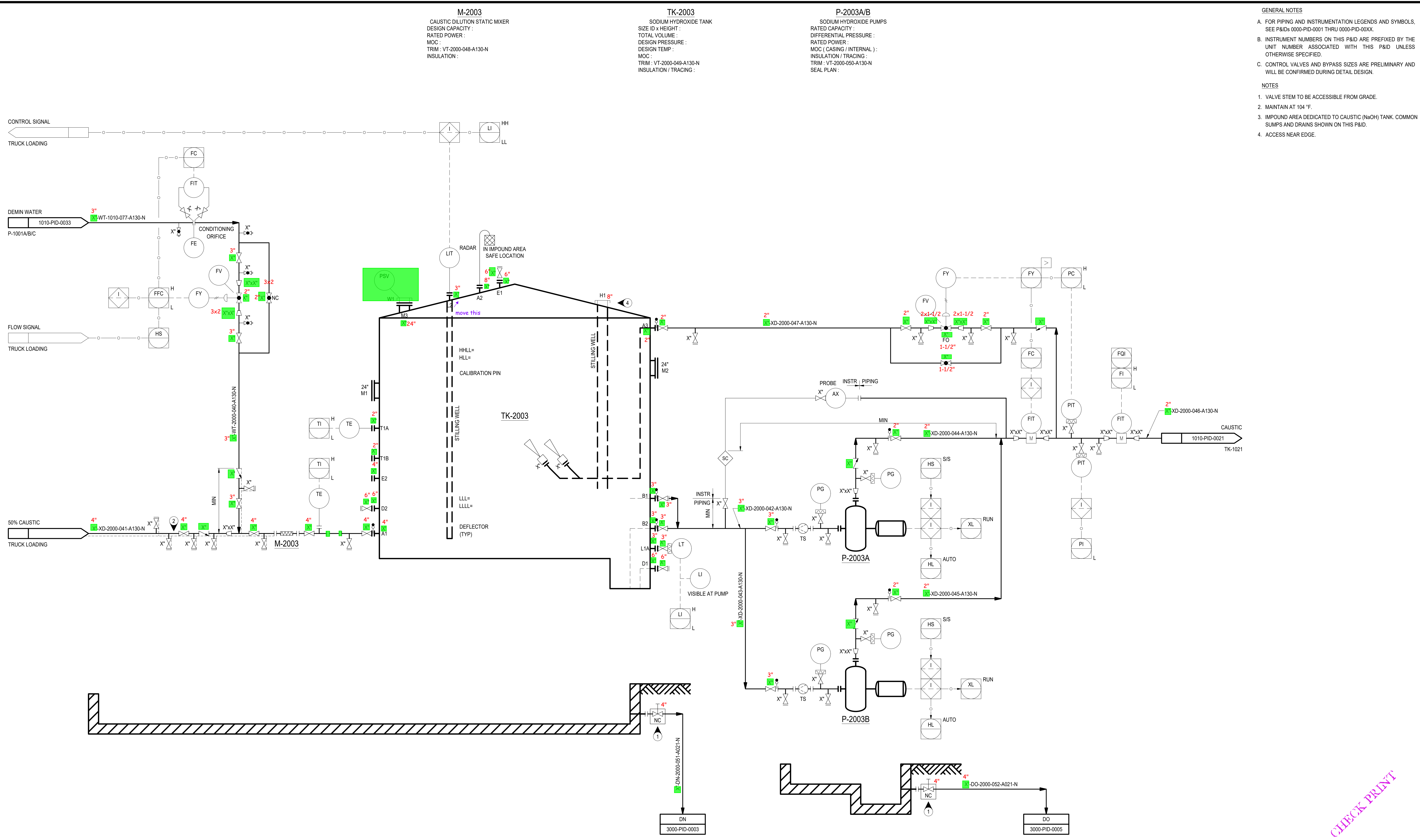
SCALE: NONE

CHECKED: APB

ENG APVL: PDN

DRAWING NO.: 2000-PID-0003

REV
A



- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDS 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. VALVE STEM TO BE ACCESSIBLE FROM GRADE.
2. MAINTAIN AT 104 °F.
3. IMPOUND AREA DEDICATED TO CAUSTIC (NaOH) TANK. COMMON SUMPS AND DRAINS SHOWN ON THIS P&ID.
4. ACCESS NEAR EDGE.

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

SODIUM HYDROXIDE STORAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

FIRM NUMBER #13619

2000-PID-0005

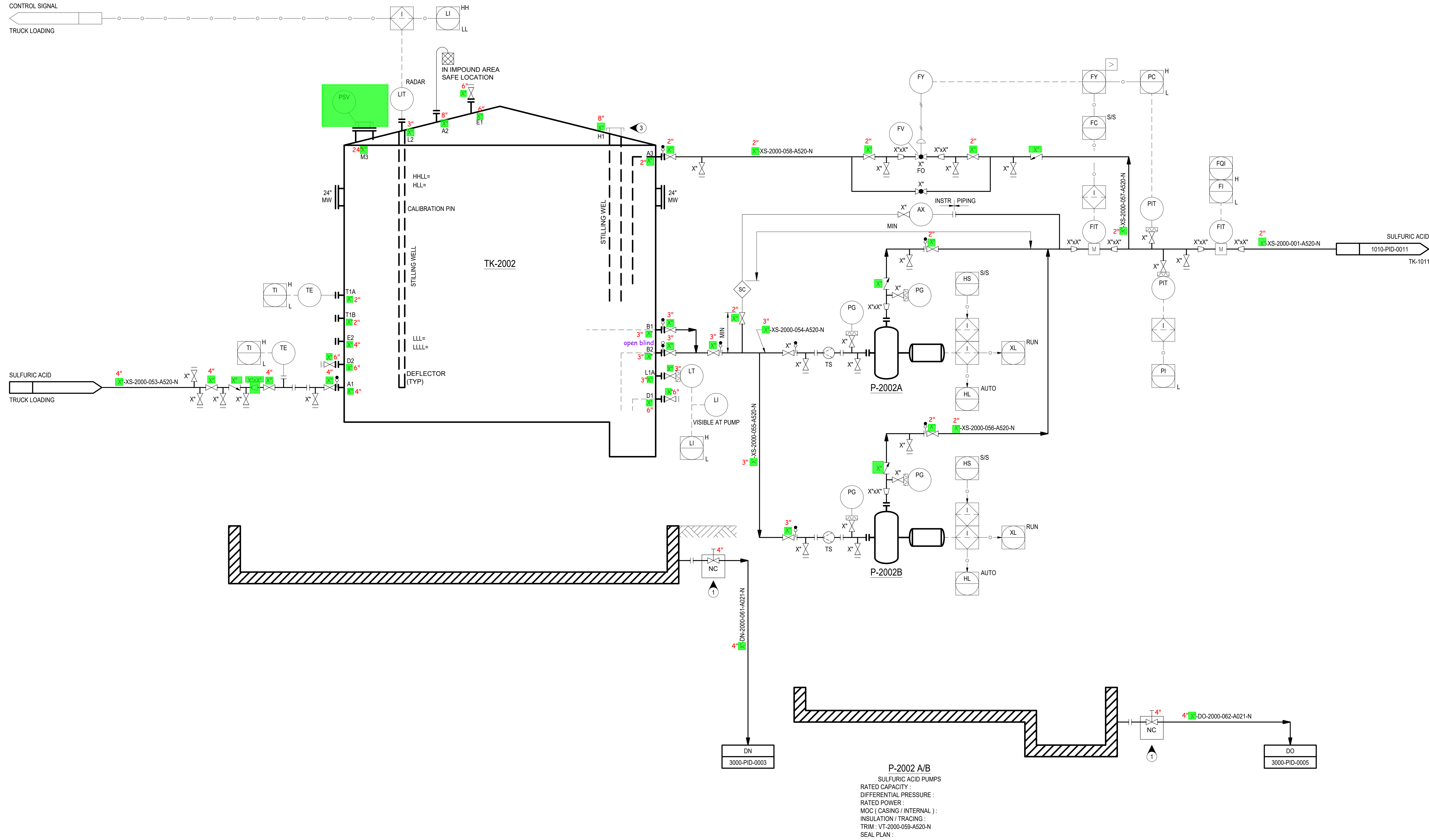
REV A

CHECK PRINT

TK-2002
SULFURIC ACID TANK
SIZE ID x HEIGHT :
TOTAL VOLUME :
DESIGN PRESSURE :
DESIGN TEMP :
MOC :
TRIM : VT-2000-059-A520-N
INSULATION / TRACING :

- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

1. VALVE STEM TO BE ACCESSIBLE FROM GRADE.
2. IMPOUND AREA DEDICATED TO ACID TANK. COMMON SUMPS AND DRAINS SHOWN ON THIS P&ID.
3. ACCESS NEAR EDGE.



CHECK PRINT

[illegible]

REFERENCE DRAWINGS

[illegible]

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO.

PIPING & INSTRUMENTATION DIAGRAM

SULFURIC ACID STORAGE

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW



**SSP ENGINEERING
& CONSULTING**
13100 NORTHWEST FREEWAY
SUITE 625,
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

**ENGINEERING
CONSULTING**
WEST FREEWAY
E 625,

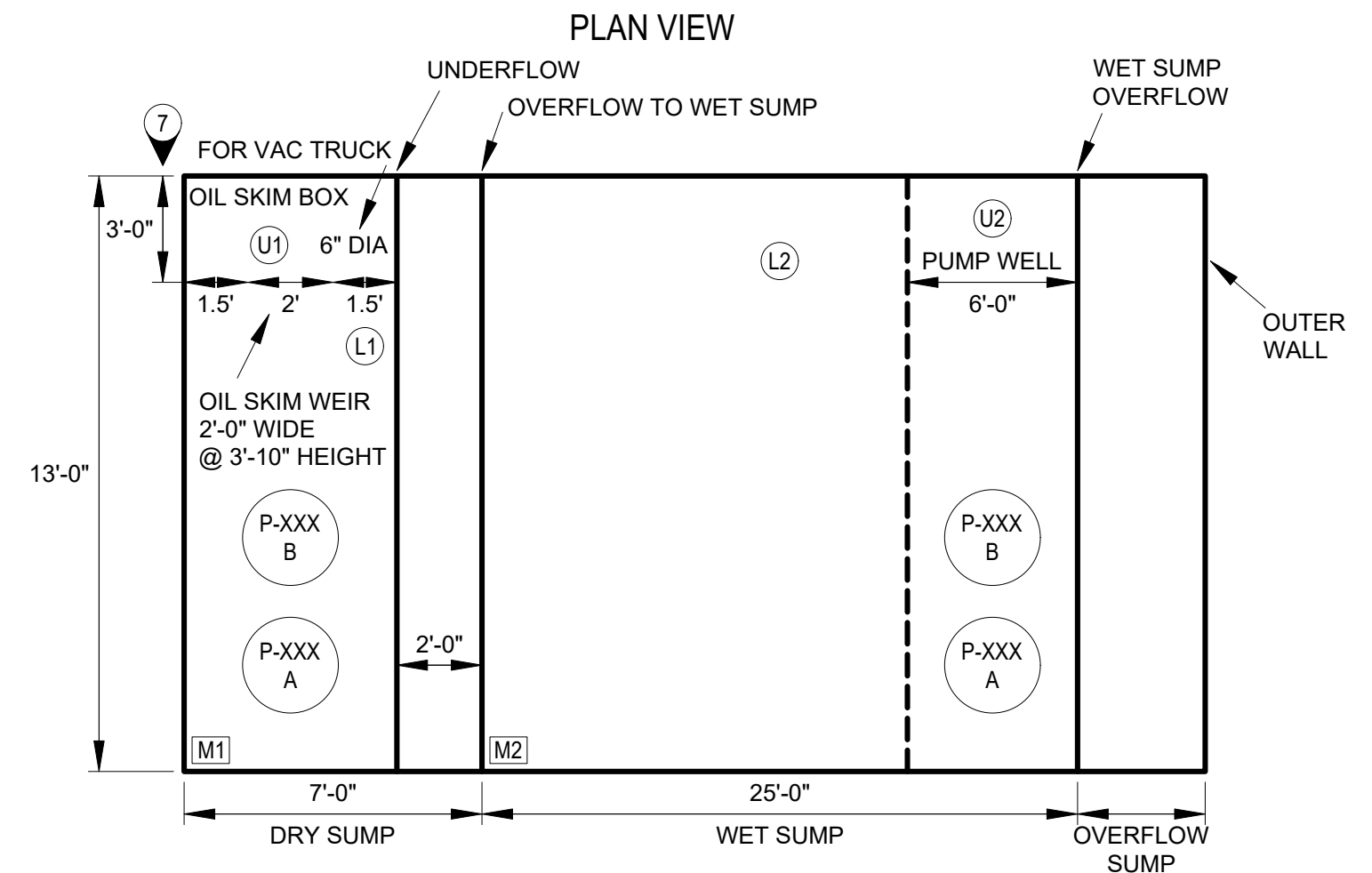
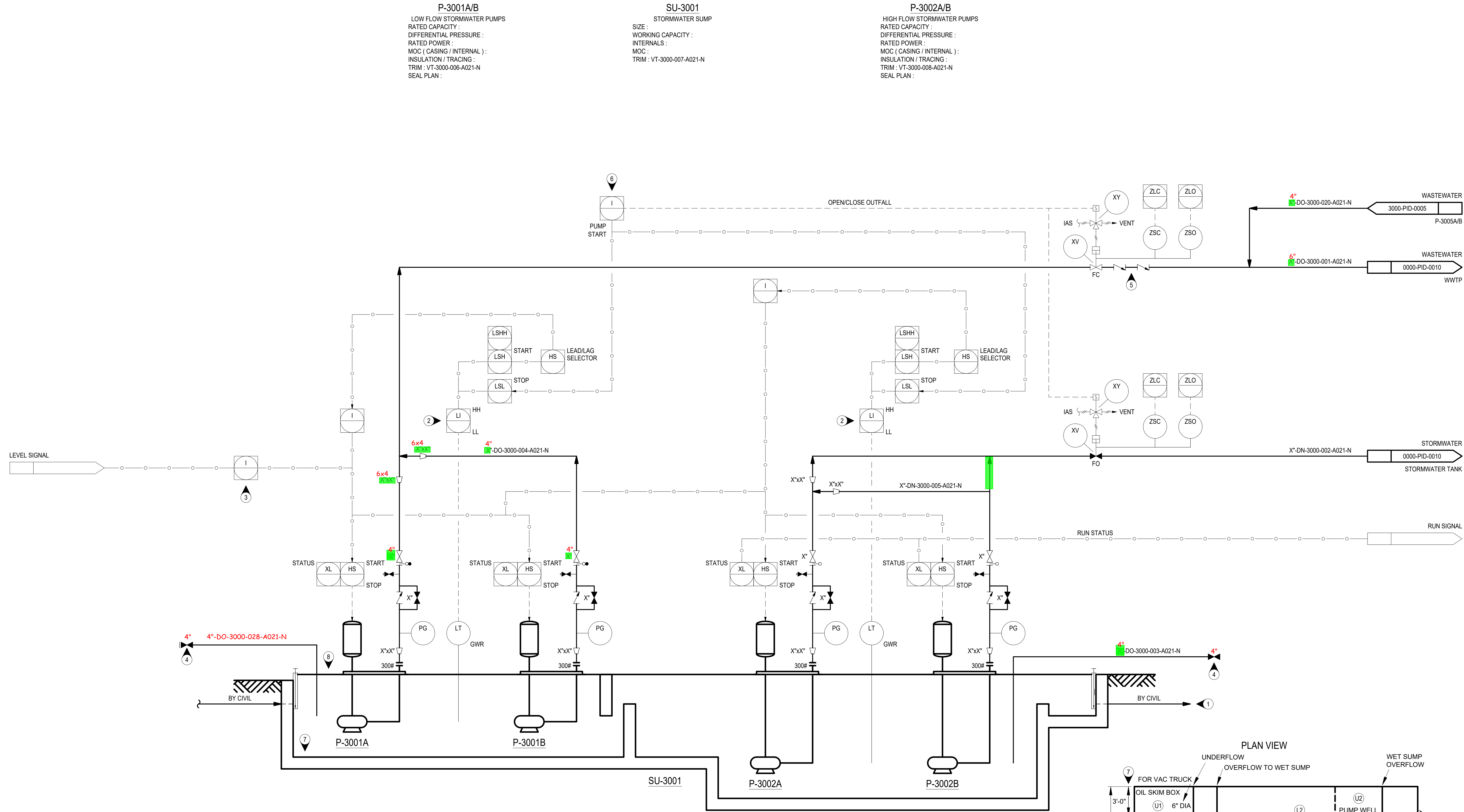
SALE: NONE

CHECKED: APR

ENG APVL: PDN

A

DRAWING LOCATION: c:\users\jason\one\drive - eng\ssps\files - 25ma02 - aromatics utility scope\old eyes\revin\04 - process\pids - process\pids, (Border Rev 1 5 18 25), PLOT DATE: ---, PLOT BY: jason, PLOTTED BY: jason, 3000-pid-0001.dwg



- GENERAL NOTES
- FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
 - INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
 - CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
- TO STORMWATER DITCH.
 - FOR EACH SET OF PUMPS, THE LEAD PUMP WILL START ON HIGH LEVEL AND THE LAG PUMP WILL START ON HIGH HIGH LEVEL IN THEIR RESPECTIVE SUMP. EACH SET OF PUMPS WILL STOP ON LOW LEVEL IN THEIR RESPECTIVE SUMP.
 - FIRST FLUSH PUMP TRIP. BOTH SETS OF PUMPS WILL STOP ONCE TOTAL FIRST FLUSH VOLUME IS COLLECTED.
 - FOR TRUCK LOADING.
 - DISSIMILAR CHECK VALVE TYPES.
 - SYSTEM RESET SEQUENCE (MANUAL RESET) IS:
 - INTERLOCK OPENS SHUTOFF VALVE TO DITCH AND CLOSSES SHUTOFF VALVE TO TANK.
 - ONE LOW AND ONE HIGH PUMP WILL START.
 - WHEN LL REACHES IN THE RESPECTIVE SUMP, THE RESPECTIVE PUMPS TRIP.
 - FINALLY, RESET SHUT OFF VALVES TO NORMAL POSITION OF PUMPING TO THE TANK.
 - SEE PLAN VIEW FOR OIL SKIM BOX.
 - 24"x 24" ACCESS HATCH.

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

STORMWATER SUMP

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER
#13619

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

DRAWING NO: **3000-PID-0001**

REV **A**

CHECK PRINT

SU-3003

NONOILY WATER SUMP

SIZE :

WORKING CAPACITY :

INTERNALS :

MOC :

TRIM : VT-3000-014-A021-N

P-3003A/B

NONOILY WATER PUMPS

RATED CAPACITY :

DIFFERENTIAL PRESSURE :

RATED POWER :

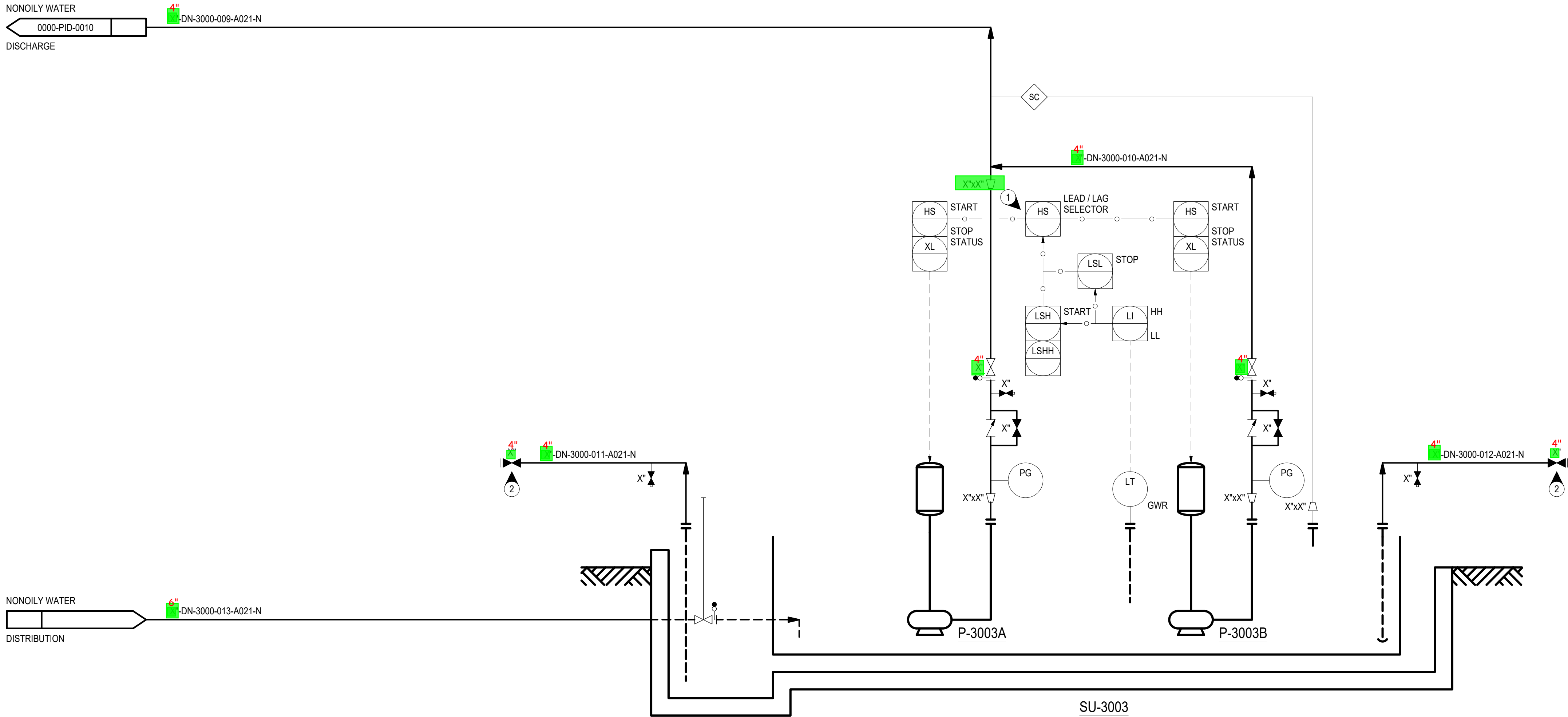
MOC (CASING / INTERNAL) :

INSULATION / TRACING :

TRIM : VT-3000-015-A021-N

SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.
- NOTES
1. EACH PUMP SET IS IN A LEAD/ LAG CONFIGURATION. THE LEAD PUMP IN THE PUMP SET WILL START ON HIGH LEVEL AND THE LAG PUMP WILL START ON HIGH HIGH LEVEL. BOTH PUMPS IN THE PUMP SET WILL STOP ON LOW LEVEL.
2. TRUCK LOADING CONNECTION.



CHECK PRINT

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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUE FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

WASTEWATER

NONOILY WATER COLLECTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY

SUITE 625

HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

SCALE: NONE

DRAWN BY: GS

CHECKED: APB

DESIGN BY:

ENG APVL: PDN

DRAWING NO:

3000-PID-0003

REV A

FIRM NUMBER

#13619

2. TRUCK LOADING CONNECTION.



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REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC

PLANT: PORT ARTHUR REFINERY

JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

WASTEWATER

OILY WATER COLLECTION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP

SSP ENGINEERING & CONSULTING

13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

FIRM NUMBER

#13619

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

3000-PID-0005

SU-3007

SANITARY LIFT STATION

DESIGN CAPACITY :

RATED POWER :

MOC :

TRIM : VT-3000-028-A021-N

INSULATION :

P-3007A/B

SANITARY LIFT STATION PUMPS

RATED CAPACITY :

DIFFERENTIAL PRESSURE :

RATED POWER :

MOC (CASING / INTERNAL) :

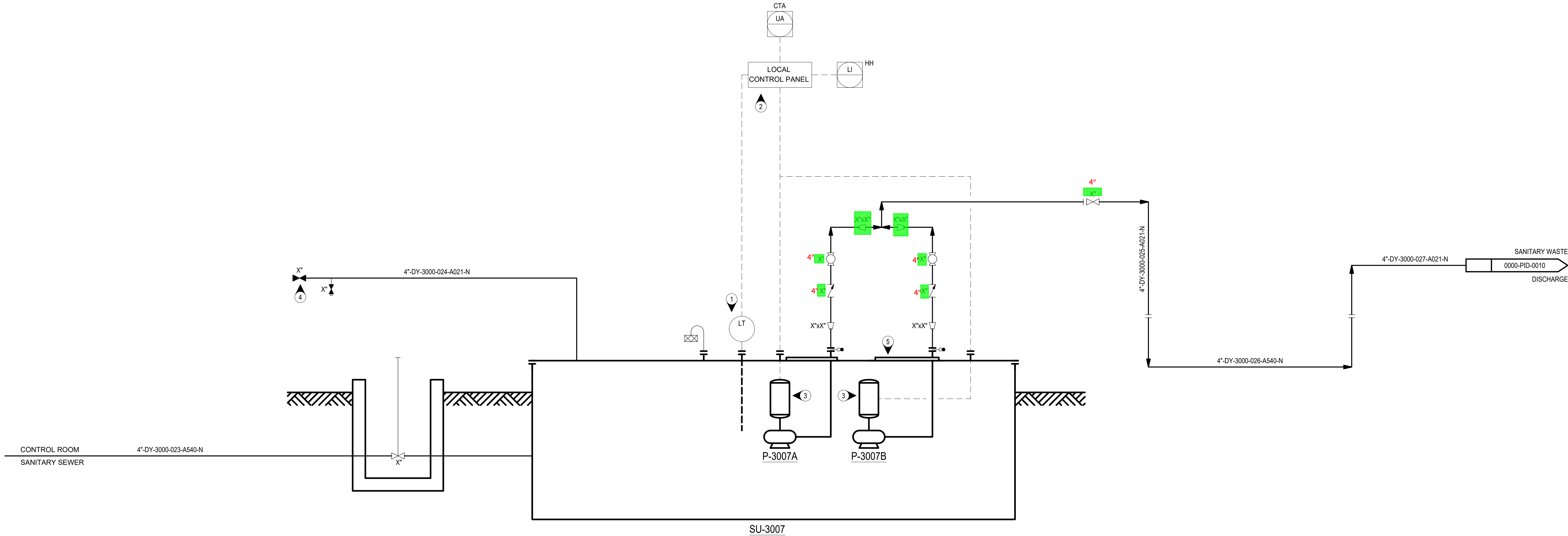
INSULATION / TRACING :

TRIM : VT-3000-029-A021-N

SEAL PLAN :

- GENERAL NOTES
- A. FOR PIPING AND INSTRUMENTATION LEGENDS AND SYMBOLS, SEE P&IDs 0000-PID-0001 THRU 0000-PID-00XX.
- B. INSTRUMENT NUMBERS ON THIS P&ID ARE PREFIXED BY THE UNIT NUMBER ASSOCIATED WITH THIS P&ID UNLESS OTHERWISE SPECIFIED.
- C. CONTROL VALVES AND BYPASS SIZES ARE PRELIMINARY AND WILL BE CONFIRMED DURING DETAIL DESIGN.

- NOTES
1. CONTROLS START ONE PUMP AT HIGH LEVEL AND THE SECOND AT HIGH HIGH LEVEL.
2. PRIMARY PUMP OPERATION ALTERNATIVES BETWEEN PUMPS.
3. LOW LIQUID LEVEL STOPS BOTH PUMPS.
4. TRUCK LOADING.
5. 38"x20" PUMP ACCESS DOOR.



CHECK PRINT

NOT PUBLISHED. ALL RIGHTS RESERVED BY THE ENGINEER. DRAWINGS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE ISSUED TO DESCRIBE DESIGN INTENT. THEY ARE NOT TO BE USED ON OTHER PROJECTS OR EXTENSIONS TO THIS PROJECT. EXCEPT BY AGREEMENT IN WRITING AND WITH APPROPRIATE COMPENSATION TO THE ENGINEER, CONTRACTOR, SUB-CONTRACTORS, & SUPPLIERS ARE RESPONSIBLE FOR CONFIRMING AND CORRELATING DIMENSIONS AND CONDITIONS AT THE JOB SITE AND SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY ERRORS OR DISCREPANCIES IN REGARD TO SUCH DRAWINGS. DESCRIBING DESIGN INTENT SHALL NOT INDICATE EVERY DETAIL OR SYSTEM FOR CONSTRUCTION. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE TO PROVIDE ADDITIONAL DRAWINGS AS NECESSARY FOR PERMITS, ERECTION, INSTALLATION, AND COORDINATION OF ALL WORK. CONTRACTOR, SUB-CONTRACTORS, AND SUPPLIERS SHALL BE RESPONSIBLE FOR ALL LABOR AND MATERIALS REQUIRED IN ACCORDANCE WITH DESIGN INTENT. ALL MATERIALS SHALL BE NEW UNLESS SPECIFIED OTHERWISE AND SHALL BE INSTALLED LEVEL, PLUMB, STRAIGHT, AND TRUE TO LINE, IN ACCORDANCE WITH THE BEST PRACTICES OF THE TRADES. THE ENGINEER SHALL NOT BE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES, OR FOR SAFETY PRECAUTIONS AND PROGRAMS IN CONNECTION WITH THE WORK. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH APPLICABLE CODES AND ORDINANCES.

REFERENCE DRAWINGS:

REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD	REV #	DATE	DESCRIPTION	DRWN	CHKD	APVD
A	12-05-25	ISSUED FOR APPROVAL	GS	APB	PDN						

MOTIVA

CLIENT: MOTIVA ENTERPRISES LLC
PLANT: PORT ARTHUR REFINERY
JOB NO:

PIPING & INSTRUMENTATION DIAGRAM

WASTEWATER

SANITARY LIFT STATION

PROJ TITLE: AROMATICS UTILITY SCOPE REVIEW

SSP ENGINEERING & CONSULTING
13100 NORTHWEST FREEWAY
SUITE 625
HOUSTON, TEXAS 77040

PROJECT NO: 25MA02

DRAWN BY: GS

DESIGN BY:

DRAWING NO:

SCALE: NONE

CHECKED: APB

ENG APVL: PDN

3000-PID-0007

REV A

FIRM NUMBER #13619